Active Directory Certificate Services
Hardened PKI Design

Audit-ready architecture, configuration evidence, and governance controls.

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Table of Contents

1. Executive Summary ***4***

2. PKI Overview and Enterprise Importance ***4***

3. Document Control ***4***

4. Change Log ***4***

5. Governance, Change Control, and Ownership ***4***

6. Risk Assessment and Cryptographic Design Decisions ***4***

7. Control Framework Alignment ***5***

7.1 NIST SP 800-53 Rev. 5 (Selected) *5*

7.2 ISO/IEC 27001:2022 (Selected) *5*

8. Roles and Responsibilities (RACI) ***5***

9. Cryptographic Reassessment Triggers ***5***

10. Standards Alignment and Numbering Convention ***6***

Appendix A – Certificate Server Configuration ***7***

Background *7*

Configuration for Template Issuance *7*

Appendix B – Certificate Template Configuration Artifacts ***8***

Template: Computer – Hardened *9*

Template: Web Server – Hardened *15*

Template: Domain Controller Authentication – Hardened *22*

Template: Domain Controller – Hardened *28*

Template: RDP-WinRM - Hardened *35*

Template: Kerberos Authentication - Hardened *42*

Template: Web Server – Appliance *49*

Template: Code Signing – Hardened *56*

Template: Directory Email Replication – Hardened *63*

Appendix C – Active Directory Security Groups ***70***

draft

Group: PKI-CodeSigners *70*

Group: PKI-ApplianceCerts *73*

Group: PKI-Admins *78*

Appendix D – Active Directory Group Policy* *78

Scheduled Task Enforcement *78*

Creating the Schedule Task 79

Create a New Group Policy 79

Appendix E – Applied Certificate Sampling by Intent* *84

Domain Controller Authentication *84*

Directory Email Replication *86*

Kerberos Authentication *87*

PKI Appliance Certs *89*

PKI Code Signers *89*

RDS WinRM *89*

Web Server *89*

Web Server Appliance *89*

Appendix F – Auditor Walkthrough Script* *90

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1. Executive Summary

This document defines the hardened Active Directory Certificate Services (AD CS) baseline implemented for the environment. The design enforces modern cryptography, least-privilege enrollment, and repeatable certificate lifecycle management. External-facing services continue to use commercial/public certificates; this PKI supports internal trust and secure management channels.

2. PKI Overview and Enterprise Importance

Public Key Infrastructure (PKI) provides enterprise authentication, encryption, integrity, and non-repudiation. In Active Directory environments, PKI underpins Kerberos (including PKINIT), LDAPS, TLS-protected management (WinRM/RDP), and service-to-service trust. A hardened PKI reduces operational risk and limits trust sprawl.

3. Document Control

Document Title: Active Directory Certificate Services – Hardened PKI Design

Author: realworldtec

Organization: realworldtec

Document Version: 3.6

Date Created: 2026-04-14

Last Reviewed: 2026-04-14

Reviewed By: Internal Security & Infrastructure

Approval Status: Approved for Audit Presentation

4. Change Log

Version	Date	Author	Description
3.5	2025-12-17	realworldtec	Reformatted into audit-ready structure with appendices and control mapping.

5. Governance, Change Control, and Ownership

PKI configuration is security-critical and governed through explicit change control, peer review, and periodic reassessment. Certificate template changes, enrollment scope changes, and CA configuration changes require documented review prior to implementation.

6. Risk Assessment and Cryptographic Design Decisions

Cryptographic selections balance security strength, interoperability, and operational stability. RSA was selected for broad enterprise compatibility. RSA 3072 is used for domain controllers, Kerberos, WinRM, and RDP to exceed baseline requirements while avoiding RSA 4096 operational cost. RSA 2048 is used for internal web services where TLS session security is primarily provided by ephemeral key exchange and symmetric ciphers. SHA-256

was selected for standards compliance and interoperability. Private keys are non-exportable by default; exportability is permitted only by exception.

7. Control Framework Alignment

7.1 NIST SP 800-53 Rev. 5 (Selected)

IA-7: Approved cryptographic mechanisms (KSP/CNG, RSA, SHA-256).

IA-5: Authenticator management (lifetimes, renewal, non-exportable keys).

SC-12 / SC-13: Key establishment and protection (enterprise PKI and TLS trust).

CM-2 / CM-6: Baseline and configuration settings (documented hardened templates).

AU-3 / AU-12: Auditability (CA DB records and revocation inclusion).

7.2 ISO/IEC 27001:2022 (Selected)

A.5.15: Access control (least-privilege enrollment).

A.5.17: Authentication information (certificate-based authentication).

A.8.24: Use of cryptography (approved algorithms and documented decisions).

A.8.28: Secure authentication (Kerberos PKINIT, LDAPS, TLS management).

A.8.32: Change management (versioning, review, approval).

8. Roles and Responsibilities (RACI)

R = Responsible, A = Accountable, C = Consulted, I = Informed

Activity	Security	Infrastructure	Operations	Audit
PKI Architecture & Design	A	R	C	I
Certificate Template Management	A	R	C	I
CA Operations & Monitoring	A	R	R	I
Cryptographic Policy Decisions	A	C	I	I
Audit Support & Evidence	R	C	I	I

9. Cryptographic Reassessment Triggers

Reassessment will occur upon any of the following:

Updated NIST guidance impacting RSA or SHA-256

New regulatory or compliance requirements

Material changes to the enterprise threat landscape

Introduction of services requiring ECC

Major OS / Active Directory functional level changes

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10. Standards Alignment and Numbering Convention

This document uses numbered sections and appendices to support audit traceability and ISO-style navigation. Headings are intended to be referenced directly during walkthroughs and evidence requests.

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Appendix A – Certificate Server Configuration

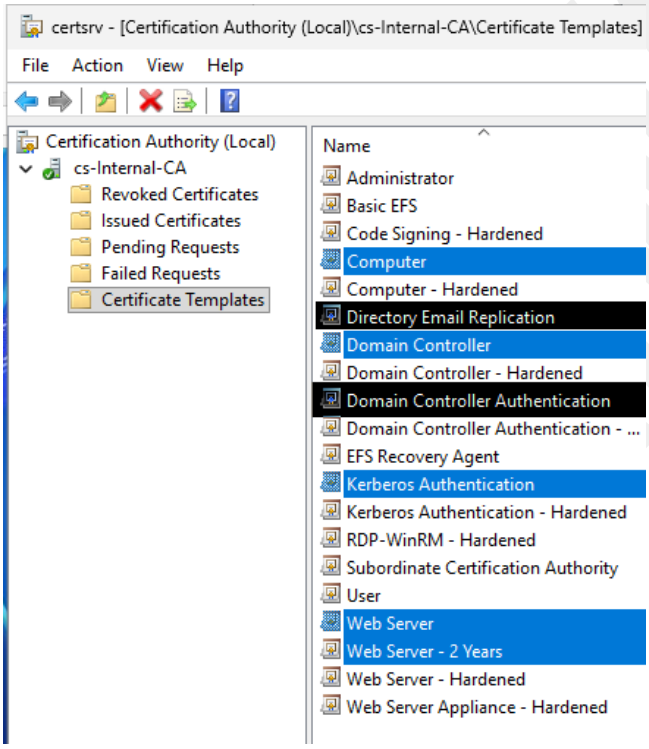
Background

There are no known scripting commands to create certificate templates. The process is manual and laborious that involves *duplicating* like kind certificates and modifying them accordingly. Appendix B demonstrates the settings needed for the various intended purposes. The Certificate Server template application is located at: C:\Windows\System32\certtmpl.msc

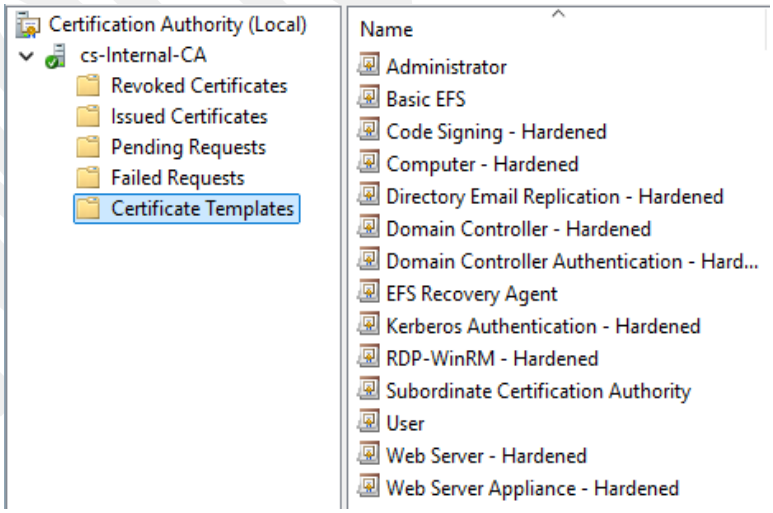
Configuration for Template Issuance

Certificates will use explicit definitions as defined in Appendix B. The newly created templates must now be added for issuance using C:\Windows\system32\certsrv.msc.
To add the template for issuance you right click on Certificate Templates/New/Certificate to Issue, then add the defined hardened templates one by one. Cleanup of the initial configuration should be performed to remove the base install weak Cryptography template definitions from being issued by the Certificate Authority. The "* - Hardened" templates will supersede the like kind duplications. After adding, sorting the template list alphabetically then seeing the two like names makes it easy to identify what should be deleted thus retaining only the hardened labeled version. By cleaning up the base installation our hardened templates will be enforced without duplication of weaker base install cryptology thus eliminating certificate ambiguity when utilized.

Before Delete



After Delete



Appendix B – Certificate Template Configuration Artifacts

The following pages contain the original certificate template configuration artifacts, including screenshots captured during implementation. These artifacts serve as audit evidence and operational reference.

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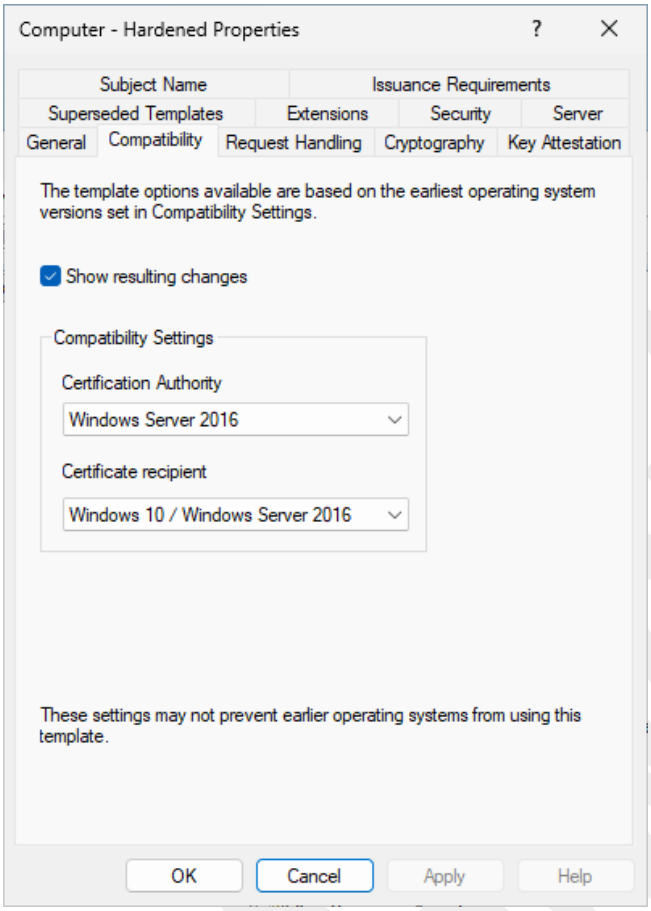
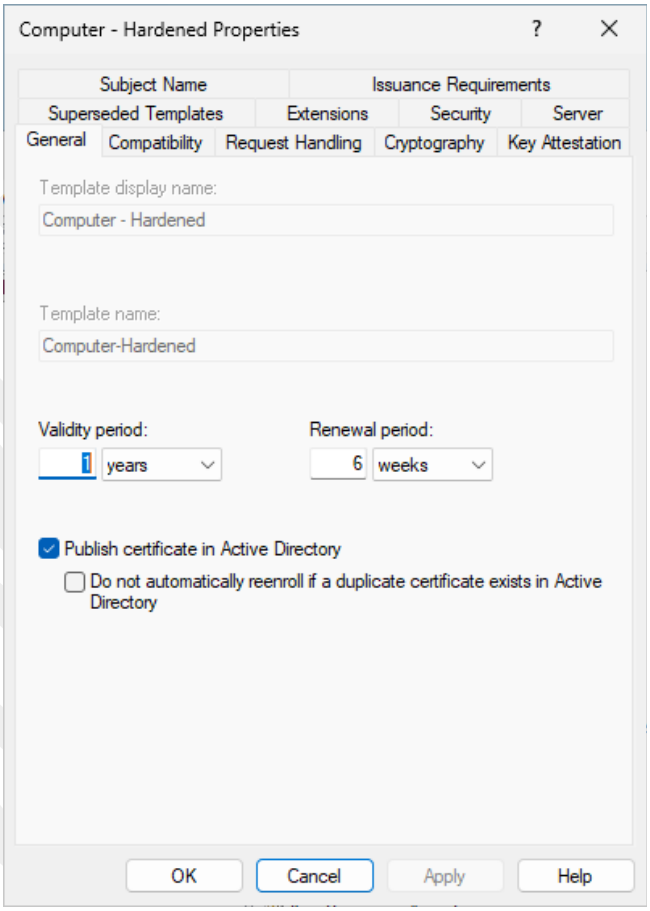
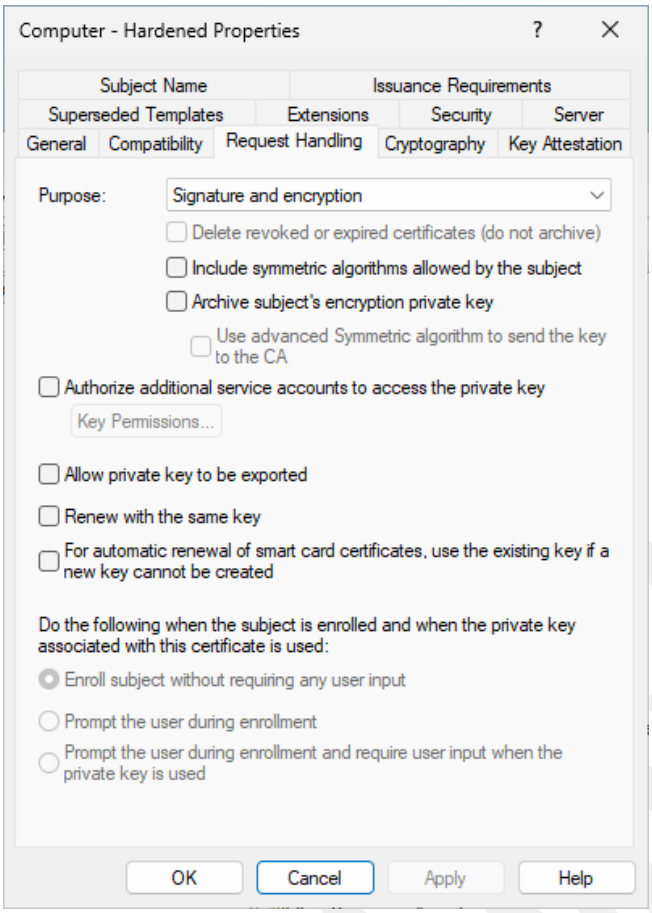
Template: Computer – Hardened

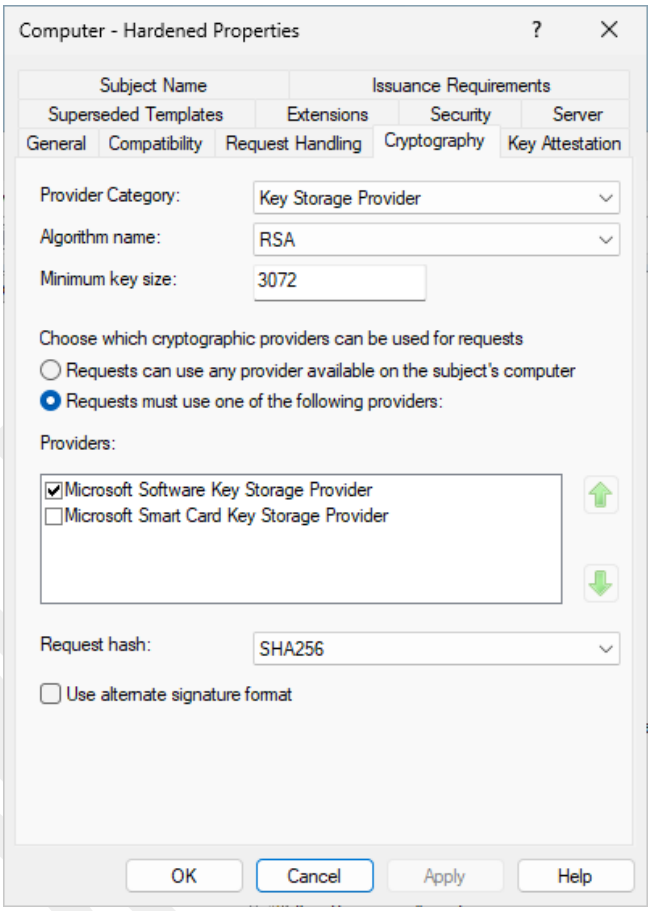
```yaml
purpose:
  - Issues the primary machine identity certificate for domain-joined computers
  - Supports mutual authentication, TLS, and machine-level trust
  - Enforces modern cryptographic standards for workstation and member servers
based_on:
  - Computer
compatibility:
  certification_authority: Windows Server 2016
  certificate_recipient: Windows 10 / Windows Server 2016
cryptography:
  provider_category: Key Storage Provider
  provider: Microsoft Software Key Storage Provider
  algorithm: RSA
  minimum_key_size: 3072
  hash_algorithm: SHA256
  private_key_exportable: false
  requests_can_use_any_provider: false
subject_name:
  source: Active Directory
  subject_name_format: None
  subject_alternative_name:
    - DNS
extensions:
  application_policies:
    - Client Authentication
    - Server Authentication
  key_usage:
    - Digital Signature
    - Key Encipherment
  basic_constraints:
    ca: false
validity:
  validity_period: 1 year
  renewal_period: 6 weeks
  publish_to_active_directory: true
enrollment:
  domain_computers:
    enroll: true
    autoenroll: true
  domain_admins:
    read: true
issuance_requirements:
  ca_manager_approval: false
  authorized_signatures_required: 0
request_handling:
  purpose: Signature and Encryption
  archive_private_key: false
key_attestation:
  required: false

security:
  certificate_database_storage: enabled
```

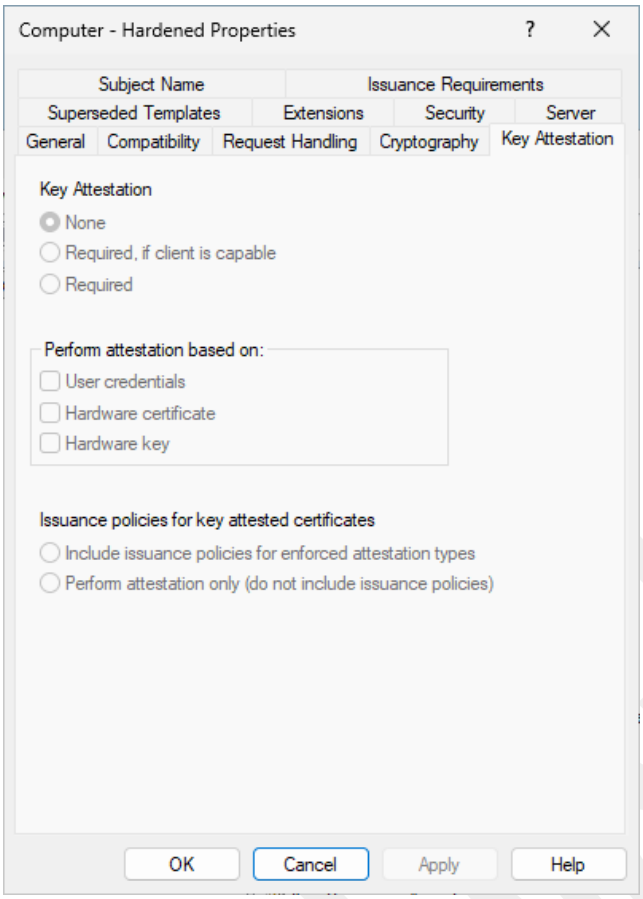
```
  revocation_information: included
supersedes:
  - Computer
notes:
  - Replaces the default Computer template
  - Ensures consistent machine authentication posture across the domain
```

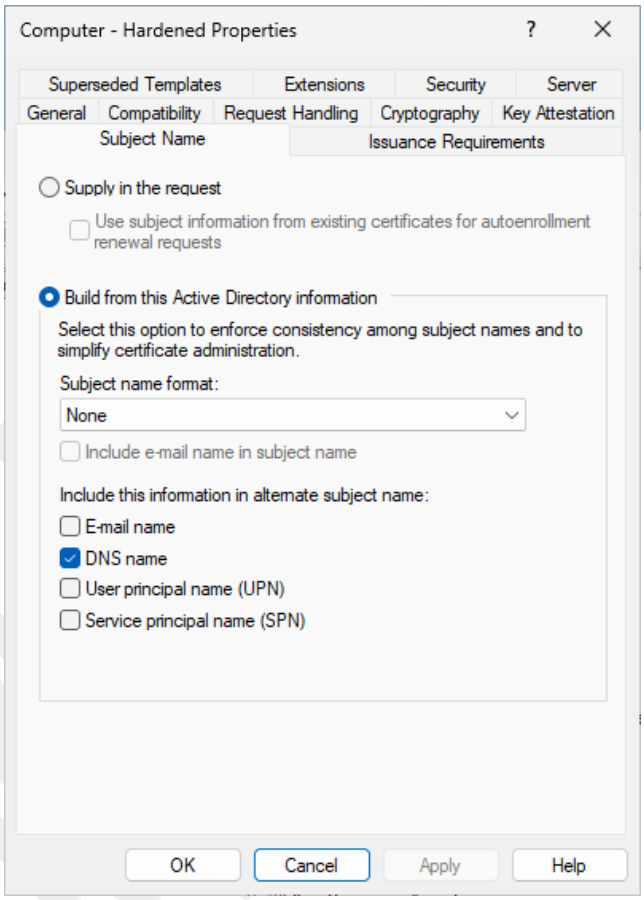


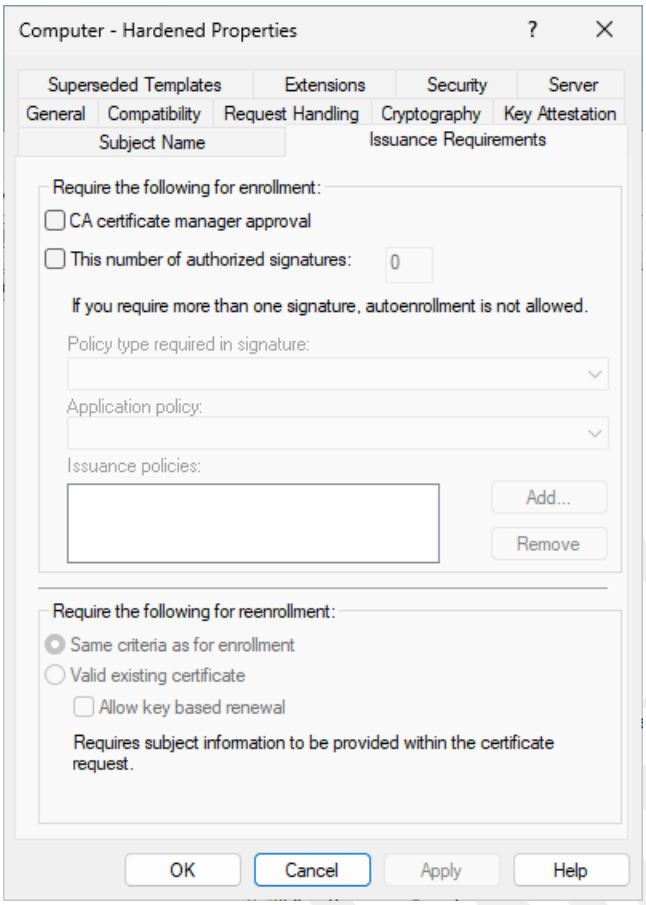


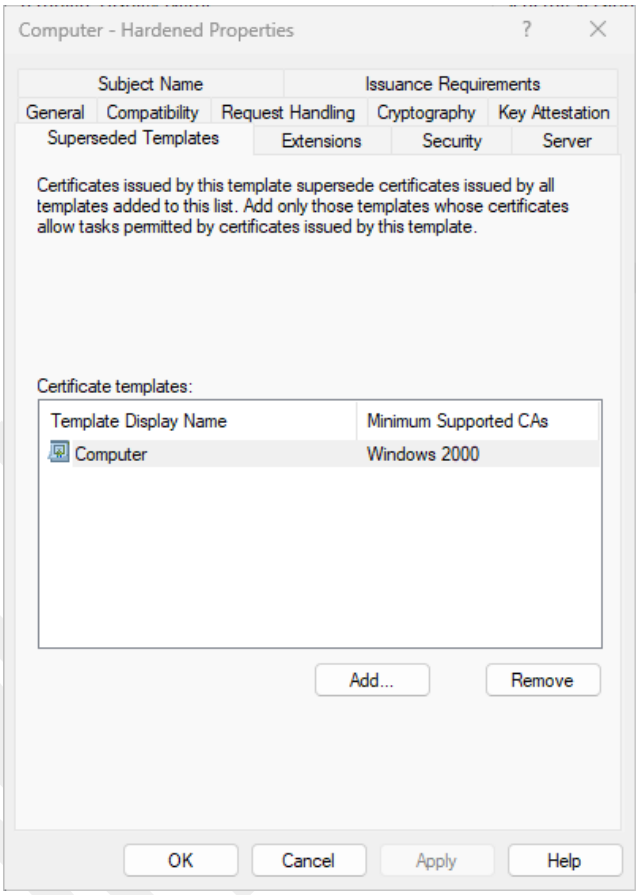

Computer - Hardened Properties
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
General
Compatibility
Request Handling
Cryptography
Key Attestation
Purpose:
Signature and encryption
Delete revoked or expired certificates (do not archive)
Include symmetric algorithms allowed by the subject
Archive subject's encryption private key
Use advanced Symmetric algorithm to send the key to the CA
Authorize additional service accounts to access the private key
Key Permissions...
Allow private key to be exported
Renew with the same key
For automatic renewal of smart card certificates, use the existing key if a new key cannot be created
Do the following when the subject is enrolled and when the private key associated with this certificate is used:
Enroll subject without requiring any user input
Prompt the user during enrollment
Prompt the user during enrollment and require user input when the private key is used
OK
Cancel
Apply
Help


Computer - Hardened Properties
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
General
Compatibility
Request Handling
Cryptography
Key Attestation
Provider Category:
Key Storage Provider
Algorithm name:
RSA
Minimum key size:
3072
Choose which cryptographic providers can be used for requests
Requests can use any provider available on the subject's computer
Requests must use one of the following providers:
Providers:
Microsoft Software Key Storage Provider
Microsoft Smart Card Key Storage Provider
Request hash:
SHA256
Use alternate signature format
OK
Cancel
Apply
Help


Computer - Hardened Properties
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
General
Compatibility
Request Handling
Cryptography
Key Attestation
Key Attestation
None
Required, if client is capable
Required
Perform attestation based on:
User credentials
Hardware certificate
Hardware key
Issuance policies for key attested certificates
Include issuance policies for enforced attestation types
Perform attestation only (do not include issuance policies)
OK
Cancel
Apply
Help


Computer - Hardened Properties
Superseded Templates
Extensions
Security
Server
General
Compatibility
Request Handling
Cryptography
Key Attestation
Subject Name
Issuance Requirements
Supply in the request
Use subject information from existing certificates for autoenrollment renewal requests
Build from this Active Directory information
Select this option to enforce consistency among subject names and to simplify certificate administration.
Subject name format:
None
Include e-mail name in subject name
Include this information in alternate subject name:
E-mail name
DNS name
User principal name (UPN)
Service principal name (SPN)
OK
Cancel
Apply
Help





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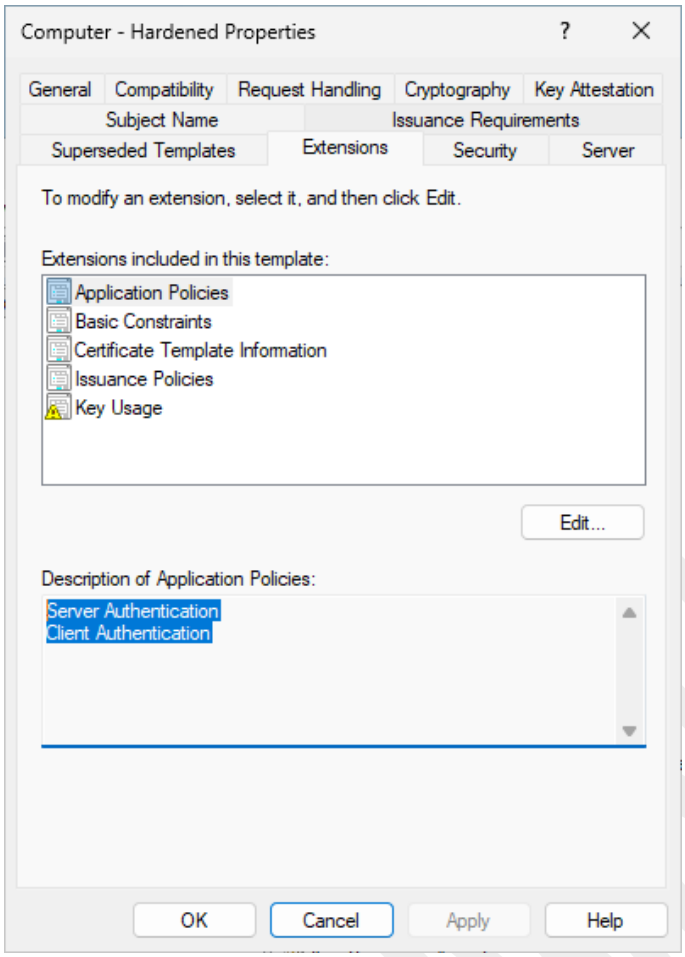
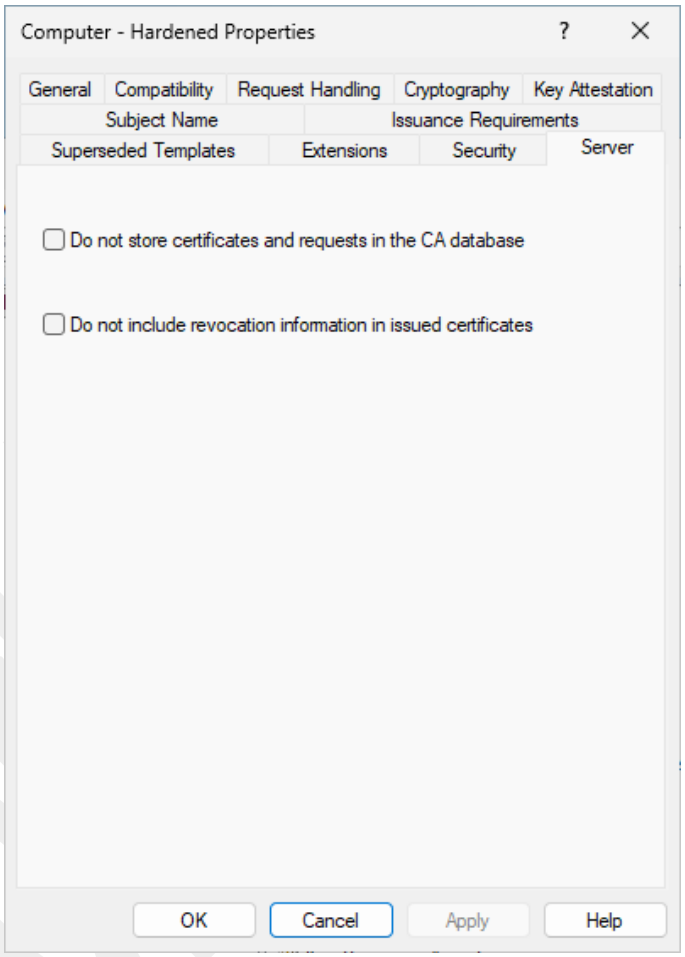
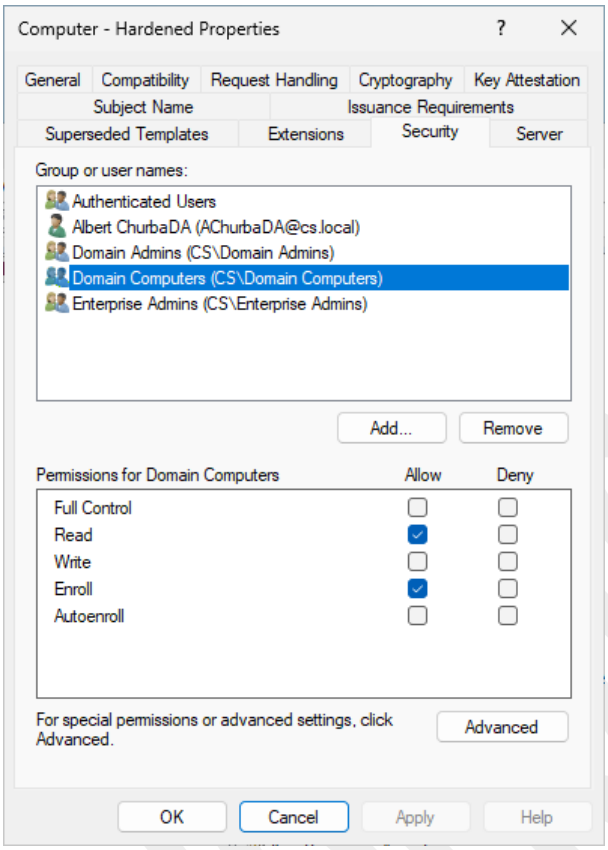

Computer - Hardened Properties
General
Compatibility
Request Handling
Cryptography
Key Attestation
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
To modify an extension, select it, and then click Edit.
Extensions included in this template:
Application Policies
Basic Constraints
Certificate Template Information
Issuance Policies
Key Usage
Edit...
Description of Application Policies:
Server Authentication
Client Authentication
OK
Cancel
Apply
Help


Computer - Hardened Properties
General
Compatibility
Request Handling
Cryptography
Key Attestation
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
Do not store certificates and requests in the CA database
Do not include revocation information in issued certificates
OK
Cancel
Apply
Help



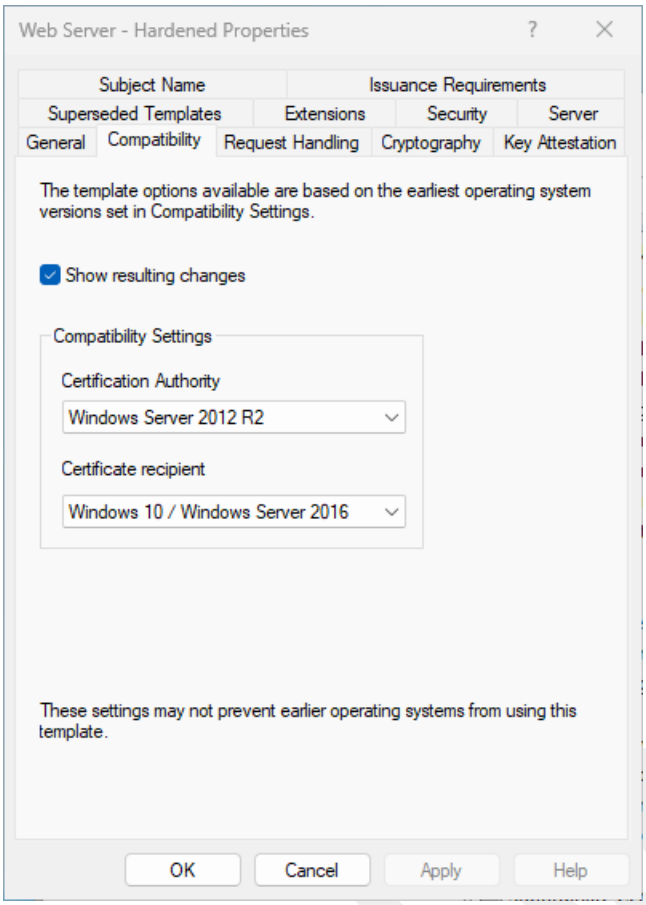
Template: Web Server – Hardened

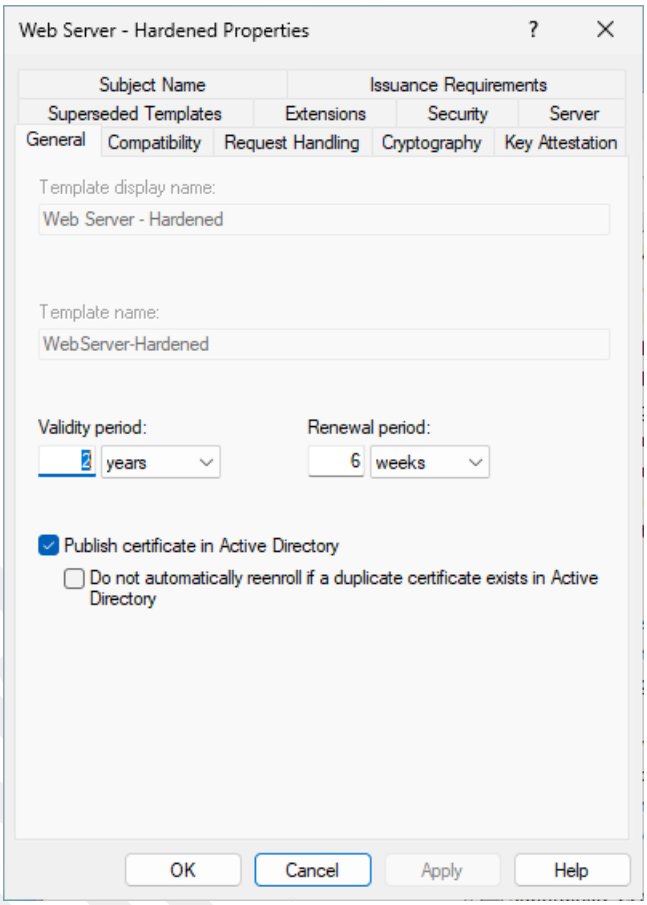
```
purpose:
  - Issues TLS certificates for internal web services
  - Secures IIS, WinRM, and application-layer HTTPS endpoints
  - Prevents private key export and weak cryptographic usage
based_on:
  - Web Server – 2 Year
compatibility:
  certification_authority: Windows Server 2016
  certificate_recipient: Windows Server 2016
cryptography:
  provider_category: Key Storage Provider
  provider: Microsoft Software Key Storage Provider
  algorithm: RSA
  minimum_key_size: 2048
  hash_algorithm: SHA256
  private_key_exportable: false
  requests_can_use_any_provider: false
subject_name:
  source: Request
  subject_alternative_name:
    - DNS
    - IP
extensions:
  application_policies:
```

```yaml
    - Server Authentication
  key_usage:
    - Digital Signature
    - Key Encipherment
  basic_constraints:
    ca: false
validity:
  validity_period: 2 years
  renewal_period: 6 weeks
  publish_to_active_directory: false
enrollment:
  domain_admins:
    enroll: true
  enterprise_admins:
    enroll: true
issuance_requirements:
  ca_manager_approval: false
  authorized_signatures_required: 0
request_handling:
  purpose: Signature and Encryption
  archive_private_key: false
key_attestation:
  required: false
security:
  certificate_database_storage: enabled
  revocation_information: included

supersedes:
  - Web Server
  - Web Server - 2 Years

notes:
  - Intended for internal HTTPS services only
  - SAN enforcement prevents CN-only misuse
```





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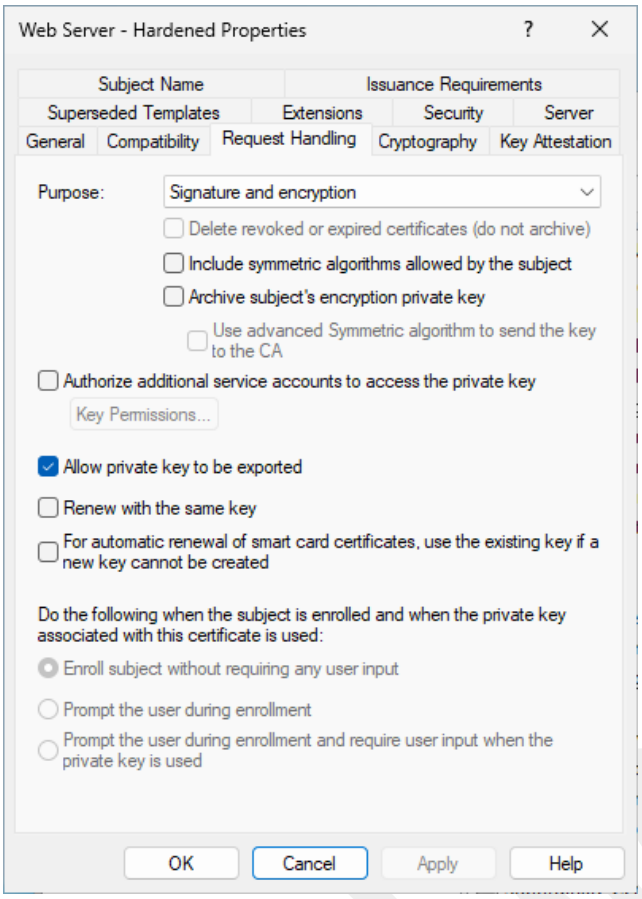

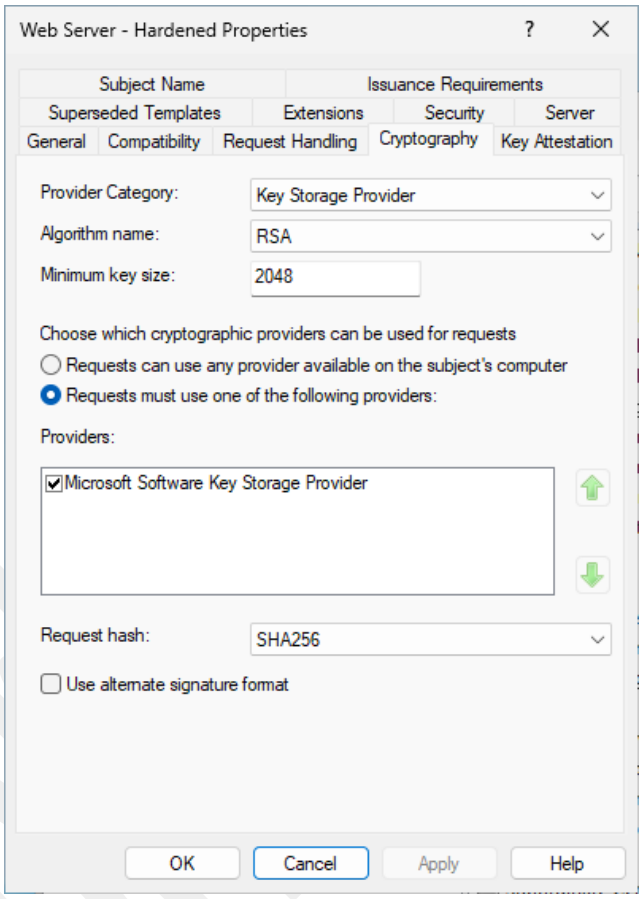

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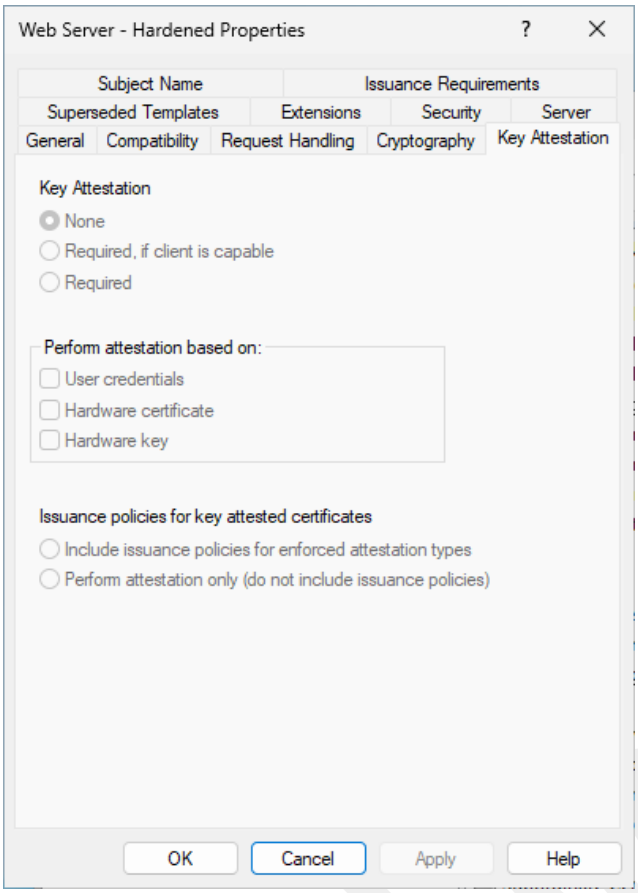
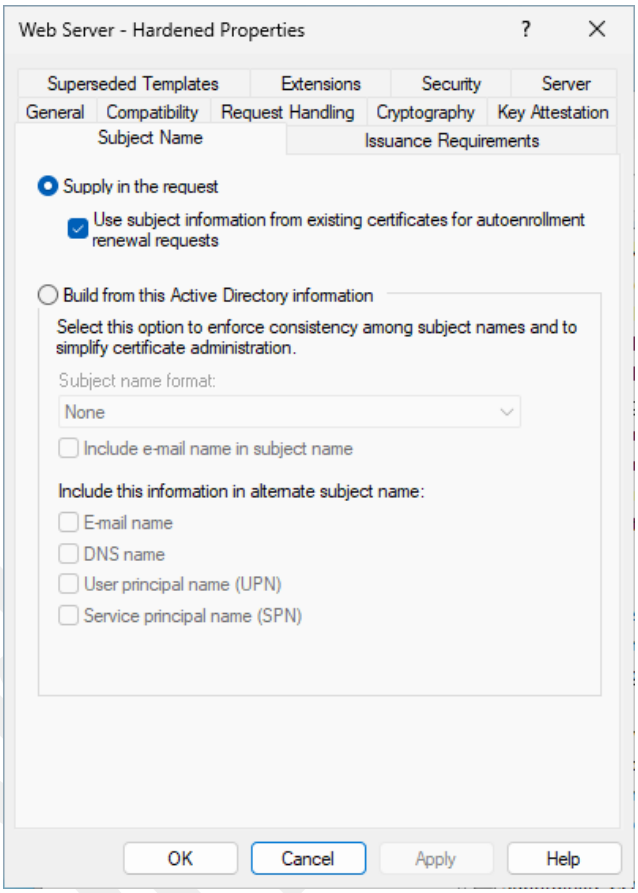
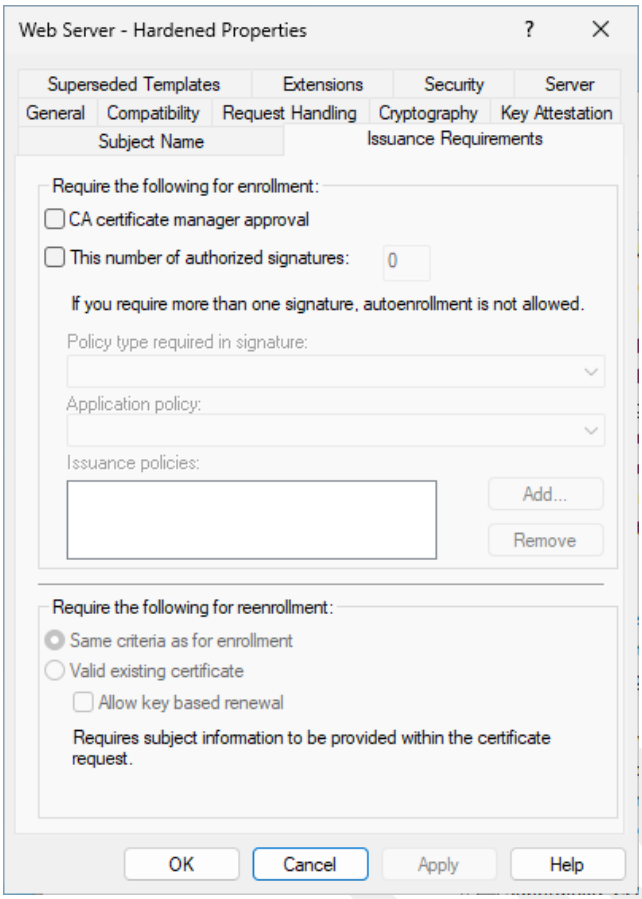
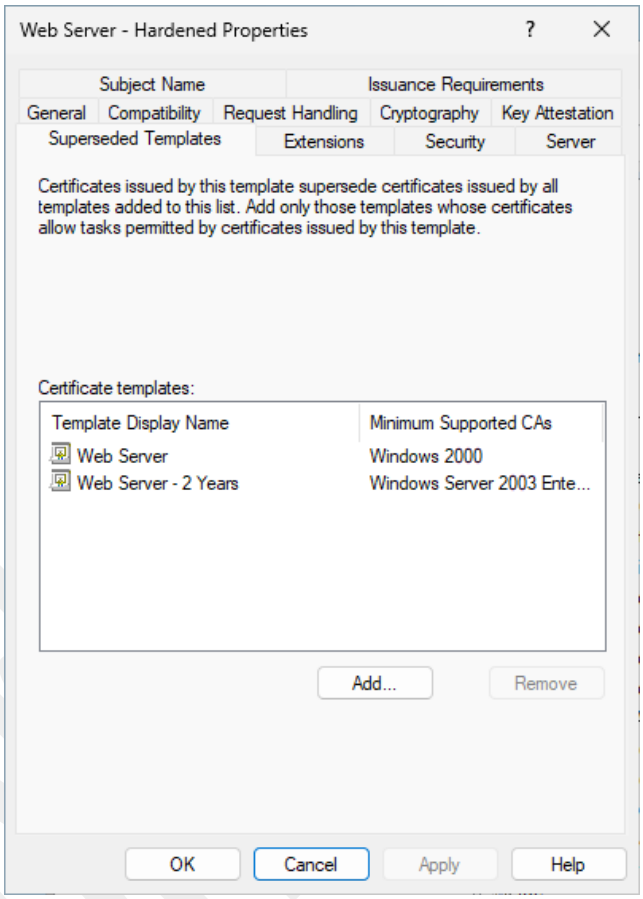
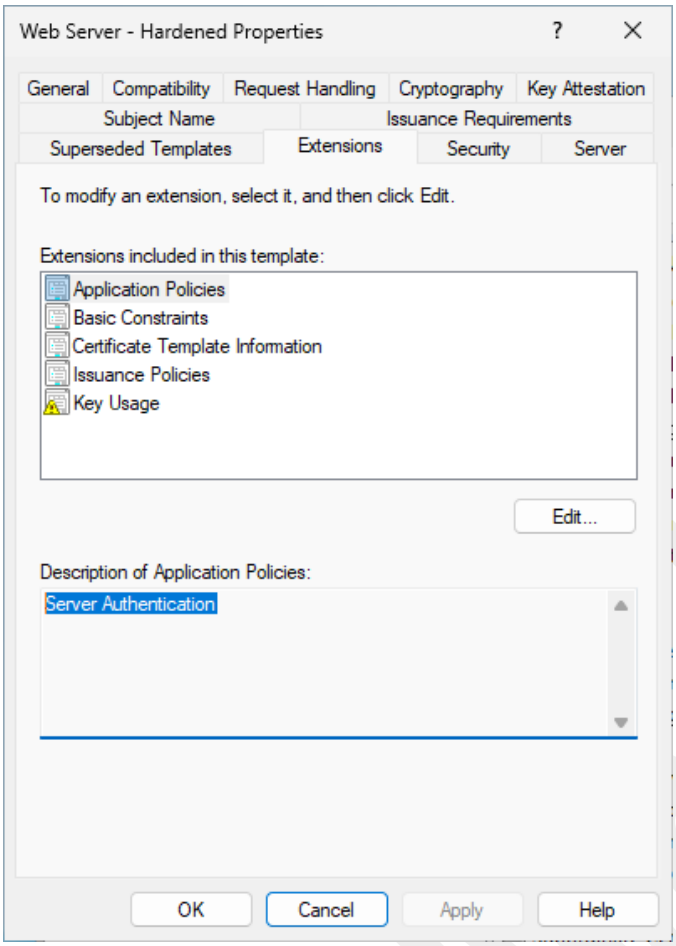
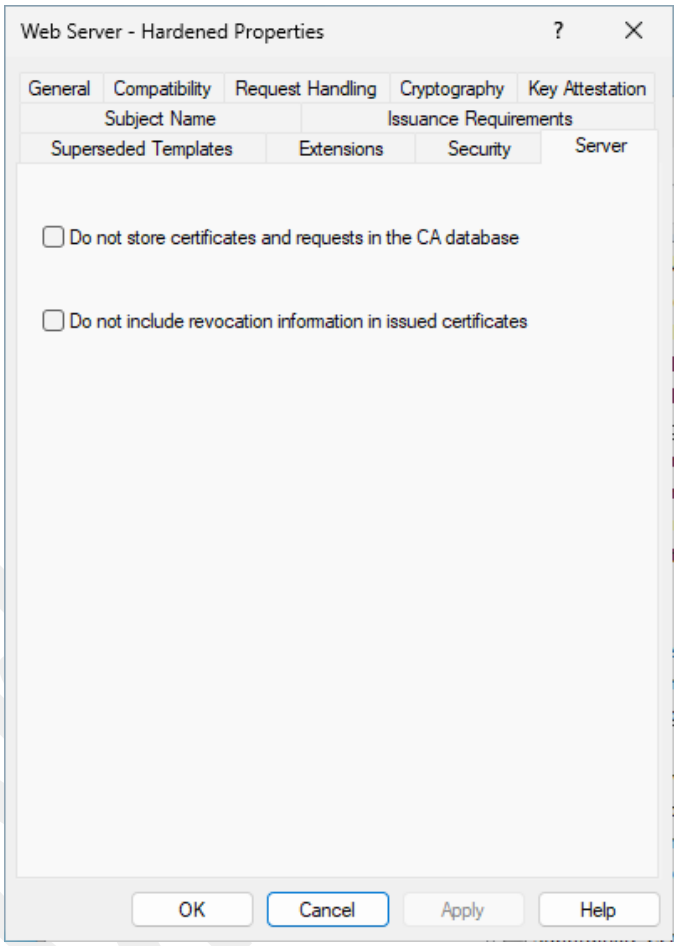

Web Server - Hardened Properties
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
General
Compatibility
Request Handling
Cryptography
Key Attestation
Key Attestation
None
Required, if client is capable
Required
Perform attestation based on:
User credentials
Hardware certificate
Hardware key
Issuance policies for key attested certificates
Include issuance policies for enforced attestation types
Perform attestation only (do not include issuance policies)
OK
Cancel
Apply
Help

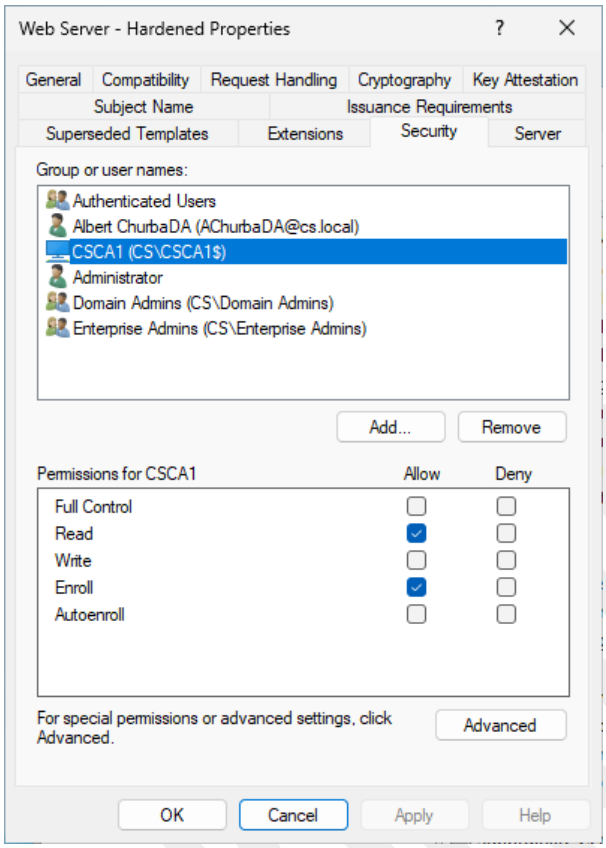

Web Server - Hardened Properties
Superseded Templates
Extensions
Security
Server
General
Compatibility
Request Handling
Cryptography
Key Attestation
Subject Name
Issuance Requirements
Supply in the request
Use subject information from existing certificates for autoenrollment renewal requests
Build from this Active Directory information
Select this option to enforce consistency among subject names and to simplify certificate administration.
Subject name format:
None
Include e-mail name in subject name
Include this information in alternate subject name:
E-mail name
DNS name
User principal name (UPN)
Service principal name (SPN)
OK
Cancel
Apply
Help


Web Server - Hardened Properties
Superseded Templates
Extensions
Security
Server
General
Compatibility
Request Handling
Cryptography
Key Attestation
Subject Name
Issuance Requirements
Require the following for enrollment:
CA certificate manager approval
This number of authorized signatures:
0
If you require more than one signature, autoenrollment is not allowed.
Policy type required in signature:
Application policy:
Issuance policies:
Add...
Remove
Require the following for reenrollment:
Same criteria as for enrollment
Valid existing certificate
Allow key based renewal
Requires subject information to be provided within the certificate request.
OK
Cancel
Apply
Help


Web Server - Hardened Properties
Subject Name
Issuance Requirements
General
Compatibility
Request Handling
Cryptography
Key Attestation
Superseded Templates
Extensions
Security
Server
Certificates issued by this template supersede certificates issued by all templates added to this list. Add only those templates whose certificates allow tasks permitted by certificates issued by this template.
Certificate templates:
Template Display Name
Minimum Supported CAs
Web Server
Windows 2000
Web Server - 2 Years
Windows Server 2003 Ente...
Add...
Remove
OK
Cancel
Apply
Help


Web Server - Hardened Properties
General
Compatibility
Request Handling
Cryptography
Key Attestation
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
To modify an extension, select it, and then click Edit.
Extensions included in this template:
Application Policies
Basic Constraints
Certificate Template Information
Issuance Policies
Key Usage
Edit...
Description of Application Policies:
Server Authentication
OK
Cancel
Apply
Help


Web Server - Hardened Properties
General
Compatibility
Request Handling
Cryptography
Key Attestation
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
Do not store certificates and requests in the CA database
Do not include revocation information in issued certificates
OK
Cancel
Apply
Help



Template: Domain Controller Authentication – Hardened

```
purpose:
  - Issues the primary machine identity certificate for Active Directory Domain Controllers
  - Enables secure LDAP (LDAPS), smart card logon support, and DC service authentication
  - Enforces modern cryptographic standards and correct SAN behavior for domain controllers
based_on:
  - Domain Controller Authentication
compatibility:
  certification_authority: Windows Server 2016
  certificate_recipient: Windows Server 2016
cryptography:
  provider_category: Key Storage Provider
  provider: Microsoft Software Key Storage Provider
  algorithm: RSA
  minimum_key_size: 3072
  hash_algorithm: SHA256
  private_key_exportable: false
  requests_can_use_any_provider: false
subject_name:
  source: Active Directory
  subject_name_format: None
  subject_alternative_name:
    - DNS
extensions:
  application_policies:
    - Client Authentication
```

```yaml
    - Server Authentication
    - Smart Card Logon
  key_usage:
    - Digital Signature
    - Key Encipherment
  basic_constraints:
    ca: false
validity:
  validity_period: 1 year
  renewal_period: 6 weeks
  publish_to_active_directory: true
enrollment:
  domain_controllers:
    enroll: true
    autoenroll: true
  enterprise_read_only_domain_controllers:
    enroll: true
    autoenroll: true
  domain_admins:
    read: true
issuance_requirements:
  ca_manager_approval: false
  authorized_signatures_required: 0
request_handling:
  purpose: Signature and Encryption
  archive_private_key: false
key_attestation:
  required: false
security:
  certificate_database_storage: enabled
  revocation_information: included
supersedes:
  - Domain Controller Authentication
notes:
  - Replaces the default Domain Controller Authentication template to enforce RSA 3072 / SHA256
  - SAN=DNS is required for LDAPS and modern DC identity validation
```

Domain Controller Authentication - Hardened Properti...

Subject Name | Issuance Requirements
Superseded Templates | Extensions | Security | Server
General | Compatibility | Request Handling | Cryptography | Key Attestation

The template options available are based on the earliest operating system versions set in Compatibility Settings.

☑ Show resulting changes

Compatibility Settings

Certification Authority
Windows Server 2016

Certificate recipient
Windows 10 / Windows Server 2016

These settings may not prevent earlier operating systems from using this template.

OK | Cancel | Apply | Help

Domain Controller Authentication - Hardened Properti...

Subject Name | Issuance Requirements
Superseded Templates | Extensions | Security | Server
General | Compatibility | Request Handling | Cryptography | Key Attestation

Template display name:
Domain Controller Authentication - Hardened

Template name:
DomainControllerAuthentication-Hardened

Validity period: 1 years
Renewal period: 6 weeks

☐ Publish certificate in Active Directory
☐ Do not automatically reenroll if a duplicate certificate exists in Active Directory

OK | Cancel | Apply | Help

Domain Controller Authentication - Hardened Properti...

Subject Name | Issuance Requirements
Superseded Templates | Extensions | Security | Server
General | Compatibility | Request Handling | Cryptography | Key Attestation

Purpose: Signature and encryption
☐ Delete revoked or expired certificates (do not archive)
☐ Include symmetric algorithms allowed by the subject
☐ Archive subject's encryption private key
☐ Use advanced Symmetric algorithm to send the key to the CA
☐ Authorize additional service accounts to access the private key
Key Permissions...
☐ Allow private key to be exported
☐ Renew with the same key
☐ For automatic renewal of smart card certificates, use the existing key if a new key cannot be created

Do the following when the subject is enrolled and when the private key associated with this certificate is used:
◉ Enroll subject without requiring any user input
○ Prompt the user during enrollment
○ Prompt the user during enrollment and require user input when the private key is used

OK | Cancel | Apply | Help

Domain Controller Authentication - Hardened Properti...

Subject Name | Issuance Requirements
Superseded Templates | Extensions | Security | Server
General | Compatibility | Request Handling | Cryptography | Key Attestation

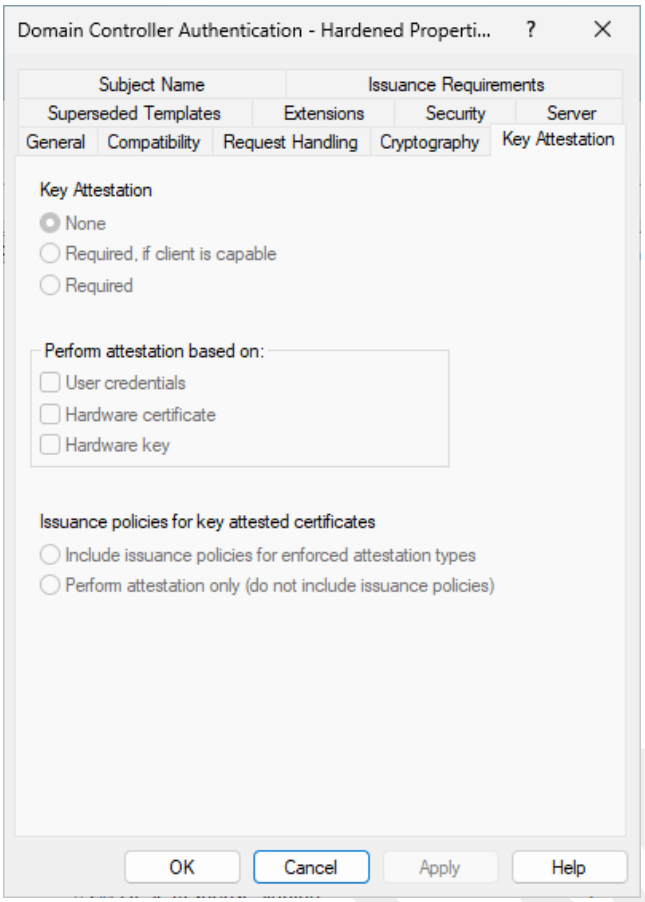
Provider Category: Key Storage Provider
Algorithm name: RSA
Minimum key size: 3072

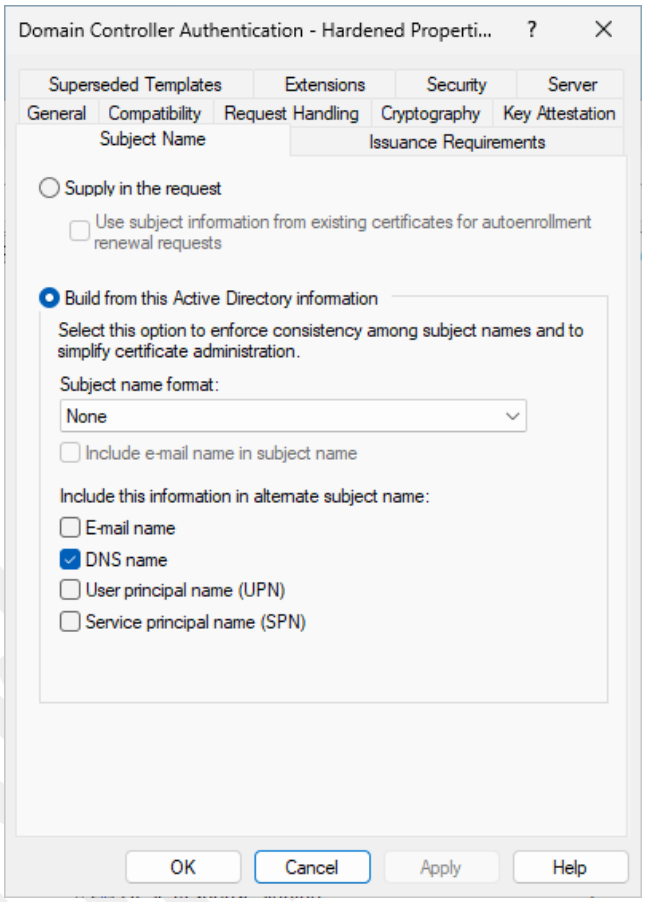
Choose which cryptographic providers can be used for requests
○ Requests can use any provider available on the subject's computer
◉ Requests must use one of the following providers:

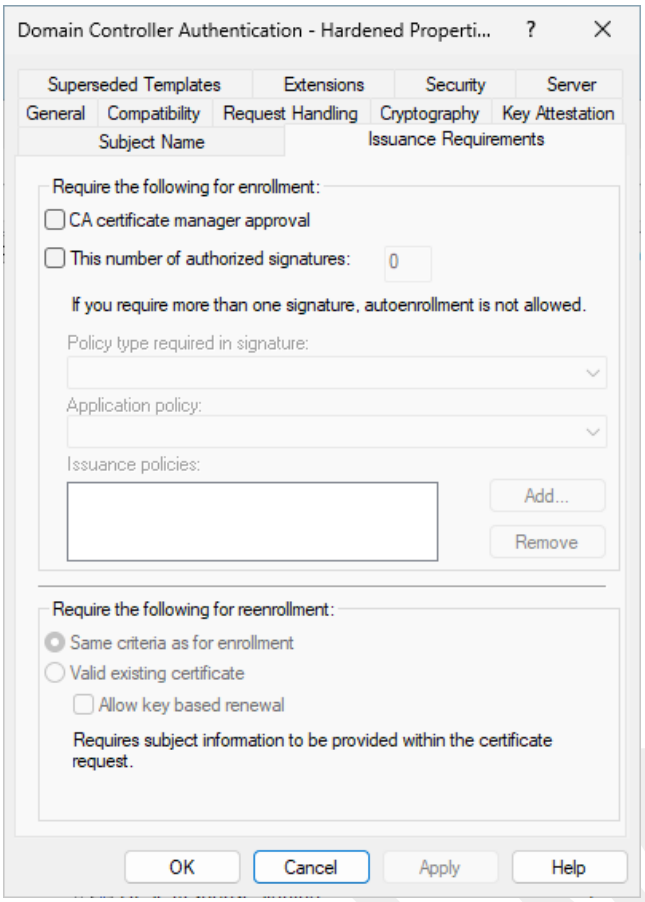
Providers:
☑ Microsoft Software Key Storage Provider
☐ Microsoft Smart Card Key Storage Provider

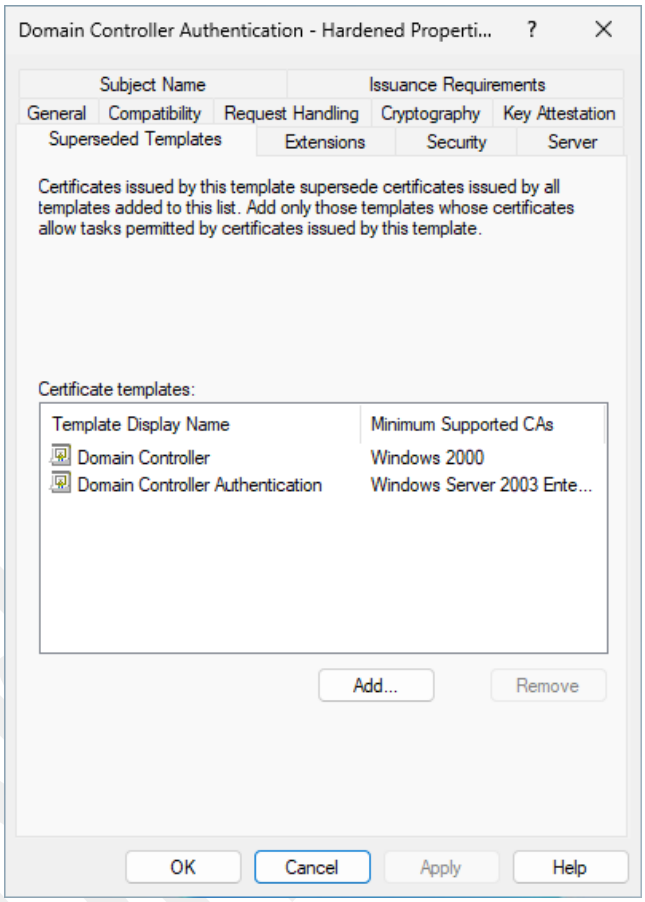
Request hash: SHA256
☐ Use alternate signature format

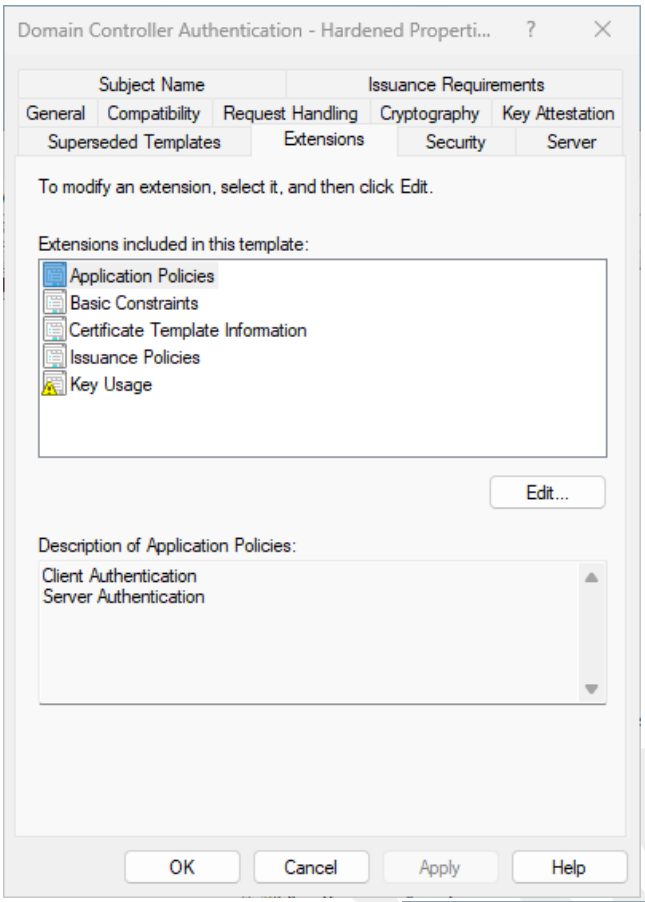
OK | Cancel | Apply | Help

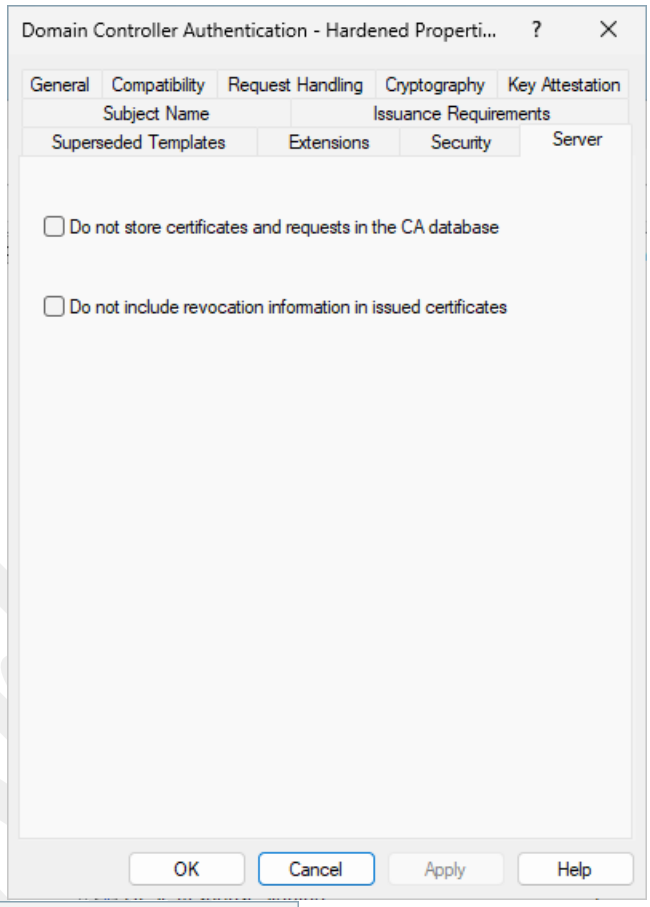

Domain Controller Authentication - Hardened Properti...
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
General
Compatibility
Request Handling
Cryptography
Key Attestation
Key Attestation
None
Required, if client is capable
Required
Perform attestation based on:
User credentials
Hardware certificate
Hardware key
Issuance policies for key attested certificates
Include issuance policies for enforced attestation types
Perform attestation only (do not include issuance policies)
OK
Cancel
Apply
Help

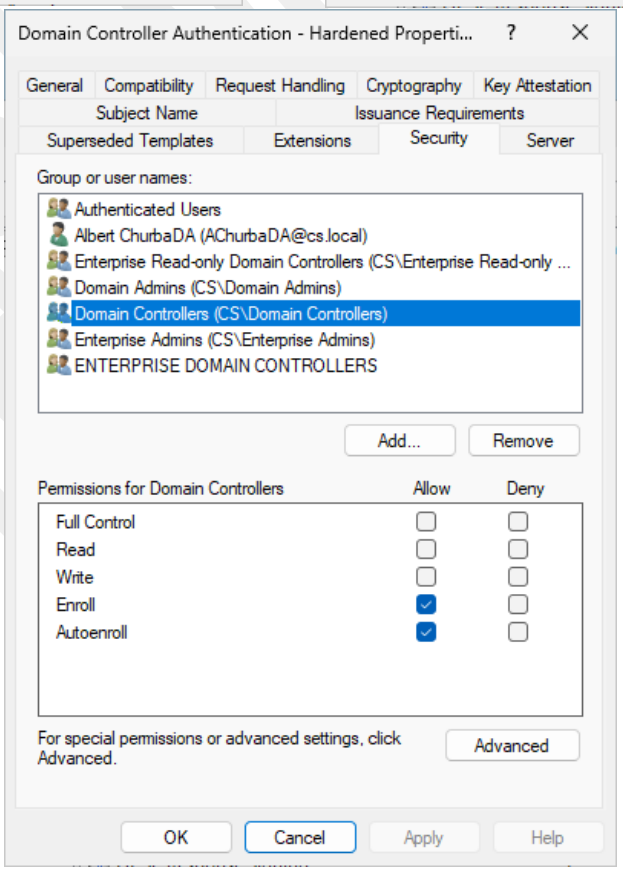

Domain Controller Authentication - Hardened Properti...
Superseded Templates
Extensions
Security
Server
General
Compatibility
Request Handling
Cryptography
Key Attestation
Subject Name
Issuance Requirements
Supply in the request
Use subject information from existing certificates for autoenrollment renewal requests
Build from this Active Directory information
Select this option to enforce consistency among subject names and to simplify certificate administration.
Subject name format:
None
Include e-mail name in subject name
Include this information in alternate subject name:
E-mail name
DNS name
User principal name (UPN)
Service principal name (SPN)
OK
Cancel
Apply
Help


Domain Controller Authentication - Hardened Properti...
Superseded Templates
Extensions
Security
Server
General
Compatibility
Request Handling
Cryptography
Key Attestation
Subject Name
Issuance Requirements
Require the following for enrollment:
CA certificate manager approval
This number of authorized signatures:
0
If you require more than one signature, autoenrollment is not allowed.
Policy type required in signature:
Application policy:
Issuance policies:
Add...
Remove
Require the following for reenrollment:
Same criteria as for enrollment
Valid existing certificate
Allow key based renewal
Requires subject information to be provided within the certificate request.
OK
Cancel
Apply
Help


Domain Controller Authentication - Hardened Properti...
Subject Name
Issuance Requirements
General
Compatibility
Request Handling
Cryptography
Key Attestation
Superseded Templates
Extensions
Security
Server
Certificates issued by this template supersede certificates issued by all templates added to this list. Add only those templates whose certificates allow tasks permitted by certificates issued by this template.
Certificate templates:
Template Display Name
Minimum Supported CAs
Domain Controller
Windows 2000
Domain Controller Authentication
Windows Server 2003 Ente...
Add...
Remove
OK
Cancel
Apply
Help


Domain Controller Authentication - Hardened Properti...
Subject Name
Issuance Requirements
General
Compatibility
Request Handling
Cryptography
Key Attestation
Superseded Templates
Extensions
Security
Server
To modify an extension, select it, and then click Edit.
Extensions included in this template:
Application Policies
Basic Constraints
Certificate Template Information
Issuance Policies
Key Usage
Edit...
Description of Application Policies:
Client Authentication
Server Authentication
OK
Cancel
Apply
Help

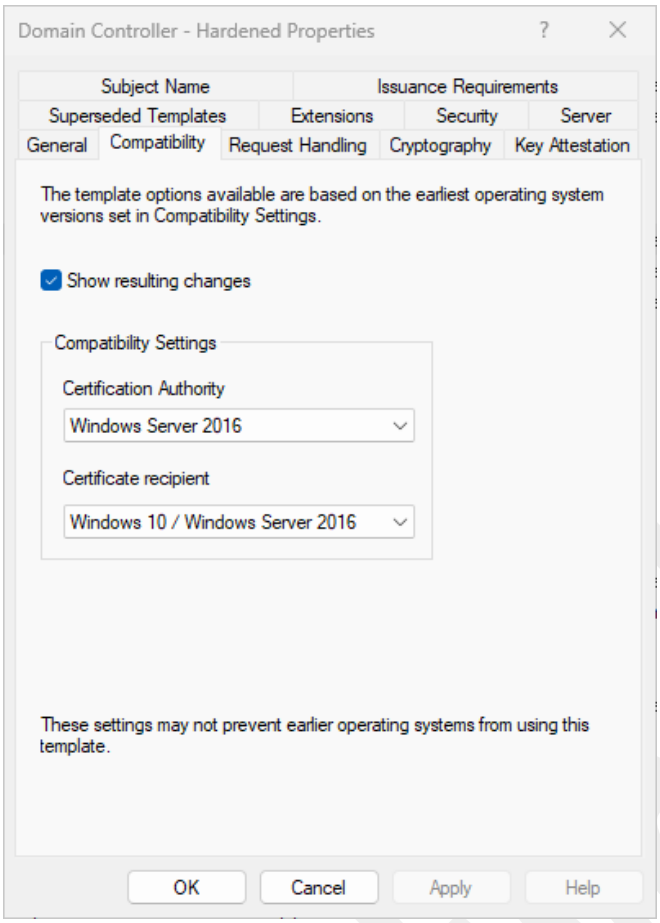

Domain Controller Authentication - Hardened Properti...
General
Compatibility
Request Handling
Cryptography
Key Attestation
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
Do not store certificates and requests in the CA database
Do not include revocation information in issued certificates
OK
Cancel
Apply
Help

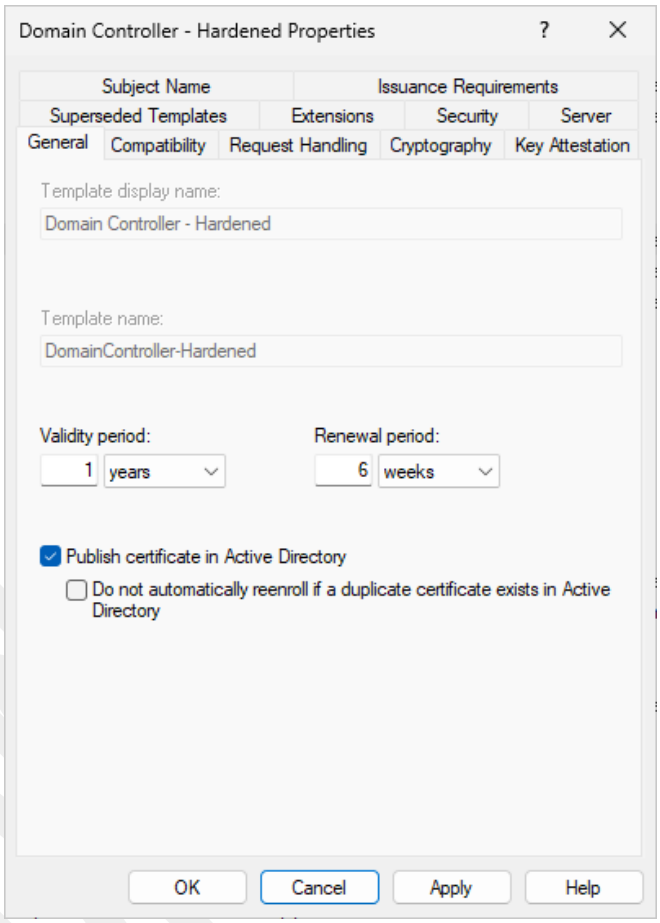

Domain Controller Authentication - Hardened Properti...
General
Compatibility
Request Handling
Cryptography
Key Attestation
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
Group or user names:
Authenticated Users
Albert ChurbaDA (AChurbaDA@cs.local)
Enterprise Read-only Domain Controllers (CS\Enterprise Read-only ...
Domain Admins (CS\Domain Admins)
Domain Controllers (CS\Domain Controllers)
Enterprise Admins (CS\Enterprise Admins)
ENTERPRISE DOMAIN CONTROLLERS
Add...
Remove
Permissions for Domain Controllers
Allow
Deny
Full Control
Read
Write
Enroll
Autoenroll
For special permissions or advanced settings, click Advanced.
Advanced
OK
Cancel
Apply
Help

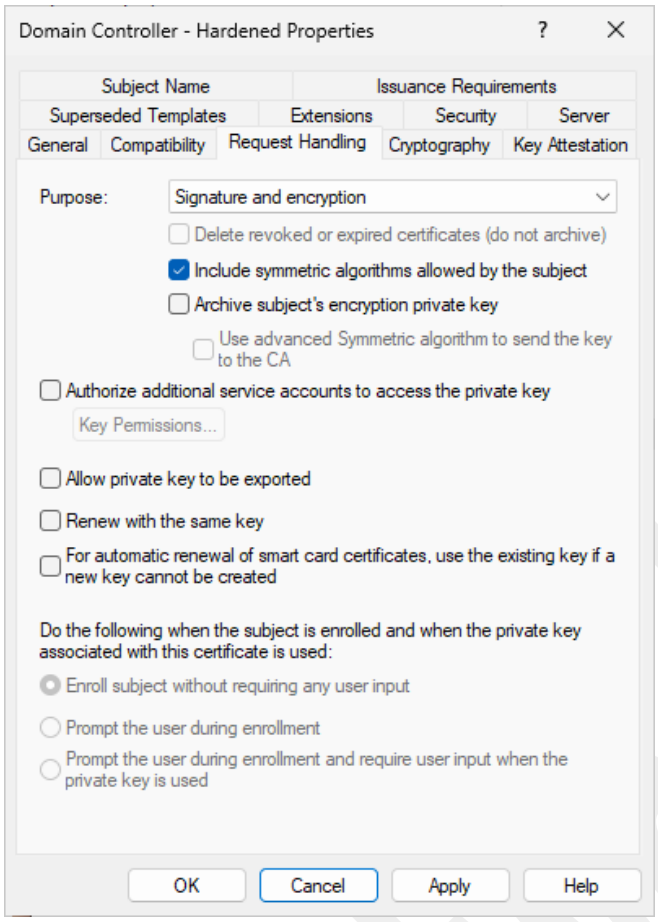
Template: Domain Controller – Hardened

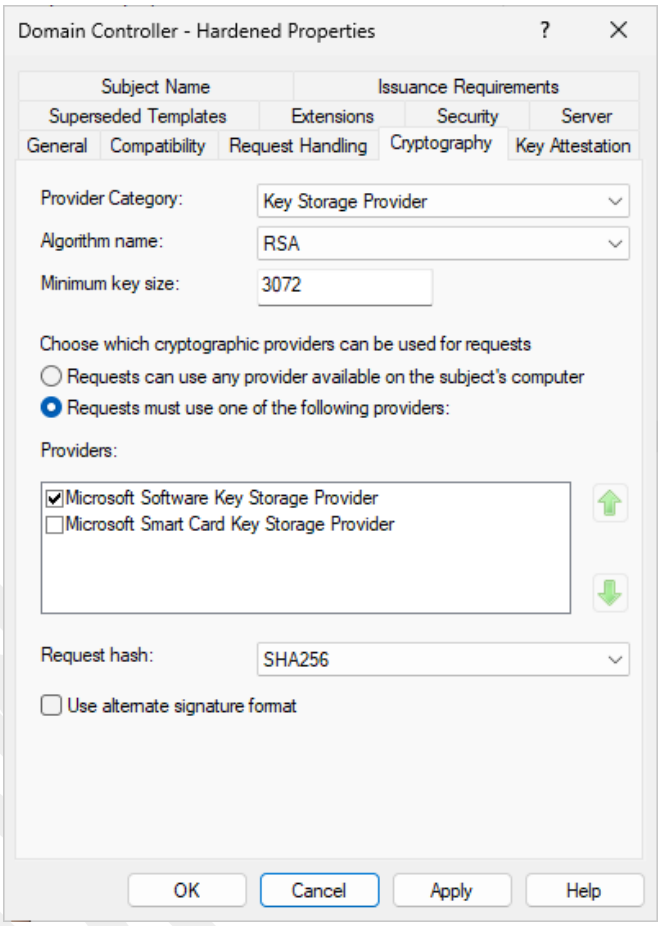

```yaml
based_on: Domain Controller Authentication
purpose:
  - Server Authentication
  - Client Authentication
  - Issues the primary machine identity certificate for Active Directory Domain Controllers. This certificate supports secure inter-domain
      controller communications, LDAP over TLS (LDAPS), SYSVOL replication, and general domain controller service authentication.
      The template enforces modern cryptography, non-exportable private keys, and controlled enrollment to domain controller computer
      accounts only.
compatibility:
  certification_authority: Windows Server 2016
  certificate_recipient: Windows 10 / Windows Server 2016
cryptography:
  provider_category: Key Storage Provider
  algorithm: RSA
  minimum_key_size: 3072
hash_algorithm: SHA256
private_key_exportable: false
  allowed_providers:
    - Microsoft Software Key Storage Provider
    - Microsoft Smart Card Key Storage Provider
  request_hash: SHA256
  allow_alternate_signature_format: false
validity:
  validity_period: 1 year
  renewal_period: 6 weeks
  publish_to_active_directory: true
subject_name:
  source: Active Directory
  subject_name_format: None
  include_in_subject_alternate_name:
    - DNS
extensions:
  application_policies:
    - Server Authentication
    - Client Authentication
  key_usage:
    - Digital Signature
    - Key Encipherment
  basic_constraints:
    ca: false
enrollment:
  require_ca_manager_approval: false
  authorized_signatures_required: 0
  reenrollment:
    require_valid_existing_certificate: true
    allow_key_based_renewal: false
private_key:
```

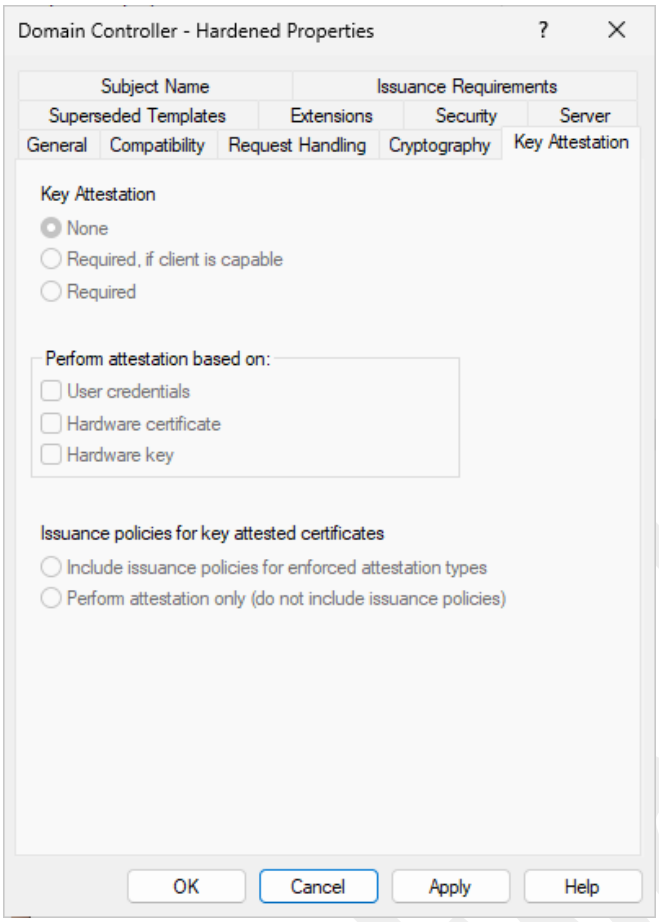
```yaml
    purpose: Signature and Encryption
    exportable: false
    archive_key: false
    include_symmetric_algorithms: true
  security:
    permissions:
      Domain Controllers:
        enroll: true
        autoenroll: false
      Enterprise Domain Controllers:
        enroll: true
        autoenroll: false
      Domain Admins:
        read: true
      Enterprise Admins:
        read: true
  supersedes:
    - Domain Controller

notes:
- Supersedes the legacy "Domain Controller" template.
- Complements the Domain Controller Authentication and Kerberos Authentication templates; each template serves a distinct
  authentication purpose.
- Enforced separation of certificates reduces misuse and simplifies audit validation.
```

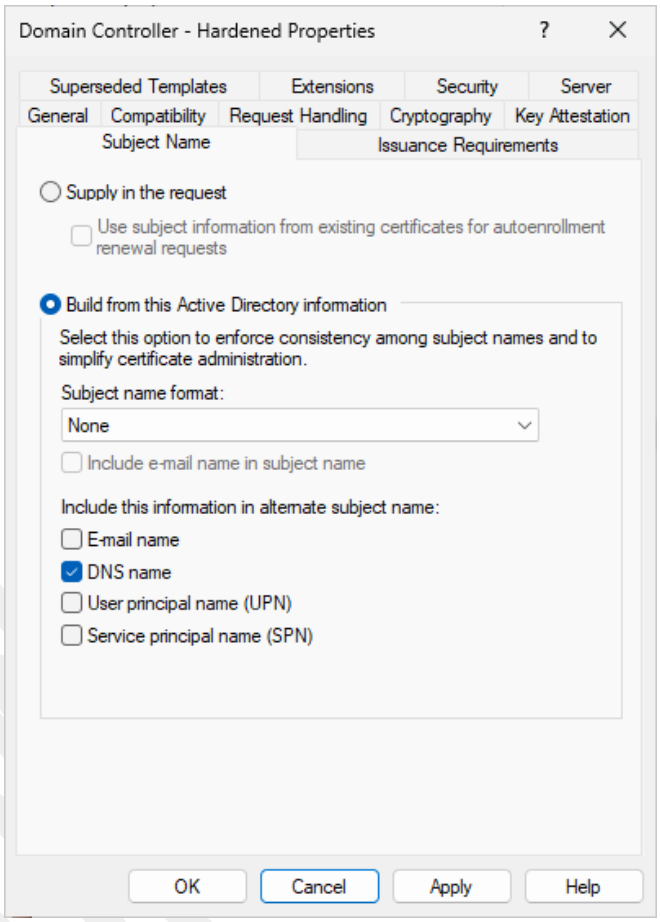

Domain Controller - Hardened Properties
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
General
Compatibility
Request Handling
Cryptography
Key Attestation
The template options available are based on the earliest operating system versions set in Compatibility Settings.
Show resulting changes
Compatibility Settings
Certification Authority
Windows Server 2016
Certificate recipient
Windows 10 / Windows Server 2016
These settings may not prevent earlier operating systems from using this template.
OK
Cancel
Apply
Help

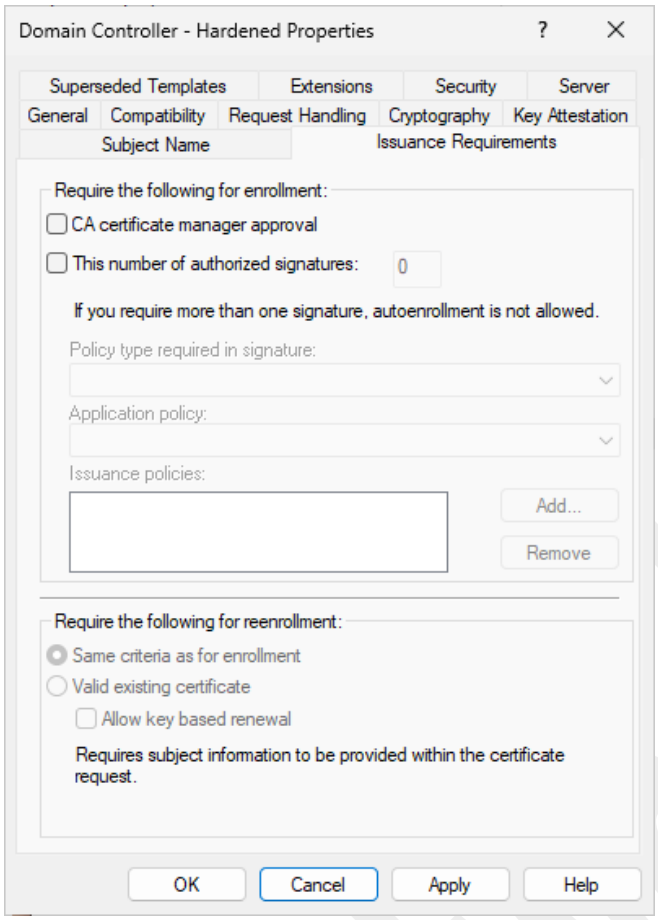

Domain Controller - Hardened Properties
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
General
Compatibility
Request Handling
Cryptography
Key Attestation
Template display name:
Domain Controller - Hardened
Template name:
DomainController-Hardened
Validity period:
1
years
Renewal period:
6
weeks
Publish certificate in Active Directory
Do not automatically reenroll if a duplicate certificate exists in Active Directory
OK
Cancel
Apply
Help

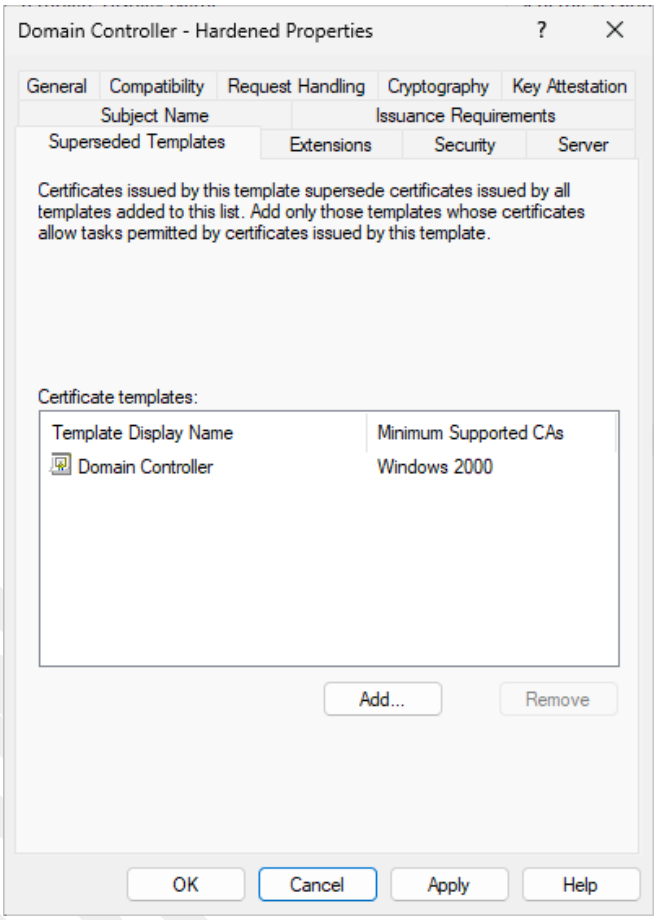

Domain Controller - Hardened Properties
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
General
Compatibility
Request Handling
Cryptography
Key Attestation
Purpose:
Signature and encryption
Delete revoked or expired certificates (do not archive)
Include symmetric algorithms allowed by the subject
Archive subject's encryption private key
Use advanced Symmetric algorithm to send the key to the CA
Authorize additional service accounts to access the private key
Key Permissions...
Allow private key to be exported
Renew with the same key
For automatic renewal of smart card certificates, use the existing key if a new key cannot be created
Do the following when the subject is enrolled and when the private key associated with this certificate is used:
Enroll subject without requiring any user input
Prompt the user during enrollment
Prompt the user during enrollment and require user input when the private key is used
OK
Cancel
Apply
Help

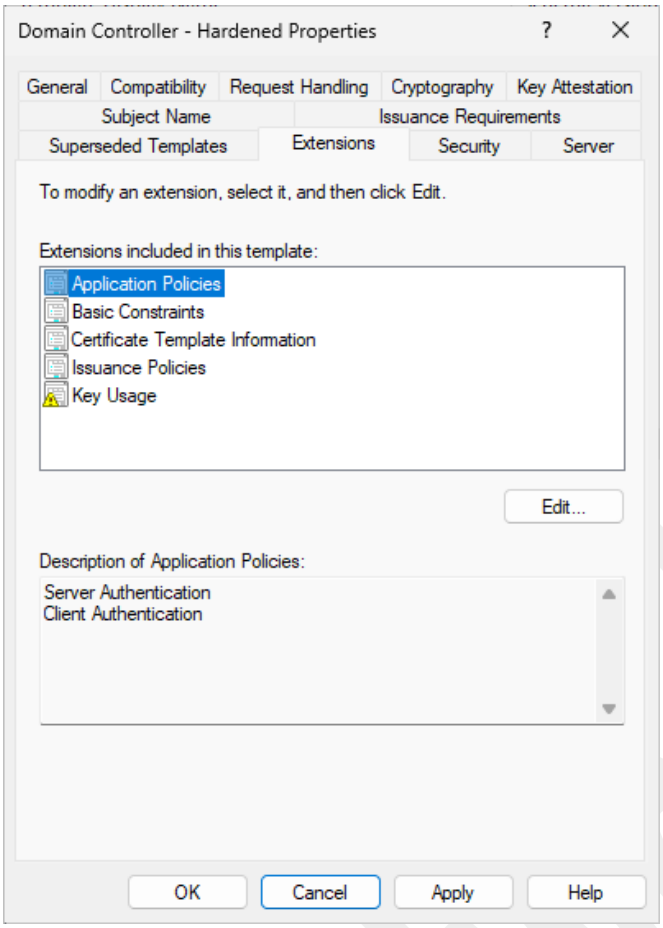

Domain Controller - Hardened Properties
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
General
Compatibility
Request Handling
Cryptography
Key Attestation
Provider Category:
Key Storage Provider
Algorithm name:
RSA
Minimum key size:
3072
Choose which cryptographic providers can be used for requests
Requests can use any provider available on the subject's computer
Requests must use one of the following providers:
Providers:
Microsoft Software Key Storage Provider
Microsoft Smart Card Key Storage Provider
Request hash:
SHA256
Use alternate signature format
OK
Cancel
Apply
Help

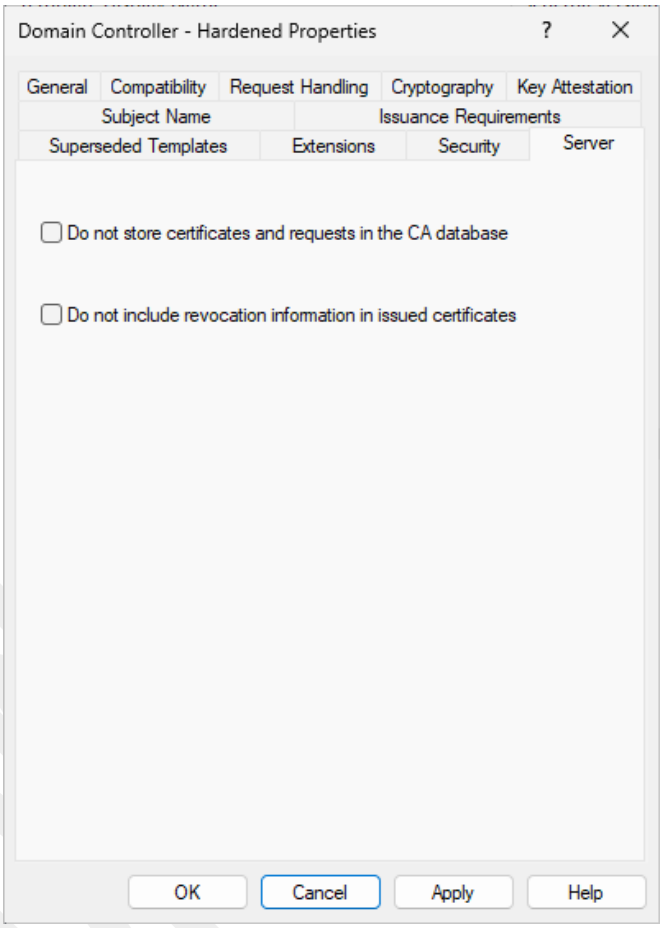

Domain Controller - Hardened Properties
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
General
Compatibility
Request Handling
Cryptography
Key Attestation
Key Attestation
None
Required, if client is capable
Required
Perform attestation based on:
User credentials
Hardware certificate
Hardware key
Issuance policies for key attested certificates
Include issuance policies for enforced attestation types
Perform attestation only (do not include issuance policies)
OK
Cancel
Apply
Help

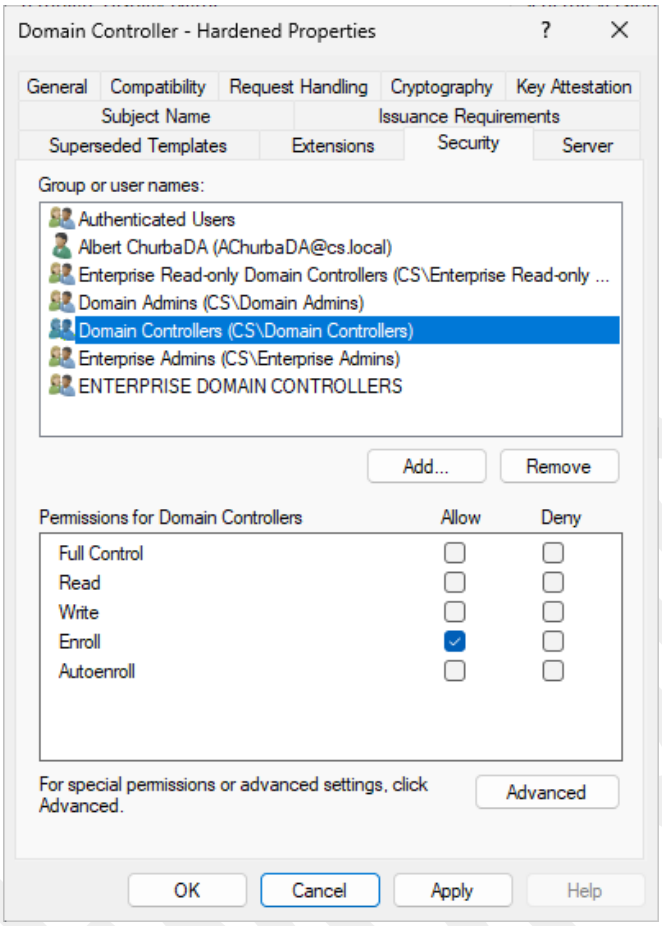

Domain Controller - Hardened Properties
Superseded Templates
Extensions
Security
Server
General
Compatibility
Request Handling
Cryptography
Key Attestation
Subject Name
Issuance Requirements
Supply in the request
Use subject information from existing certificates for autoenrollment renewal requests
Build from this Active Directory information
Select this option to enforce consistency among subject names and to simplify certificate administration.
Subject name format:
None
Include e-mail name in subject name
Include this information in alternate subject name:
E-mail name
DNS name
User principal name (UPN)
Service principal name (SPN)
OK
Cancel
Apply
Help


Domain Controller - Hardened Properties
Superseded Templates
Extensions
Security
Server
General
Compatibility
Request Handling
Cryptography
Key Attestation
Subject Name
Issuance Requirements
Require the following for enrollment:
CA certificate manager approval
This number of authorized signatures:
0
If you require more than one signature, autoenrollment is not allowed.
Policy type required in signature:
Application policy:
Issuance policies:
Add...
Remove
Require the following for reenrollment:
Same criteria as for enrollment
Valid existing certificate
Allow key based renewal
Requires subject information to be provided within the certificate request.
OK
Cancel
Apply
Help


Domain Controller - Hardened Properties
General
Compatibility
Request Handling
Cryptography
Key Attestation
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
Certificates issued by this template supersede certificates issued by all templates added to this list. Add only those templates whose certificates allow tasks permitted by certificates issued by this template.
Certificate templates:
Template Display Name
Minimum Supported CAs
Domain Controller
Windows 2000
Add...
Remove
OK
Cancel
Apply
Help


Domain Controller - Hardened Properties
General
Compatibility
Request Handling
Cryptography
Key Attestation
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
To modify an extension, select it, and then click Edit.
Extensions included in this template:
Application Policies
Basic Constraints
Certificate Template Information
Issuance Policies
Key Usage
Edit...
Description of Application Policies:
Server Authentication
Client Authentication
OK
Cancel
Apply
Help

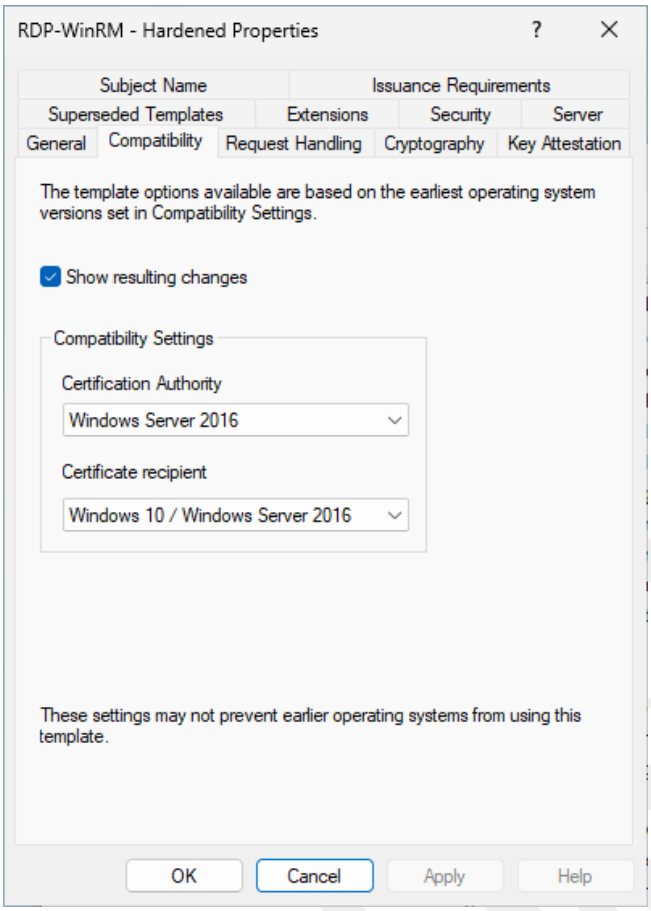

Domain Controller - Hardened Properties
General
Compatibility
Request Handling
Cryptography
Key Attestation
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
Do not store certificates and requests in the CA database
Do not include revocation information in issued certificates
OK
Cancel
Apply
Help

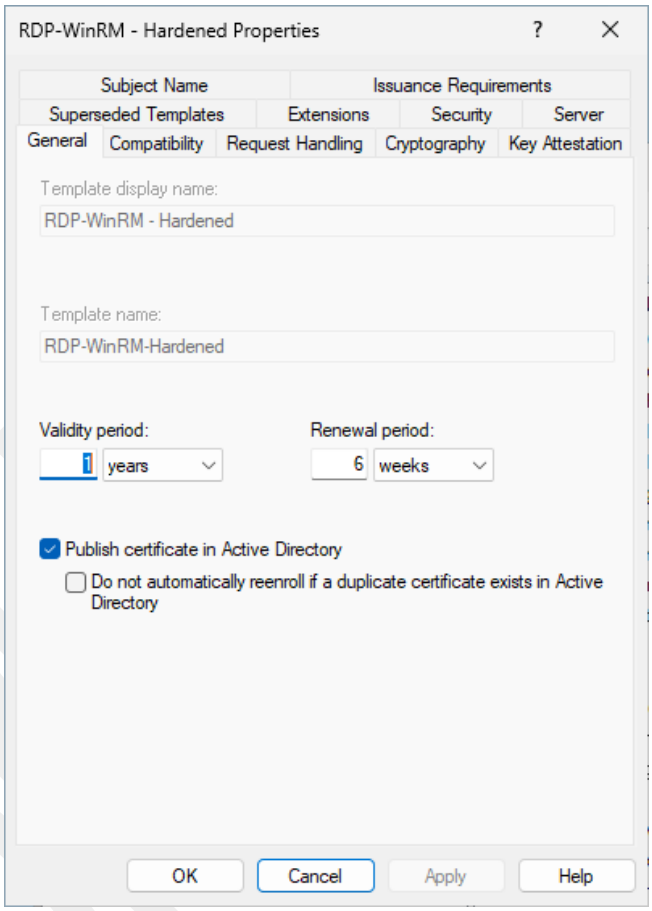


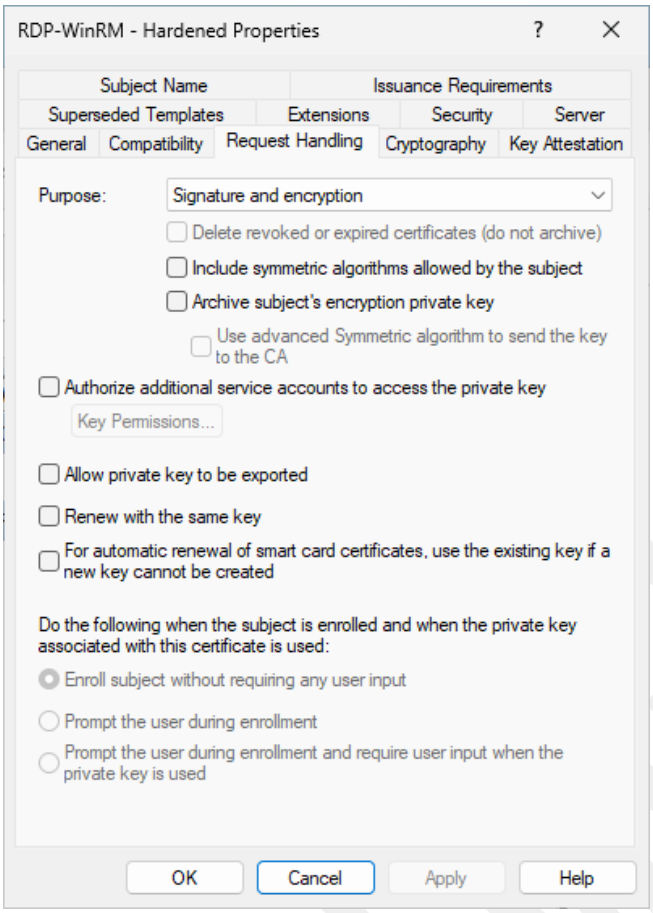
Template: RDP-WinRM- Hardened

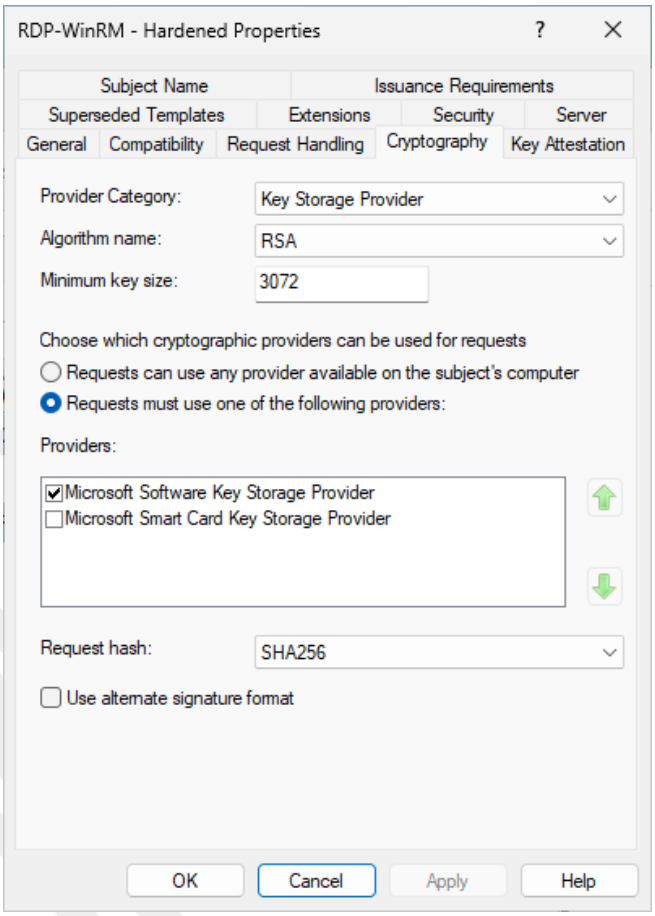
```
purpose:
  - Issues certificates for Remote Desktop and WinRM HTTPS listeners
  - Prevents self-signed certificates on management endpoints
  - Enforces TLS for administrative access
based_on:
  - Web Server
compatibility:
  certification_authority: Windows Server 2016
  certificate_recipient: Windows Server 2016
cryptography:
  provider_category: Key Storage Provider
  provider: Microsoft Software Key Storage Provider
  algorithm: RSA
  minimum_key_size: 3072
  hash_algorithm: SHA256
  private_key_exportable: false
  requests_can_use_any_provider: false
subject_name:
  source: Request
  subject_alternative_name:
    - DNS
    - IP
extensions:
```

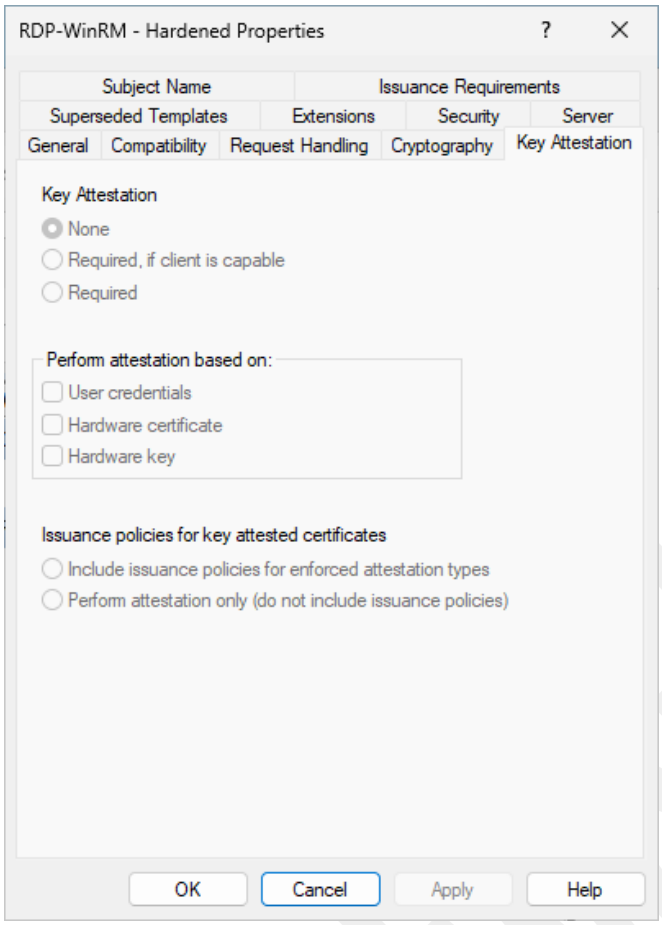
```yaml
  application_policies:
    - Server Authentication
  key_usage:
    - Digital Signature
    - Key Encipherment
  basic_constraints:
    ca: false
validity:
  validity_period: 1 year
  renewal_period: 6 weeks
  publish_to_active_directory: false
enrollment:
  domain_computers:
    enroll: true
    autoenroll: false
issuance_requirements:
  ca_manager_approval: false
  authorized_signatures_required: 0
request_handling:
  purpose: Signature and Encryption
  archive_private_key: false
key_attestation:
  required: false
security:
  certificate_database_storage: enabled
  revocation_information: included

supersedes:
  - Web Server
notes:
  - Used by RdsCertAutomation
  - Replaces default RDP self-signed certificates
```

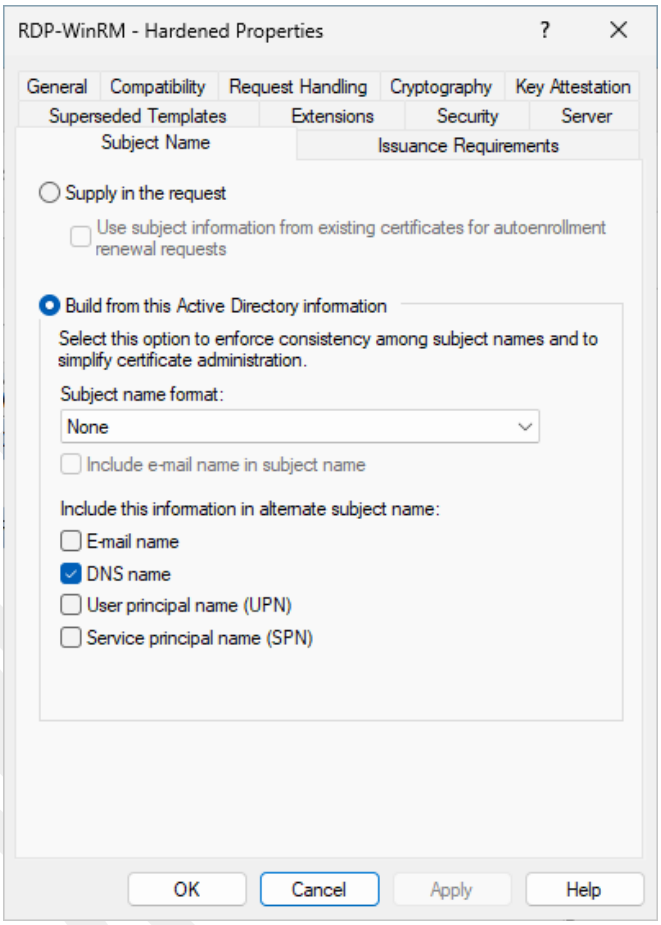

RDP-WinRM - Hardened Properties
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
General
Compatibility
Request Handling
Cryptography
Key Attestation
The template options available are based on the earliest operating system versions set in Compatibility Settings.
Show resulting changes
Compatibility Settings
Certification Authority
Windows Server 2016
Certificate recipient
Windows 10 / Windows Server 2016
These settings may not prevent earlier operating systems from using this template.
OK
Cancel
Apply
Help


RDP-WinRM - Hardened Properties
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
General
Compatibility
Request Handling
Cryptography
Key Attestation
Template display name:
RDP-WinRM - Hardened
Template name:
RDP-WinRM-Hardened
Validity period:
1
years
Renewal period:
6
weeks
Publish certificate in Active Directory
Do not automatically reenroll if a duplicate certificate exists in Active Directory
OK
Cancel
Apply
Help


RDP-WinRM - Hardened Properties
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
General
Compatibility
Request Handling
Cryptography
Key Attestation
Purpose:
Signature and encryption
Delete revoked or expired certificates (do not archive)
Include symmetric algorithms allowed by the subject
Archive subject's encryption private key
Use advanced Symmetric algorithm to send the key to the CA
Authorize additional service accounts to access the private key
Key Permissions...
Allow private key to be exported
Renew with the same key
For automatic renewal of smart card certificates, use the existing key if a new key cannot be created
Do the following when the subject is enrolled and when the private key associated with this certificate is used:
Enroll subject without requiring any user input
Prompt the user during enrollment
Prompt the user during enrollment and require user input when the private key is used
OK
Cancel
Apply
Help


RDP-WinRM - Hardened Properties
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
General
Compatibility
Request Handling
Cryptography
Key Attestation
Provider Category:
Key Storage Provider
Algorithm name:
RSA
Minimum key size:
3072
Choose which cryptographic providers can be used for requests
Requests can use any provider available on the subject's computer
Requests must use one of the following providers:
Providers:
Microsoft Software Key Storage Provider
Microsoft Smart Card Key Storage Provider
Request hash:
SHA256
Use alternate signature format
OK
Cancel
Apply
Help





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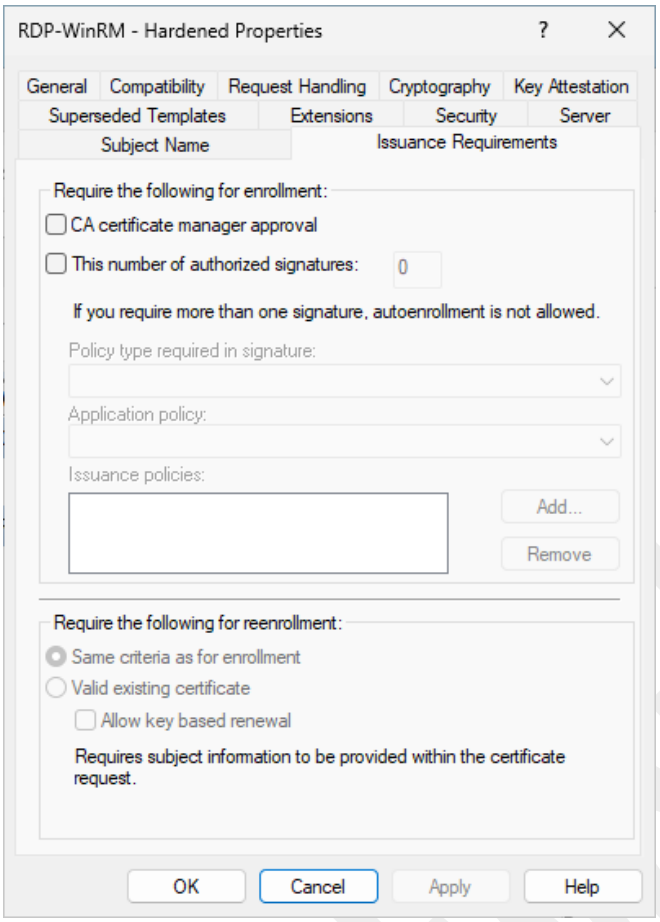
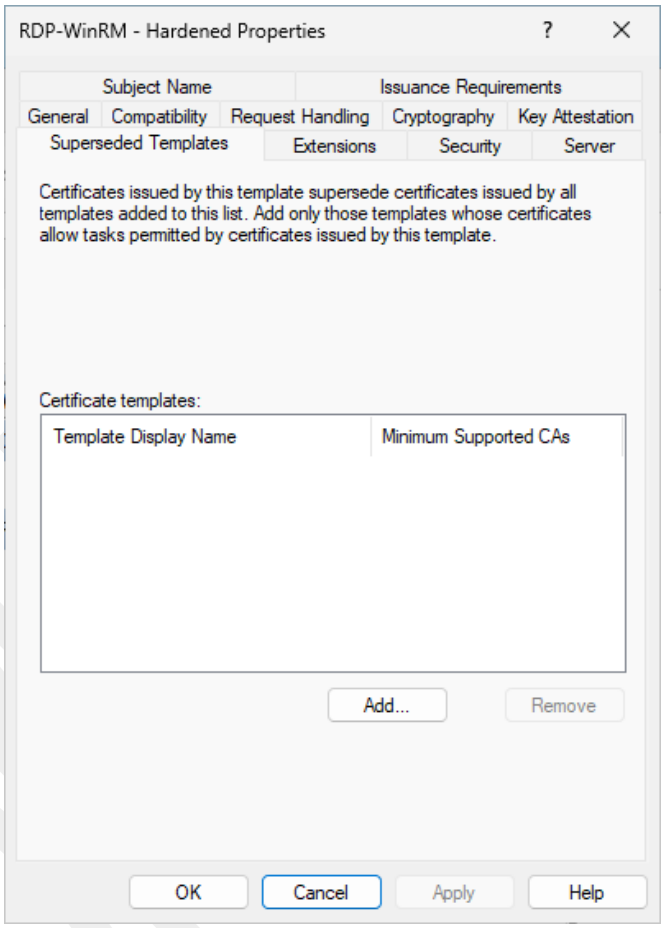
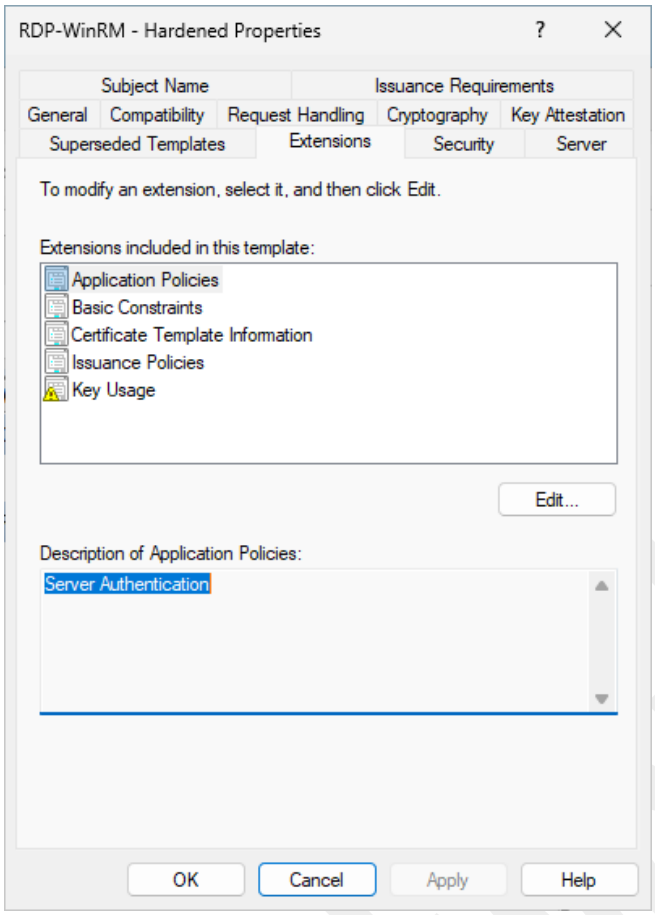
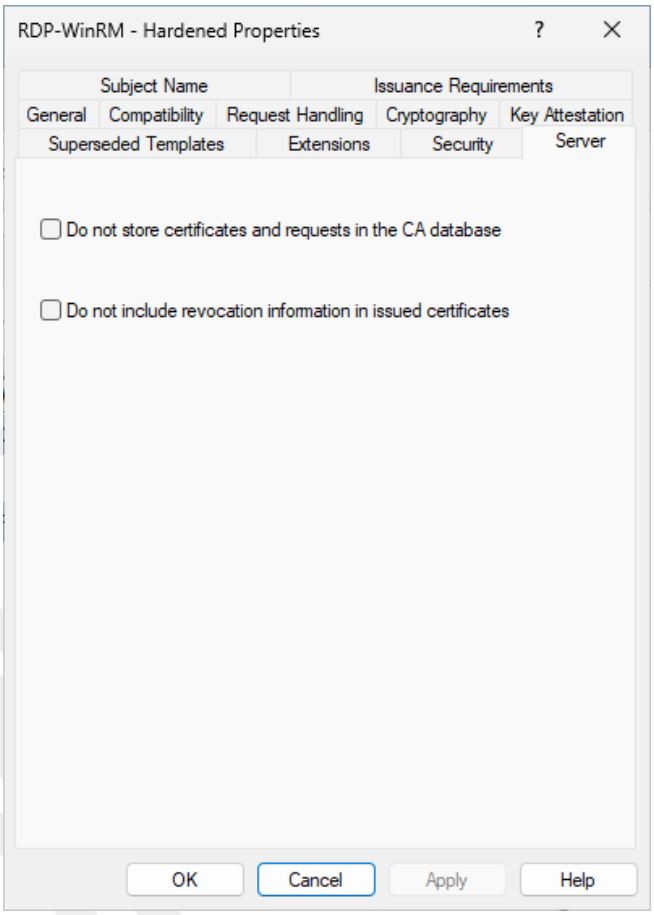
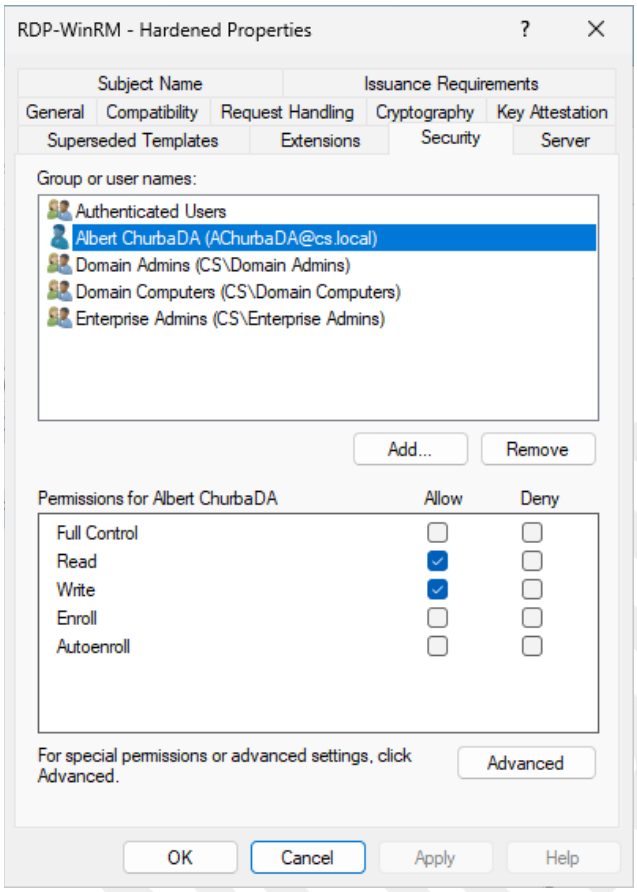

RDP-WinRM - Hardened Properties
General
Compatibility
Request Handling
Cryptography
Key Attestation
Superseded Templates
Extensions
Security
Server
Subject Name
Issuance Requirements
Require the following for enrollment:
CA certificate manager approval
This number of authorized signatures:
0
If you require more than one signature, autoenrollment is not allowed.
Policy type required in signature:
Application policy:
Issuance policies:
Add...
Remove
Require the following for reenrollment:
Same criteria as for enrollment
Valid existing certificate
Allow key based renewal
Requires subject information to be provided within the certificate request.
OK
Cancel
Apply
Help


RDP-WinRM - Hardened Properties
Subject Name
Issuance Requirements
General
Compatibility
Request Handling
Cryptography
Key Attestation
Superseded Templates
Extensions
Security
Server
Certificates issued by this template supersede certificates issued by all templates added to this list. Add only those templates whose certificates allow tasks permitted by certificates issued by this template.
Certificate templates:
Template Display Name
Minimum Supported CAs
Add...
Remove
OK
Cancel
Apply
Help


RDP-WinRM - Hardened Properties
Subject Name
Issuance Requirements
General
Compatibility
Request Handling
Cryptography
Key Attestation
Superseded Templates
Extensions
Security
Server
To modify an extension, select it, and then click Edit.
Extensions included in this template:
Application Policies
Basic Constraints
Certificate Template Information
Issuance Policies
Key Usage
Edit...
Description of Application Policies:
Server Authentication
OK
Cancel
Apply
Help

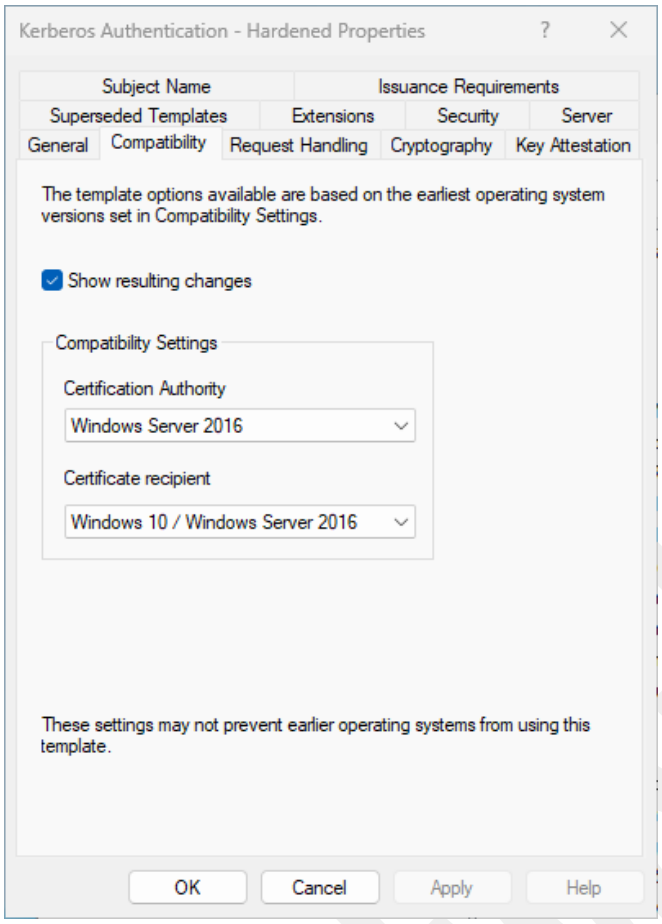

RDP-WinRM - Hardened Properties
Subject Name
Issuance Requirements
General
Compatibility
Request Handling
Cryptography
Key Attestation
Superseded Templates
Extensions
Security
Server
Do not store certificates and requests in the CA database
Do not include revocation information in issued certificates
OK
Cancel
Apply
Help

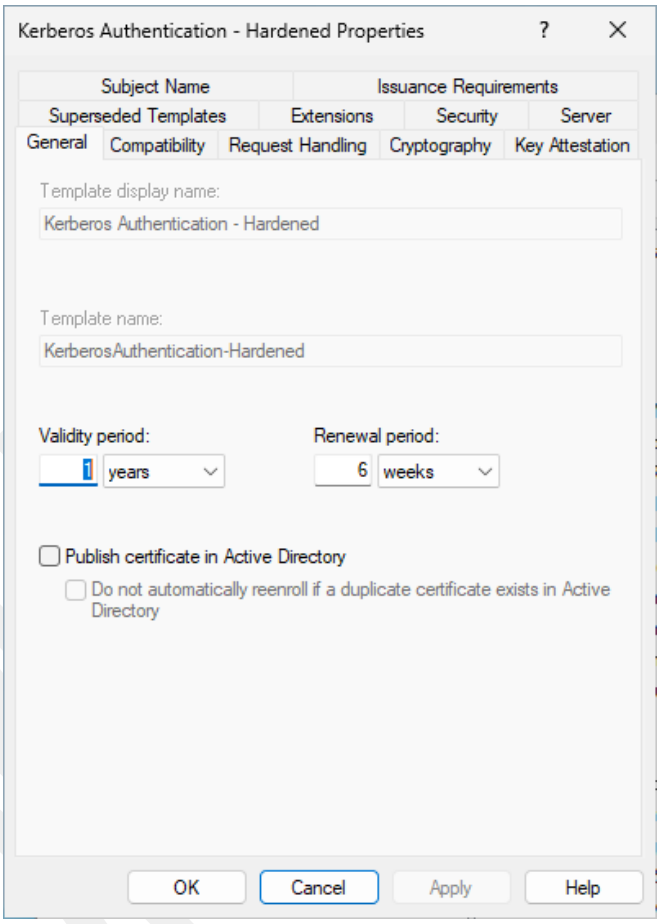


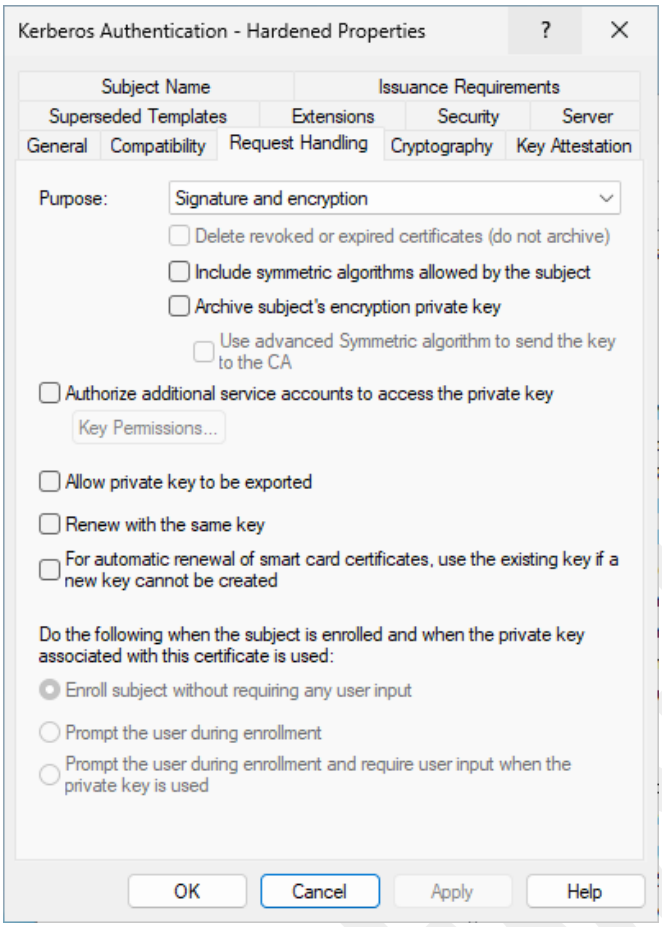
Template: Kerberos Authentication- Hardened

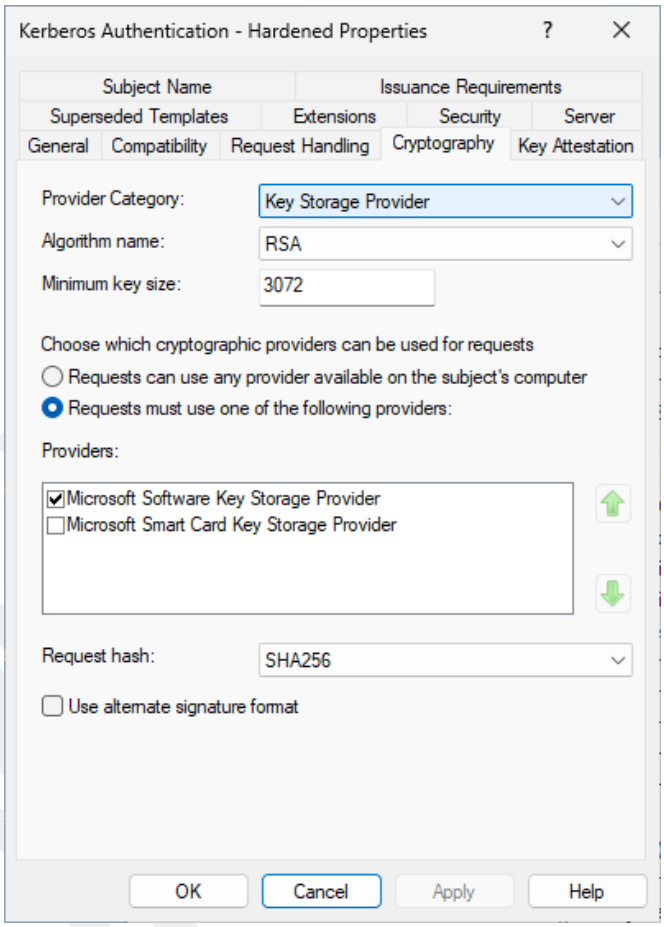
```
purpose:
  - Issues certificates used by Kerberos Key Distribution Centers
  - Supports secure Kerberos authentication and ticket issuance
  - Enforces modern cryptography for domain authentication
based_on:
  - Kerberos Authentication
compatibility:
  certification_authority: Windows Server 2016
  certificate_recipient: Windows Server 2016
cryptography:
  provider_category: Key Storage Provider
  provider: Microsoft Software Key Storage Provider
  algorithm: RSA
  minimum_key_size: 3072
  hash_algorithm: SHA256
  private_key_exportable: false
  requests_can_use_any_provider: false
subject_name:
  source: Active Directory
  subject_name_format: None
extensions:
  application_policies:
    - KDC Authentication
    - Server Authentication
```

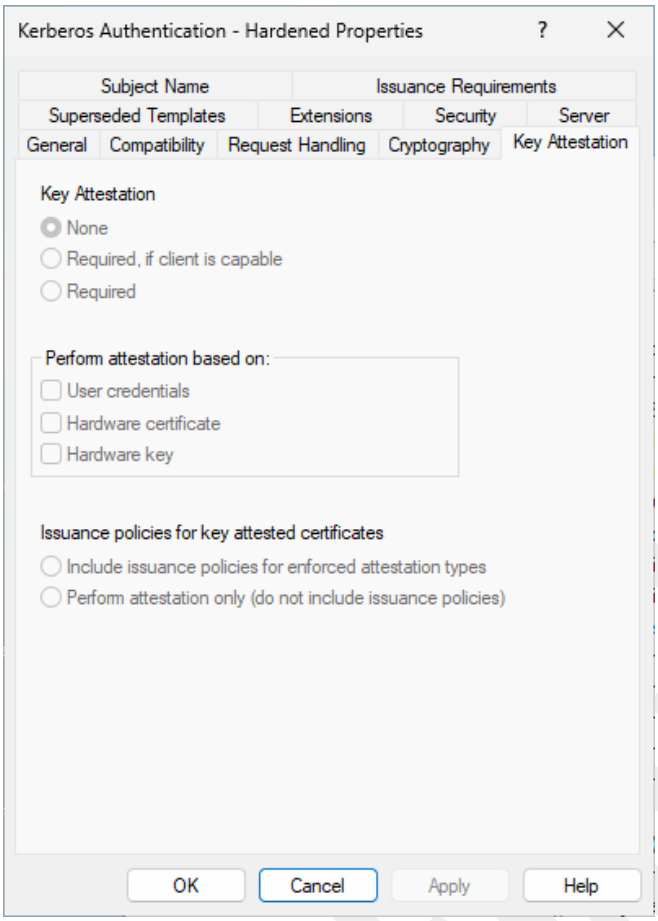
```yaml
    - Client Authentication
  key_usage:
    - Digital Signature
    - Key Encipherment
validity:
  validity_period: 1 year
  renewal_period: 6 weeks
  publish_to_active_directory: true
enrollment:
  domain_controllers:
    enroll: true
    autoenroll: true
issuance_requirements:
  ca_manager_approval: false
request_handling:
  purpose: Signature and Encryption
  archive_private_key: false
key_attestation:
  required: false
security:
  certificate_database_storage: enabled
  revocation_information: included
supersedes:
  - Kerberos Authentication

notes:
  - Required for modern Kerberos hardening
  - Must coexist with Domain Controller Authentication
```

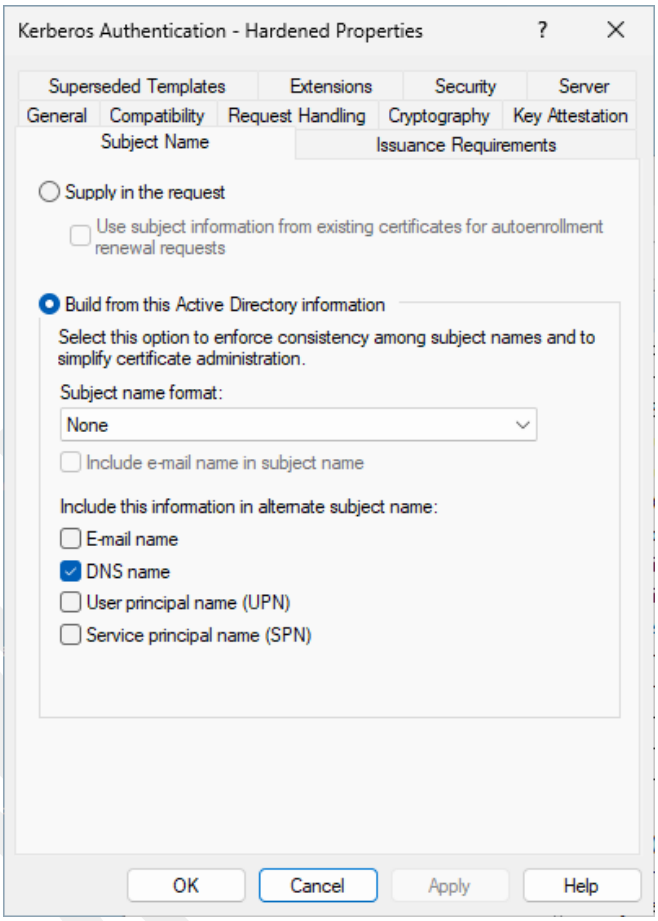

Kerberos Authentication - Hardened Properties
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
General
Compatibility
Request Handling
Cryptography
Key Attestation
The template options available are based on the earliest operating system versions set in Compatibility Settings.
Show resulting changes
Compatibility Settings
Certification Authority
Windows Server 2016
Certificate recipient
Windows 10 / Windows Server 2016
These settings may not prevent earlier operating systems from using this template.
OK
Cancel
Apply
Help

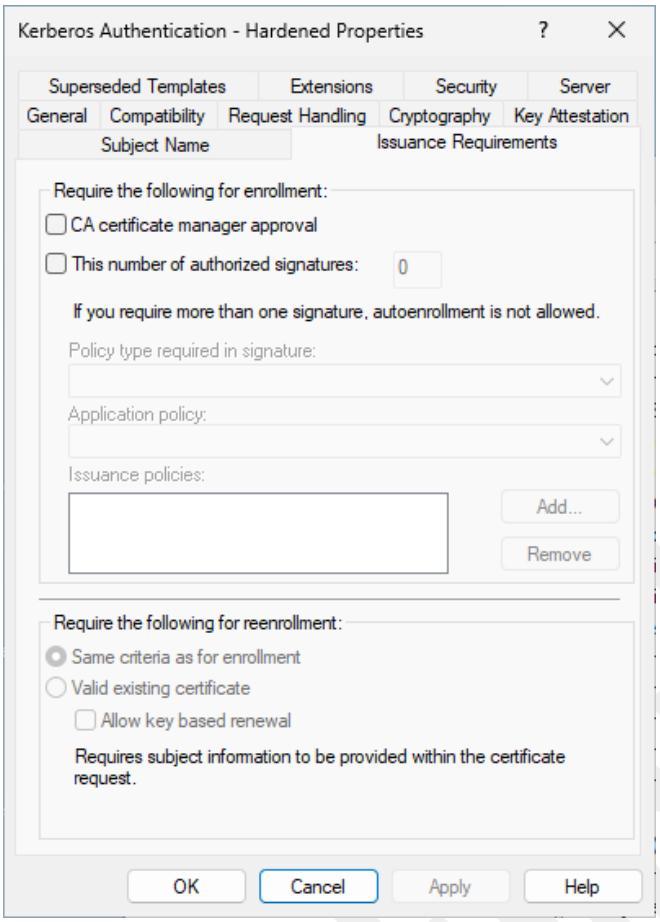

Kerberos Authentication - Hardened Properties
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
General
Compatibility
Request Handling
Cryptography
Key Attestation
Template display name:
Kerberos Authentication - Hardened
Template name:
KerberosAuthentication-Hardened
Validity period:
1
years
Renewal period:
6
weeks
Publish certificate in Active Directory
Do not automatically reenroll if a duplicate certificate exists in Active Directory
OK
Cancel
Apply
Help

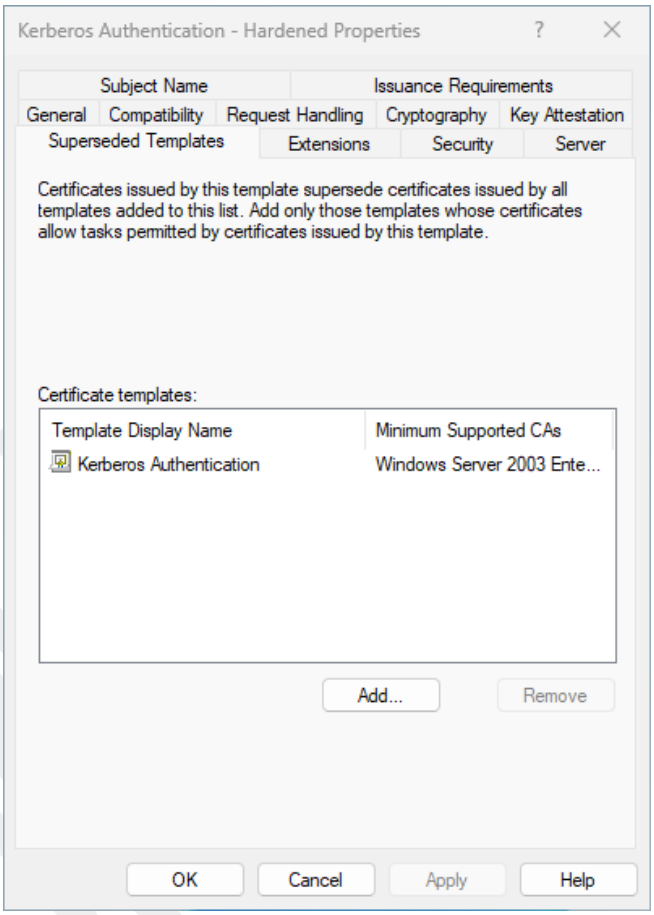

Kerberos Authentication - Hardened Properties
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
General
Compatibility
Request Handling
Cryptography
Key Attestation
Purpose:
Signature and encryption
Delete revoked or expired certificates (do not archive)
Include symmetric algorithms allowed by the subject
Archive subject's encryption private key
Use advanced Symmetric algorithm to send the key to the CA
Authorize additional service accounts to access the private key
Key Permissions...
Allow private key to be exported
Renew with the same key
For automatic renewal of smart card certificates, use the existing key if a new key cannot be created
Do the following when the subject is enrolled and when the private key associated with this certificate is used:
Enroll subject without requiring any user input
Prompt the user during enrollment
Prompt the user during enrollment and require user input when the private key is used
OK
Cancel
Apply
Help

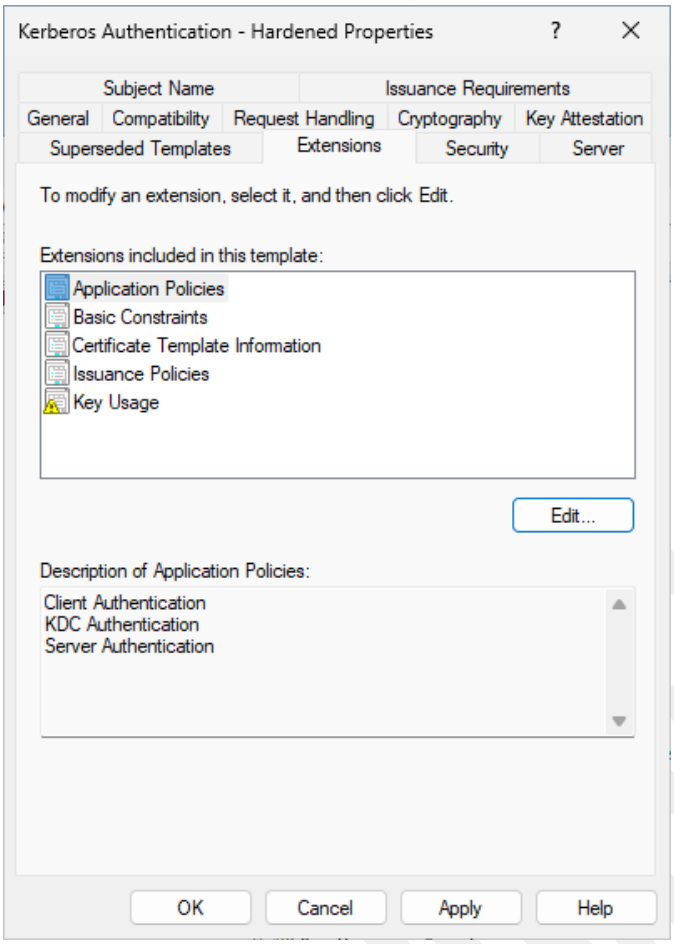

Kerberos Authentication - Hardened Properties
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
General
Compatibility
Request Handling
Cryptography
Key Attestation
Provider Category:
Key Storage Provider
Algorithm name:
RSA
Minimum key size:
3072
Choose which cryptographic providers can be used for requests
Requests can use any provider available on the subject's computer
Requests must use one of the following providers:
Providers:
Microsoft Software Key Storage Provider
Microsoft Smart Card Key Storage Provider
Request hash:
SHA256
Use alternate signature format
OK
Cancel
Apply
Help

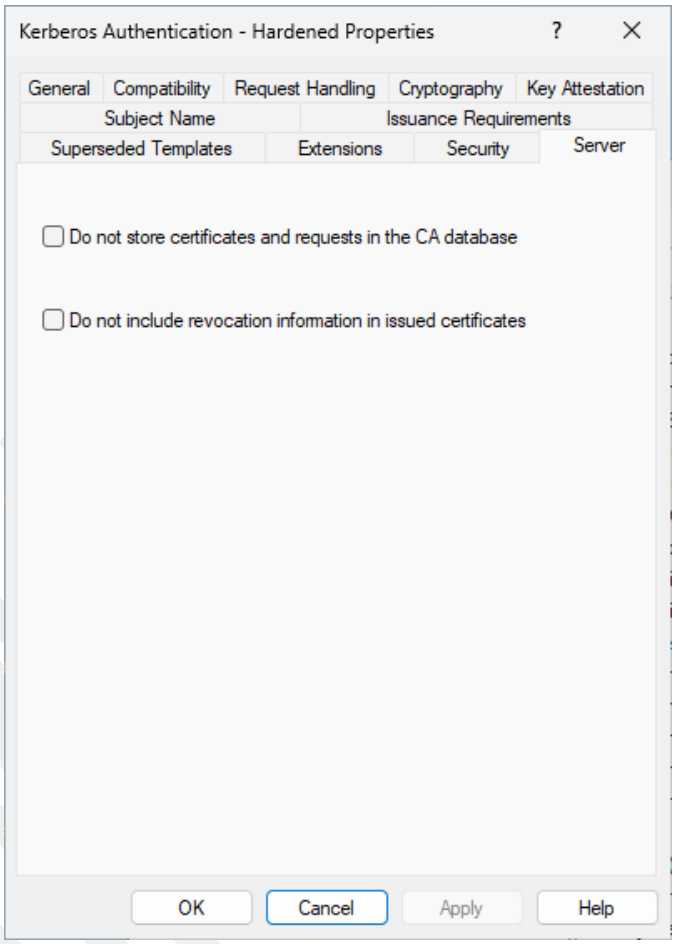

Kerberos Authentication - Hardened Properties
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
General
Compatibility
Request Handling
Cryptography
Key Attestation
Key Attestation
None
Required, if client is capable
Required
Perform attestation based on:
User credentials
Hardware certificate
Hardware key
Issuance policies for key attested certificates
Include issuance policies for enforced attestation types
Perform attestation only (do not include issuance policies)
OK
Cancel
Apply
Help

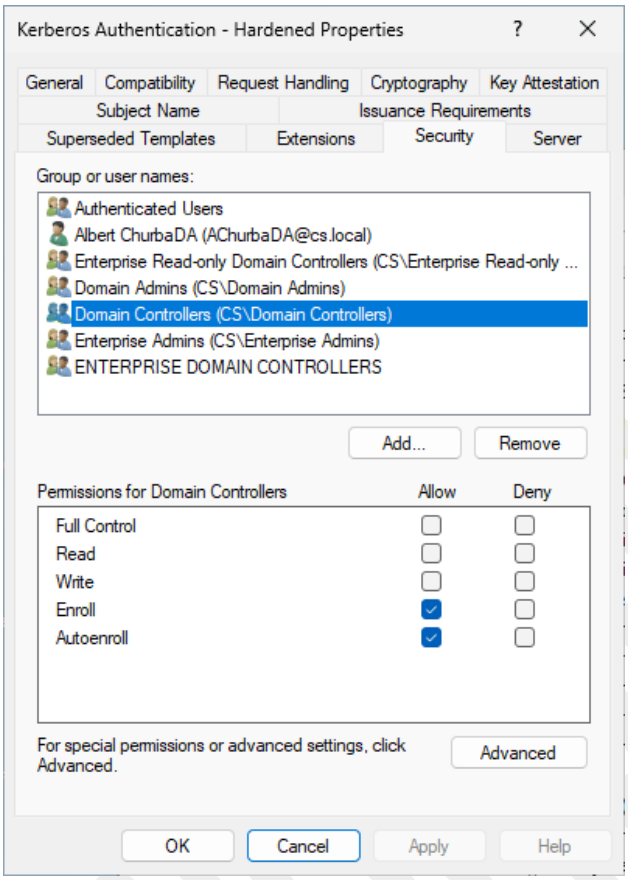

Kerberos Authentication - Hardened Properties
Superseded Templates
Extensions
Security
Server
General
Compatibility
Request Handling
Cryptography
Key Attestation
Subject Name
Issuance Requirements
Supply in the request
Use subject information from existing certificates for autoenrollment renewal requests
Build from this Active Directory information
Select this option to enforce consistency among subject names and to simplify certificate administration.
Subject name format:
None
Include e-mail name in subject name
Include this information in alternate subject name:
E-mail name
DNS name
User principal name (UPN)
Service principal name (SPN)
OK
Cancel
Apply
Help


Kerberos Authentication - Hardened Properties
Superseded Templates
Extensions
Security
Server
General
Compatibility
Request Handling
Cryptography
Key Attestation
Subject Name
Issuance Requirements
Require the following for enrollment:
CA certificate manager approval
This number of authorized signatures:
0
If you require more than one signature, autoenrollment is not allowed.
Policy type required in signature:
Application policy:
Issuance policies:
Add...
Remove
Require the following for reenrollment:
Same criteria as for enrollment
Valid existing certificate
Allow key based renewal
Requires subject information to be provided within the certificate request.
OK
Cancel
Apply
Help


Kerberos Authentication - Hardened Properties
Subject Name
Issuance Requirements
General
Compatibility
Request Handling
Cryptography
Key Attestation
Superseded Templates
Extensions
Security
Server
Certificates issued by this template supersede certificates issued by all templates added to this list. Add only those templates whose certificates allow tasks permitted by certificates issued by this template.
Certificate templates:
Template Display Name
Minimum Supported CAs
Kerberos Authentication
Windows Server 2003 Ente...
Add...
Remove
OK
Cancel
Apply
Help


Kerberos Authentication - Hardened Properties
Subject Name
Issuance Requirements
General
Compatibility
Request Handling
Cryptography
Key Attestation
Superseded Templates
Extensions
Security
Server
To modify an extension, select it, and then click Edit.
Extensions included in this template:
Application Policies
Basic Constraints
Certificate Template Information
Issuance Policies
Key Usage
Edit...
Description of Application Policies:
Client Authentication
KDC Authentication
Server Authentication
OK
Cancel
Apply
Help

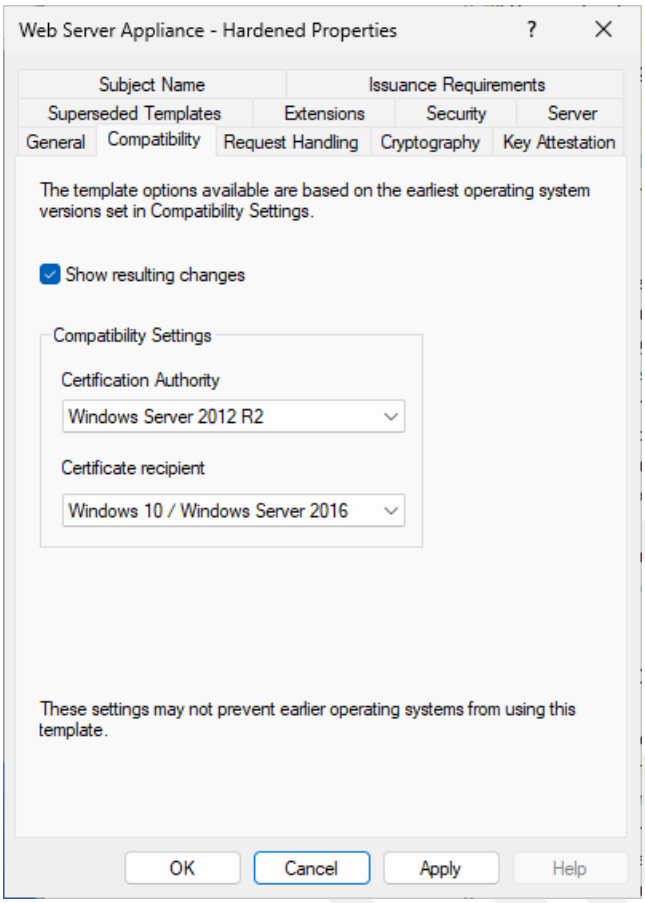

Kerberos Authentication - Hardened Properties
General
Compatibility
Request Handling
Cryptography
Key Attestation
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
Do not store certificates and requests in the CA database
Do not include revocation information in issued certificates
OK
Cancel
Apply
Help

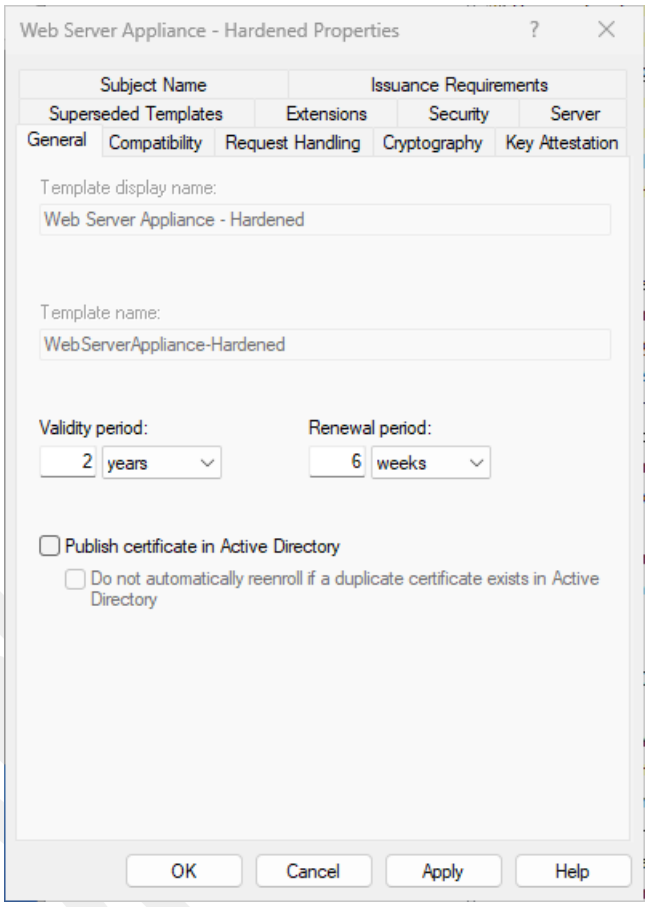


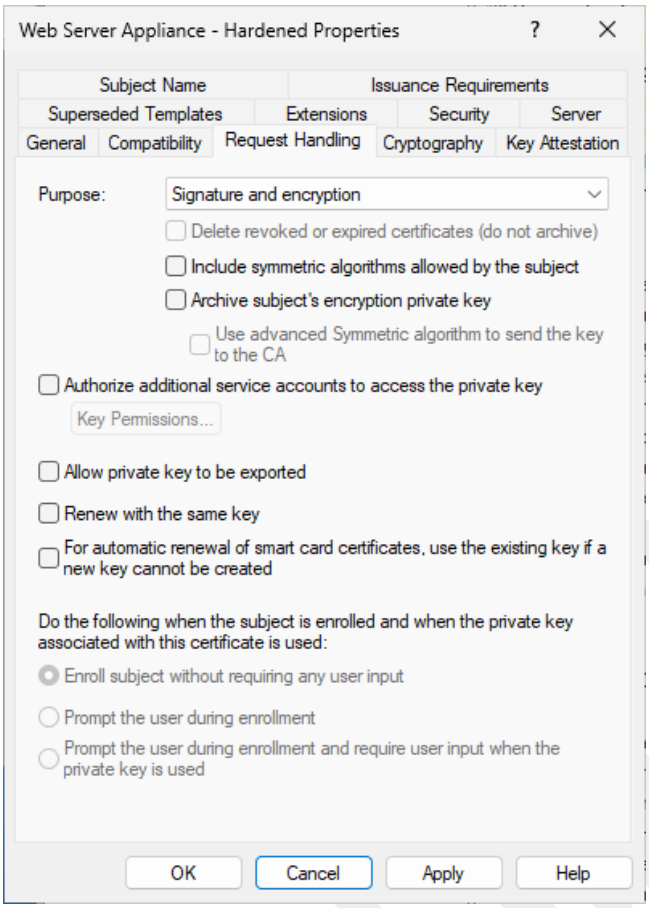
Template: Web Server – Appliance

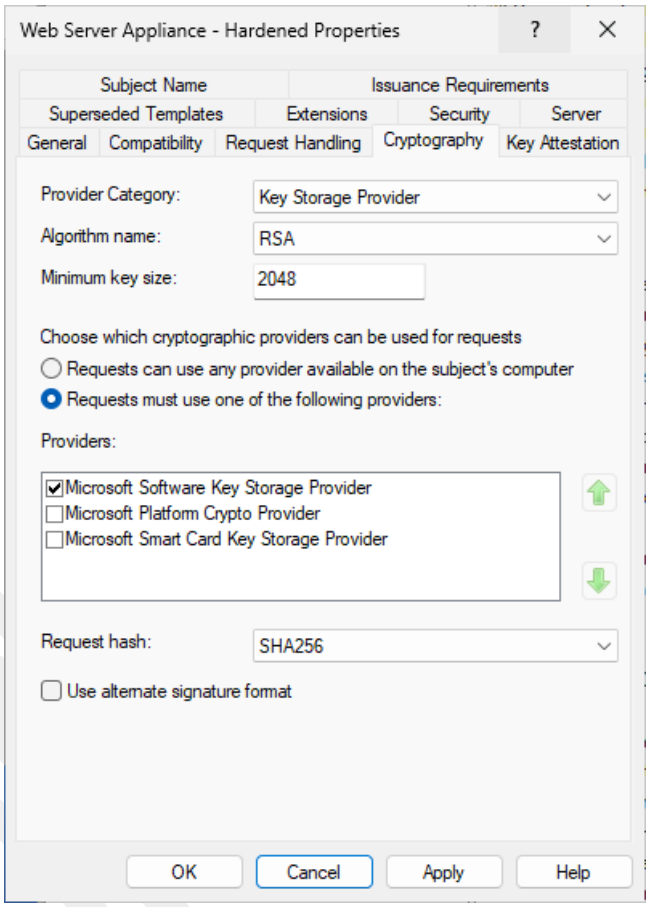
```
purpose:
  - Issues certificates for non-domain-joined appliances and infrastructure
  - Supports TLS for ESXi, iDRAC, switches, and embedded web servers
  - Allows private key export for vendor-required deployments
based_on:
  - Web Server – 2 Years
compatibility:
  certification_authority: Windows Server 2012 R2
  certificate_recipient: Windows Server 2016
cryptography:
  provider_category: Key Storage Provider
  provider: Microsoft Software Key Storage Provider
  algorithm: RSA
  minimum_key_size: 2048
  hash_algorithm: SHA256
  private_key_exportable: true
  requests_can_use_any_provider: false
subject_name:
  source: Request
  subject_alternative_name:
    - DNS
    - IP
extensions:
  application_policies:
```

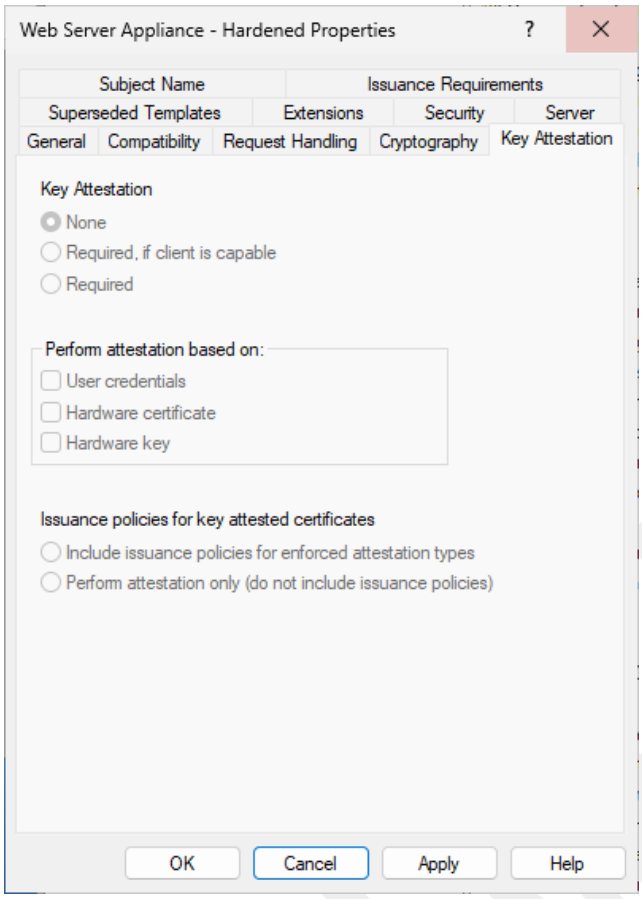
```yaml
    - Server Authentication
  key_usage:
    - Digital Signature
    - Key Encipherment
  basic_constraints:
    ca: false
validity:
  validity_period: 2 years
  renewal_period: 6 weeks
  publish_to_active_directory: false
enrollment:
  pki_appliance_certs:
    enroll: true
  domain_admins:
    read: true
issuance_requirements:
  ca_manager_approval: false
  authorized_signatures_required: 0
request_handling:
  purpose: Signature and Encryption
  archive_private_key: false
  allow_private_key_export: true

key_attestation:
  required: false
security:
  certificate_database_storage: enabled
  revocation_information: included
supersedes:
  - Web Server
notes:
  - Designed for appliances that require PEM/KEY pairs
  - Key exportability is intentional and controlled via group membership
```

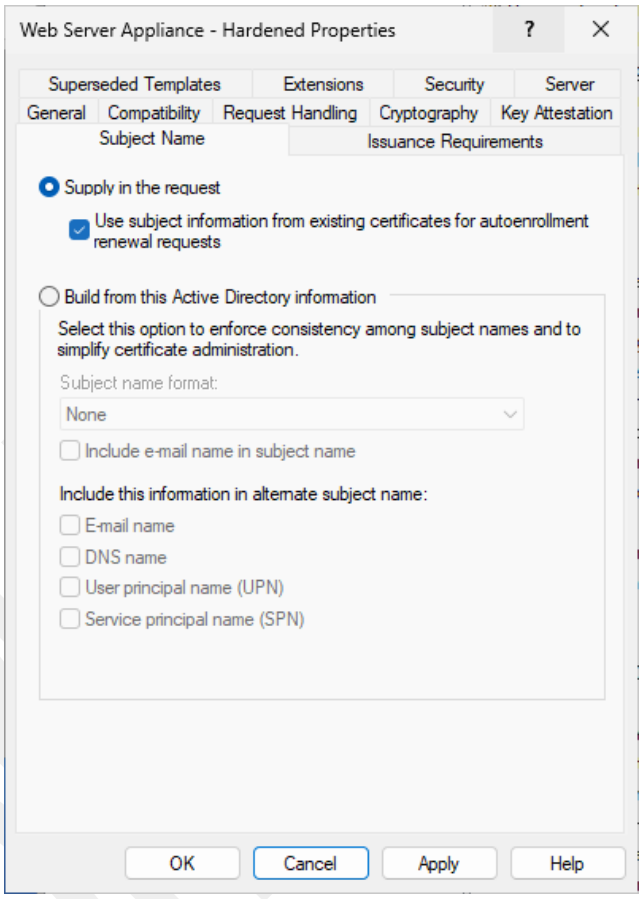

Web Server Appliance - Hardened Properties
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
General
Compatibility
Request Handling
Cryptography
Key Attestation
The template options available are based on the earliest operating system versions set in Compatibility Settings.
Show resulting changes
Compatibility Settings
Certification Authority
Windows Server 2012 R2
Certificate recipient
Windows 10 / Windows Server 2016
These settings may not prevent earlier operating systems from using this template.
OK
Cancel
Apply
Help

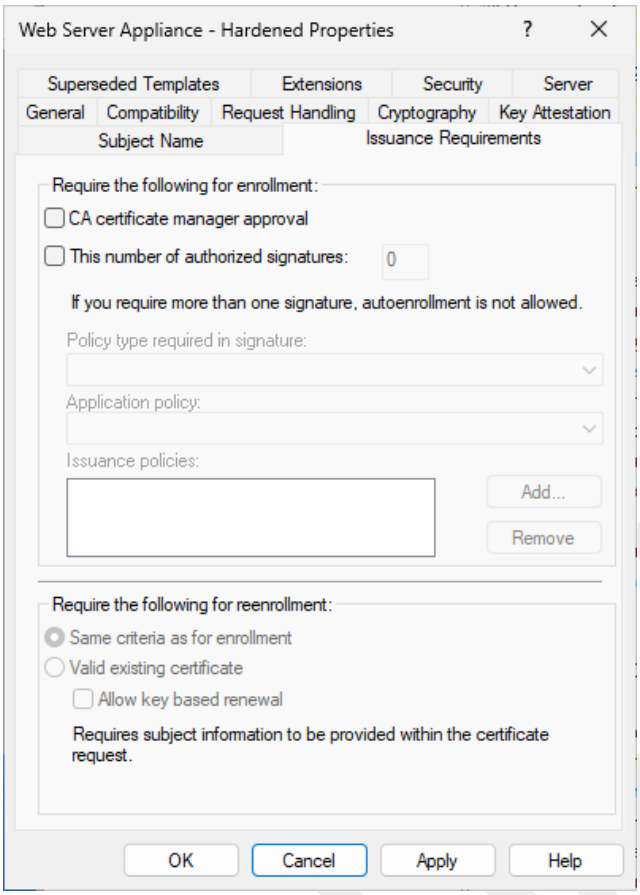

Web Server Appliance - Hardened Properties
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
General
Compatibility
Request Handling
Cryptography
Key Attestation
Template display name:
Web Server Appliance - Hardened
Template name:
WebServerAppliance-Hardened
Validity period:
2
years
Renewal period:
6
weeks
Publish certificate in Active Directory
Do not automatically reenroll if a duplicate certificate exists in Active Directory
OK
Cancel
Apply
Help

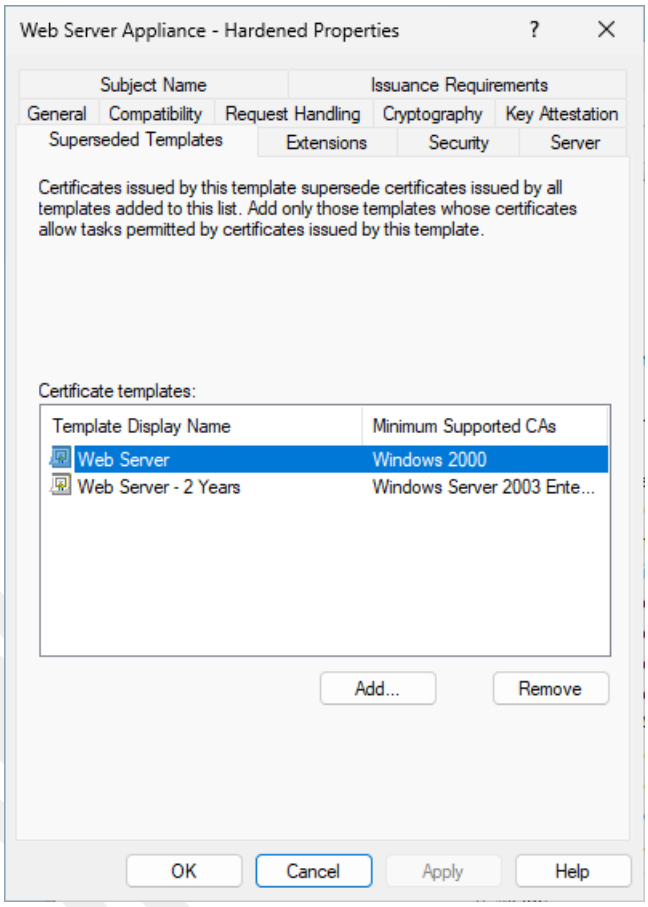

Web Server Appliance - Hardened Properties
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
General
Compatibility
Request Handling
Cryptography
Key Attestation
Purpose:
Signature and encryption
Delete revoked or expired certificates (do not archive)
Include symmetric algorithms allowed by the subject
Archive subject's encryption private key
Use advanced Symmetric algorithm to send the key to the CA
Authorize additional service accounts to access the private key
Key Permissions...
Allow private key to be exported
Renew with the same key
For automatic renewal of smart card certificates, use the existing key if a new key cannot be created
Do the following when the subject is enrolled and when the private key associated with this certificate is used:
Enroll subject without requiring any user input
Prompt the user during enrollment
Prompt the user during enrollment and require user input when the private key is used
OK
Cancel
Apply
Help

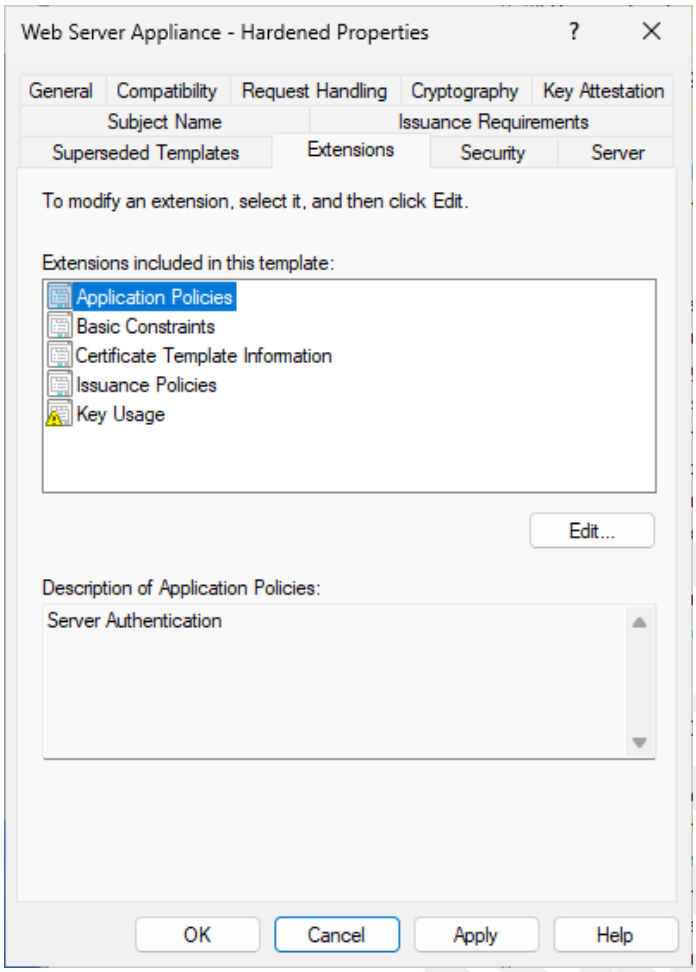

Web Server Appliance - Hardened Properties
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
General
Compatibility
Request Handling
Cryptography
Key Attestation
Provider Category:
Key Storage Provider
Algorithm name:
RSA
Minimum key size:
2048
Choose which cryptographic providers can be used for requests
Requests can use any provider available on the subject's computer
Requests must use one of the following providers:
Providers:
Microsoft Software Key Storage Provider
Microsoft Platform Crypto Provider
Microsoft Smart Card Key Storage Provider
Request hash:
SHA256
Use alternate signature format
OK
Cancel
Apply
Help

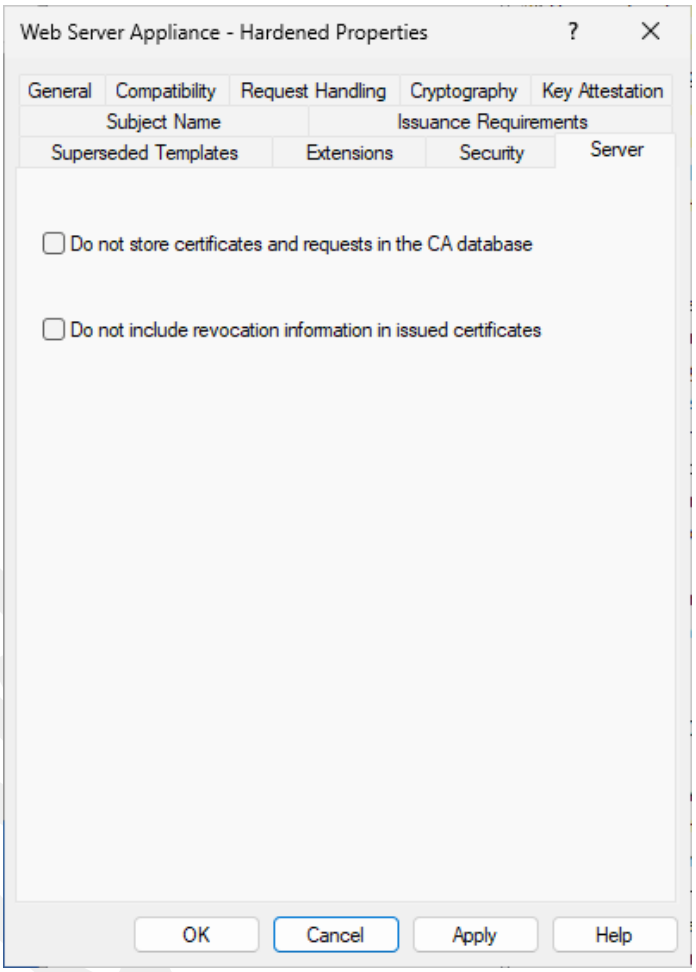

Web Server Appliance - Hardened Properties
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
General
Compatibility
Request Handling
Cryptography
Key Attestation
Key Attestation
None
Required, if client is capable
Required
Perform attestation based on:
User credentials
Hardware certificate
Hardware key
Issuance policies for key attested certificates
Include issuance policies for enforced attestation types
Perform attestation only (do not include issuance policies)
OK
Cancel
Apply
Help

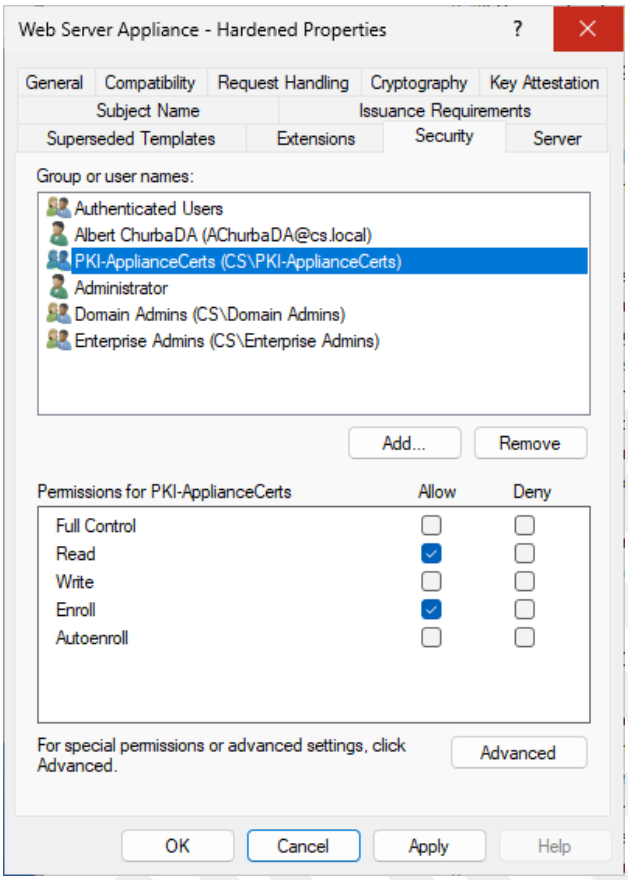

Web Server Appliance - Hardened Properties
Superseded Templates
Extensions
Security
Server
General
Compatibility
Request Handling
Cryptography
Key Attestation
Subject Name
Issuance Requirements
Supply in the request
Use subject information from existing certificates for autoenrollment renewal requests
Build from this Active Directory information
Select this option to enforce consistency among subject names and to simplify certificate administration.
Subject name format:
None
Include e-mail name in subject name
Include this information in alternate subject name:
E-mail name
DNS name
User principal name (UPN)
Service principal name (SPN)
OK
Cancel
Apply
Help


Web Server Appliance - Hardened Properties
Superseded Templates
Extensions
Security
Server
General
Compatibility
Request Handling
Cryptography
Key Attestation
Subject Name
Issuance Requirements
Require the following for enrollment:
CA certificate manager approval
This number of authorized signatures:
0
If you require more than one signature, autoenrollment is not allowed.
Policy type required in signature:
Application policy:
Issuance policies:
Add...
Remove
Require the following for reenrollment:
Same criteria as for enrollment
Valid existing certificate
Allow key based renewal
Requires subject information to be provided within the certificate request.
OK
Cancel
Apply
Help


Web Server Appliance - Hardened Properties
Subject Name
Issuance Requirements
General
Compatibility
Request Handling
Cryptography
Key Attestation
Superseded Templates
Extensions
Security
Server
Certificates issued by this template supersede certificates issued by all templates added to this list. Add only those templates whose certificates allow tasks permitted by certificates issued by this template.
Certificate templates:
Template Display Name
Minimum Supported CAs
Web Server
Windows 2000
Web Server - 2 Years
Windows Server 2003 Ente...
Add...
Remove
OK
Cancel
Apply
Help


Web Server Appliance - Hardened Properties
General
Compatibility
Request Handling
Cryptography
Key Attestation
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
To modify an extension, select it, and then click Edit.
Extensions included in this template:
Application Policies
Basic Constraints
Certificate Template Information
Issuance Policies
Key Usage
Edit...
Description of Application Policies:
Server Authentication
OK
Cancel
Apply
Help

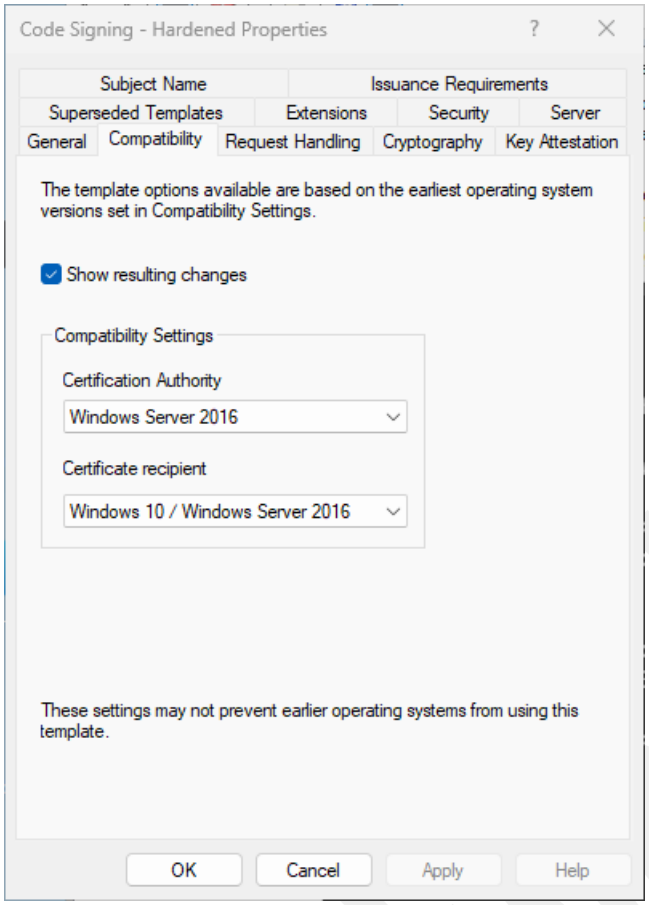

Web Server Appliance - Hardened Properties
General
Compatibility
Request Handling
Cryptography
Key Attestation
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
Do not store certificates and requests in the CA database
Do not include revocation information in issued certificates
OK
Cancel
Apply
Help

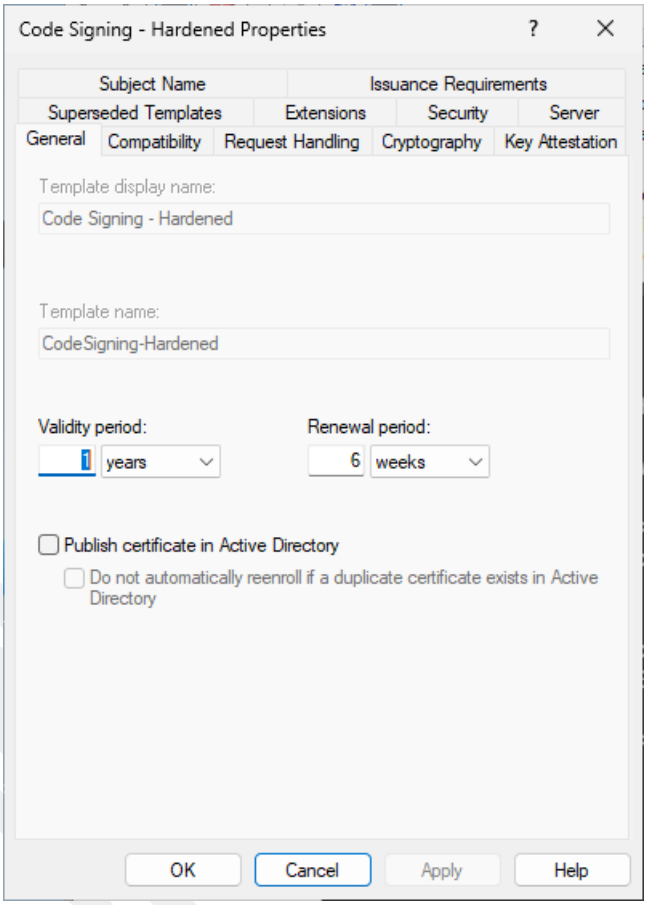


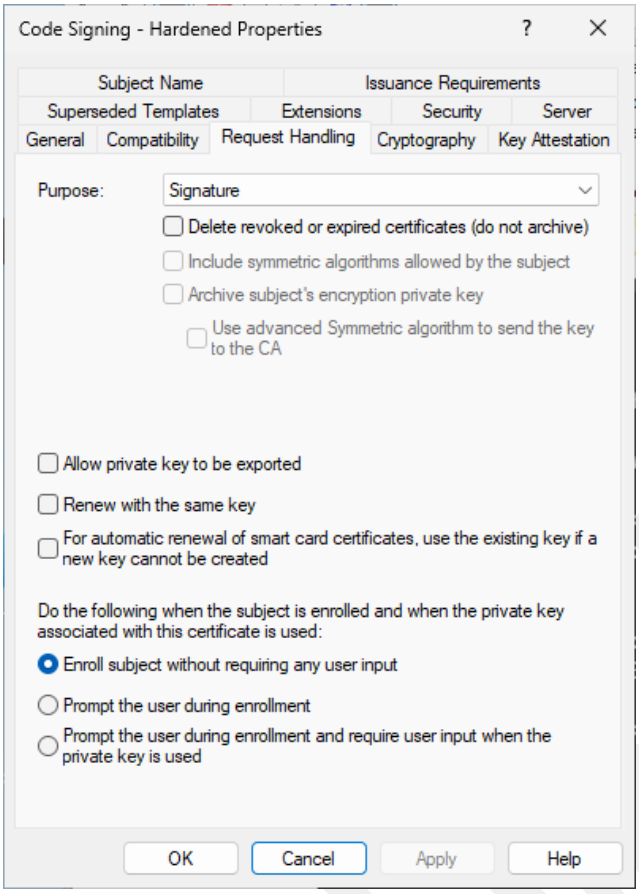
Template: Code Signing – Hardened

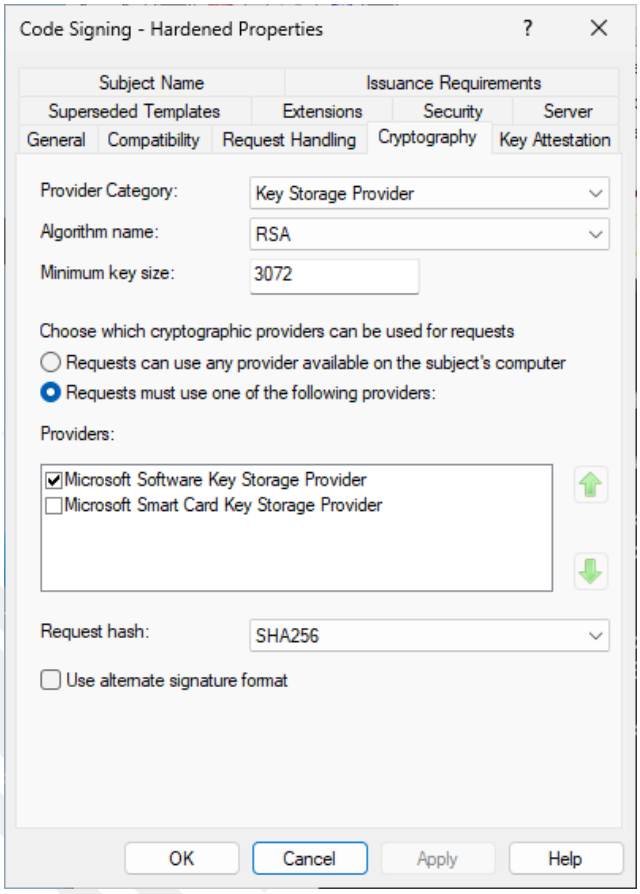
```
purpose:
  - Issues certificates for signing internal scripts and automation
  - Protects PowerShell, scheduled tasks, and administrative tooling
  - Restricts enrollment to approved signing identities
based_on:
  - Code Signing
compatibility:
  certification_authority: Windows Server 2016
  certificate_recipient: Windows Server 2016
cryptography:
  provider_category: Key Storage Provider
  provider: Microsoft Software Key Storage Provider
  algorithm: RSA
  minimum_key_size: 3072
  hash_algorithm: SHA256
  private_key_exportable: false
  requests_can_use_any_provider: false
subject_name:
  source: Request
extensions:
  application_policies:
    - Code Signing
```

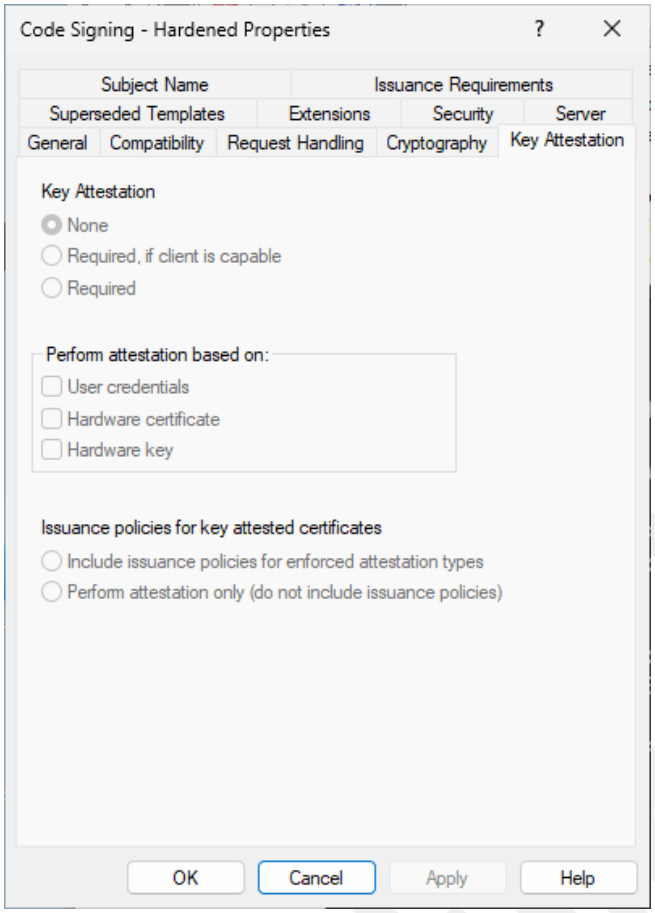
```yaml
  key_usage:
    - Digital Signature
validity:
  validity_period: 1 year
  renewal_period: 6 weeks
  publish_to_active_directory: false
enrollment:
  pki_code_signers:
    enroll: true
  domain_admins:
    read: true
issuance_requirements:
  ca_manager_approval: false
  authorized_signatures_required: 0
request_handling:
  purpose: Signature
  archive_private_key: false
key_attestation:
  required: false
security:
  certificate_database_storage: enabled
  revocation_information: included
supersedes:
  - Code Signing
notes:
  - Used to sign RdsCertAutomation and invoker scripts
  - Prevents unsigned code execution in hardened environments
```

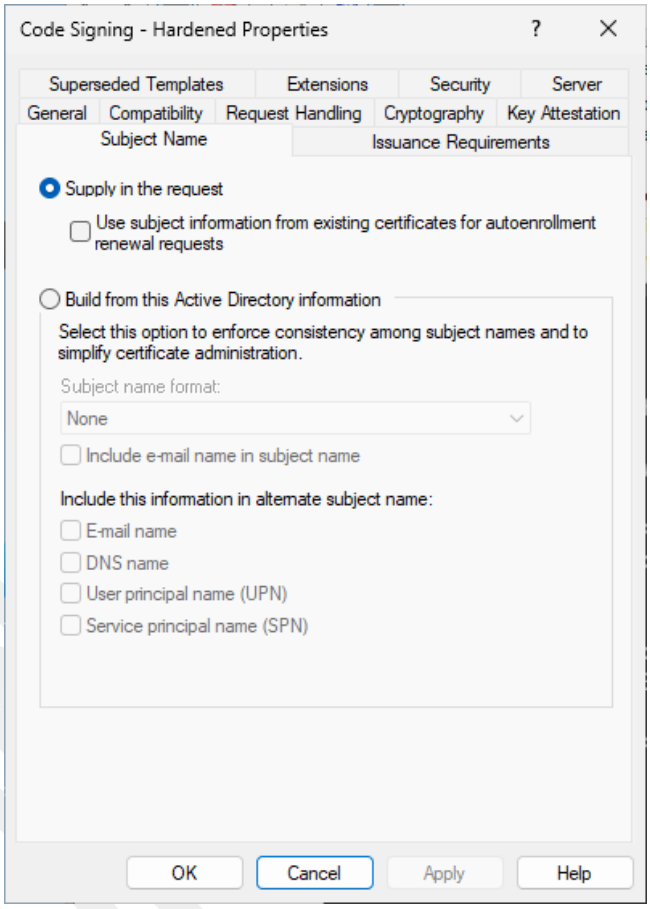

Code Signing - Hardened Properties
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
General
Compatibility
Request Handling
Cryptography
Key Attestation
The template options available are based on the earliest operating system versions set in Compatibility Settings.
Show resulting changes
Compatibility Settings
Certification Authority
Windows Server 2016
Certificate recipient
Windows 10 / Windows Server 2016
These settings may not prevent earlier operating systems from using this template.
OK
Cancel
Apply
Help


Code Signing - Hardened Properties
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
General
Compatibility
Request Handling
Cryptography
Key Attestation
Template display name:
Code Signing - Hardened
Template name:
CodeSigning-Hardened
Validity period:
1
years
Renewal period:
6
weeks
Publish certificate in Active Directory
Do not automatically reenroll if a duplicate certificate exists in Active Directory
OK
Cancel
Apply
Help


Code Signing - Hardened Properties
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
General
Compatibility
Request Handling
Cryptography
Key Attestation
Purpose:
Signature
Delete revoked or expired certificates (do not archive)
Include symmetric algorithms allowed by the subject
Archive subject's encryption private key
Use advanced Symmetric algorithm to send the key to the CA
Allow private key to be exported
Renew with the same key
For automatic renewal of smart card certificates, use the existing key if a new key cannot be created
Do the following when the subject is enrolled and when the private key associated with this certificate is used:
Enroll subject without requiring any user input
Prompt the user during enrollment
Prompt the user during enrollment and require user input when the private key is used
OK
Cancel
Apply
Help


Code Signing - Hardened Properties
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
General
Compatibility
Request Handling
Cryptography
Key Attestation
Provider Category:
Key Storage Provider
Algorithm name:
RSA
Minimum key size:
3072
Choose which cryptographic providers can be used for requests
Requests can use any provider available on the subject's computer
Requests must use one of the following providers:
Providers:
Microsoft Software Key Storage Provider
Microsoft Smart Card Key Storage Provider
Request hash:
SHA256
Use alternate signature format
OK
Cancel
Apply
Help





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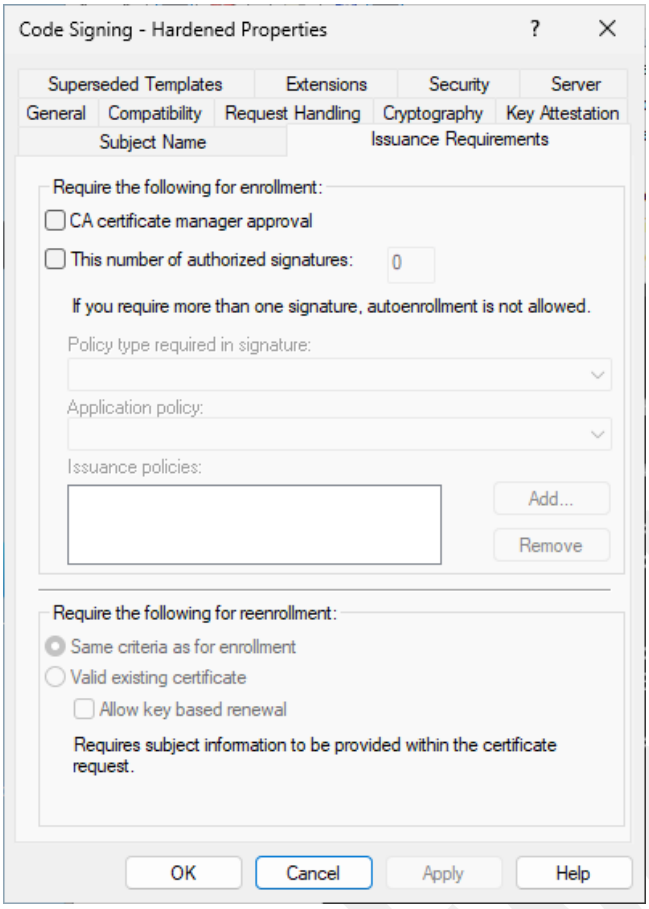
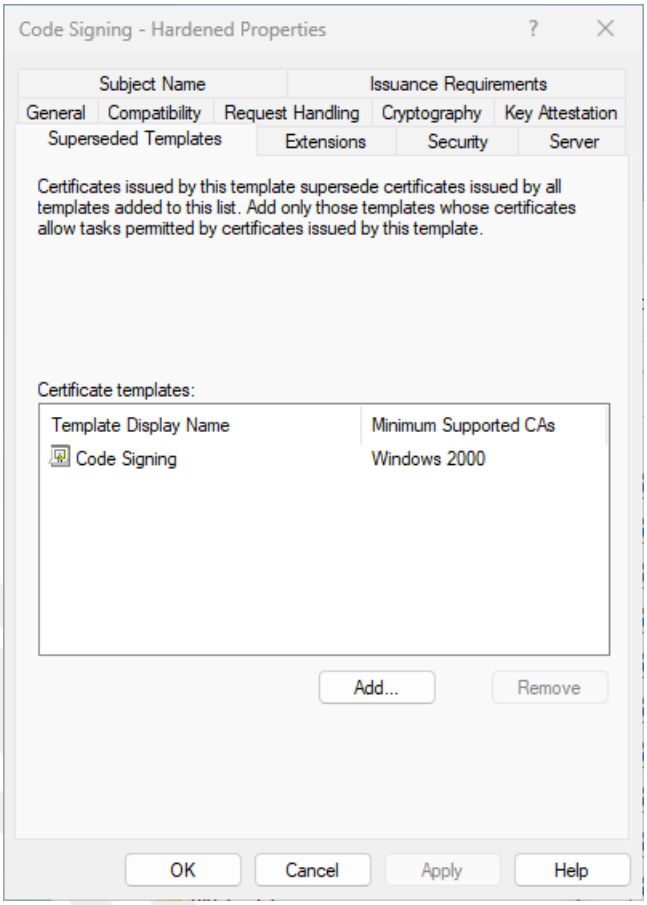
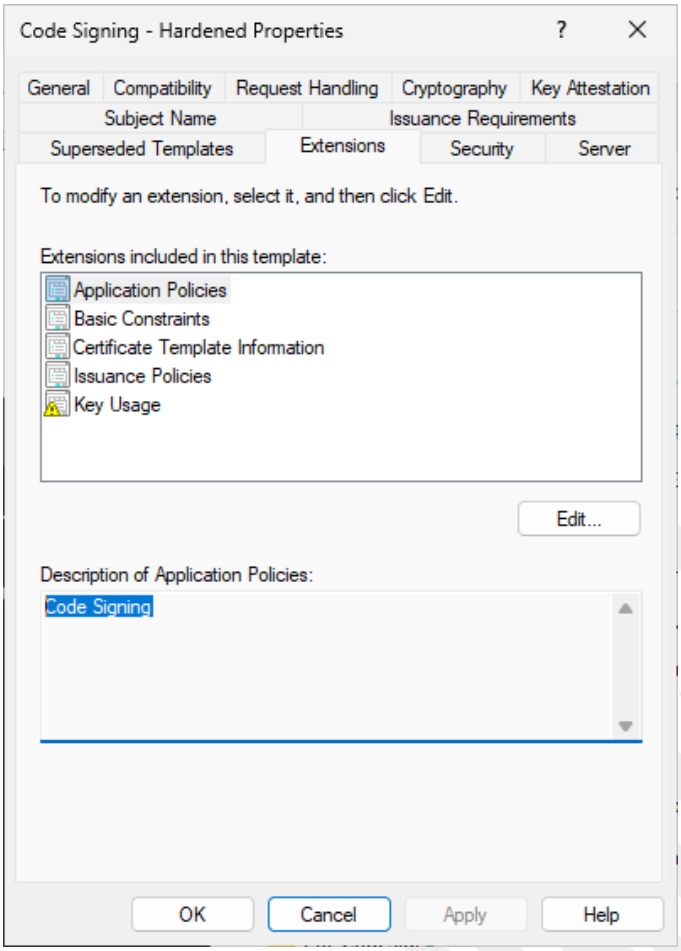
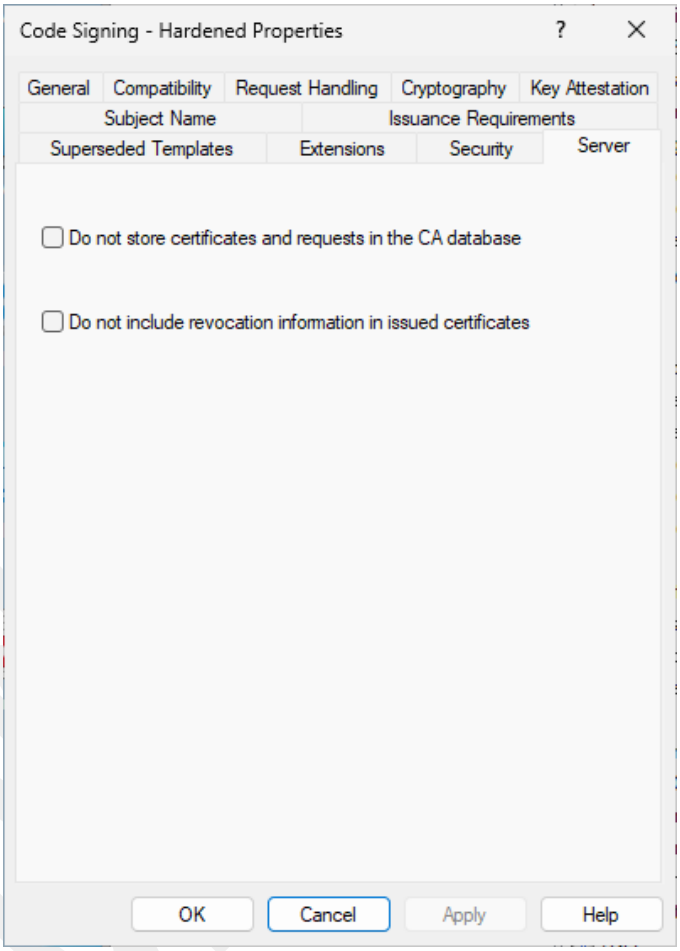
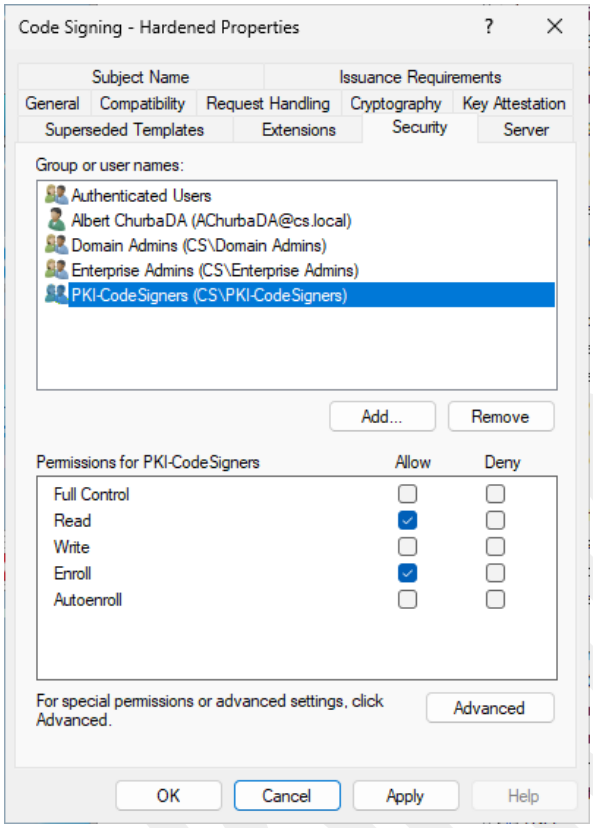

Code Signing - Hardened Properties
Superseded Templates
Extensions
Security
Server
General
Compatibility
Request Handling
Cryptography
Key Attestation
Subject Name
Issuance Requirements
Require the following for enrollment:
CA certificate manager approval
This number of authorized signatures:
0
If you require more than one signature, autoenrollment is not allowed.
Policy type required in signature:
Application policy:
Issuance policies:
Add...
Remove
Require the following for reenrollment:
Same criteria as for enrollment
Valid existing certificate
Allow key based renewal
Requires subject information to be provided within the certificate request.
OK
Cancel
Apply
Help


Code Signing - Hardened Properties
Subject Name
Issuance Requirements
General
Compatibility
Request Handling
Cryptography
Key Attestation
Superseded Templates
Extensions
Security
Server
Certificates issued by this template supersede certificates issued by all templates added to this list. Add only those templates whose certificates allow tasks permitted by certificates issued by this template.
Certificate templates:
Template Display Name
Minimum Supported CAs
Code Signing
Windows 2000
Add...
Remove
OK
Cancel
Apply
Help


Code Signing - Hardened Properties
General
Compatibility
Request Handling
Cryptography
Key Attestation
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
To modify an extension, select it, and then click Edit.
Extensions included in this template:
Application Policies
Basic Constraints
Certificate Template Information
Issuance Policies
Key Usage
Edit...
Description of Application Policies:
Code Signing
OK
Cancel
Apply
Help

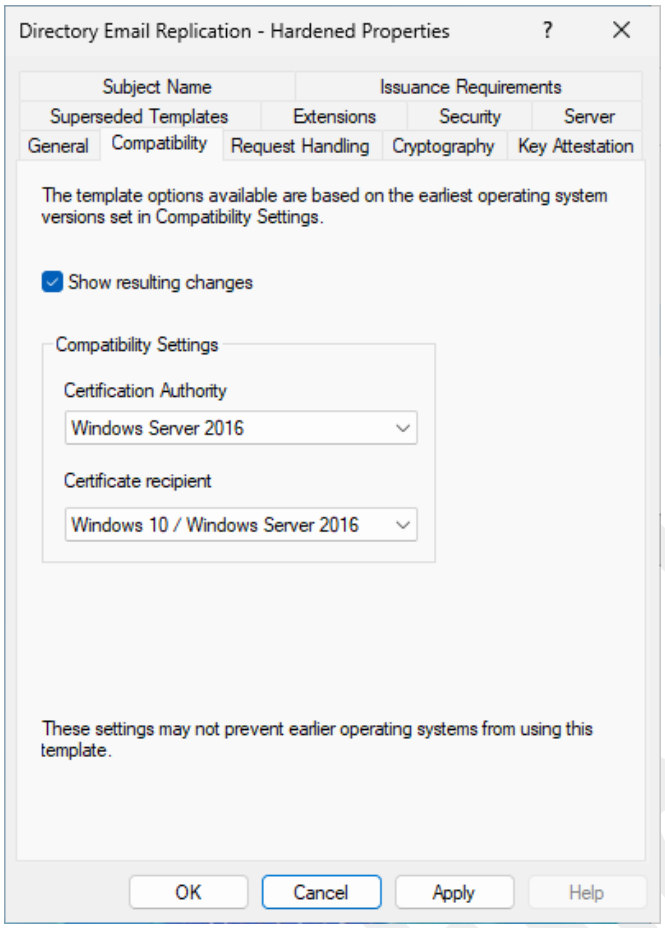

Code Signing - Hardened Properties
General
Compatibility
Request Handling
Cryptography
Key Attestation
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
Do not store certificates and requests in the CA database
Do not include revocation information in issued certificates
OK
Cancel
Apply
Help

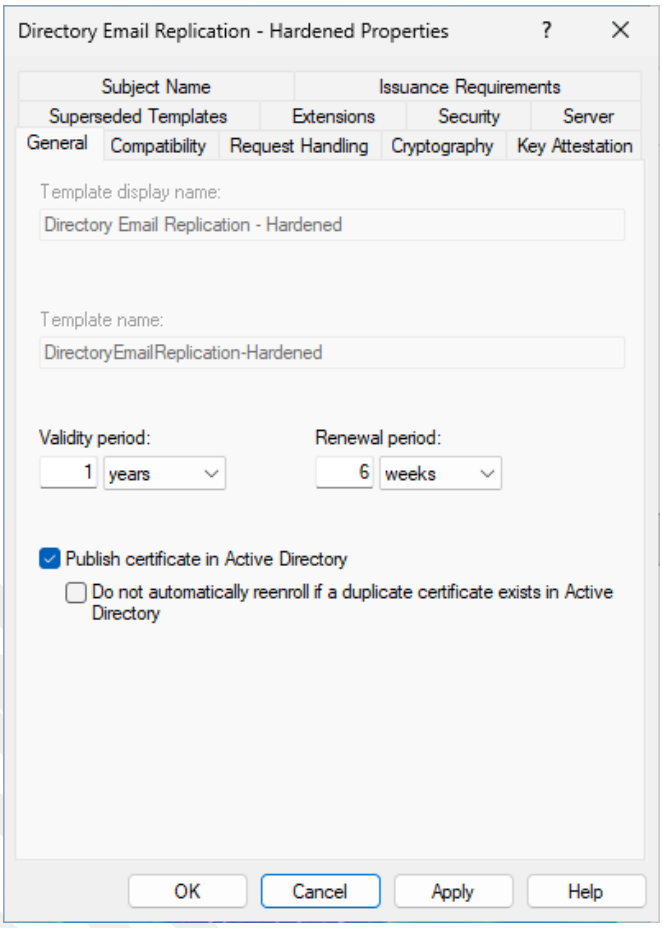


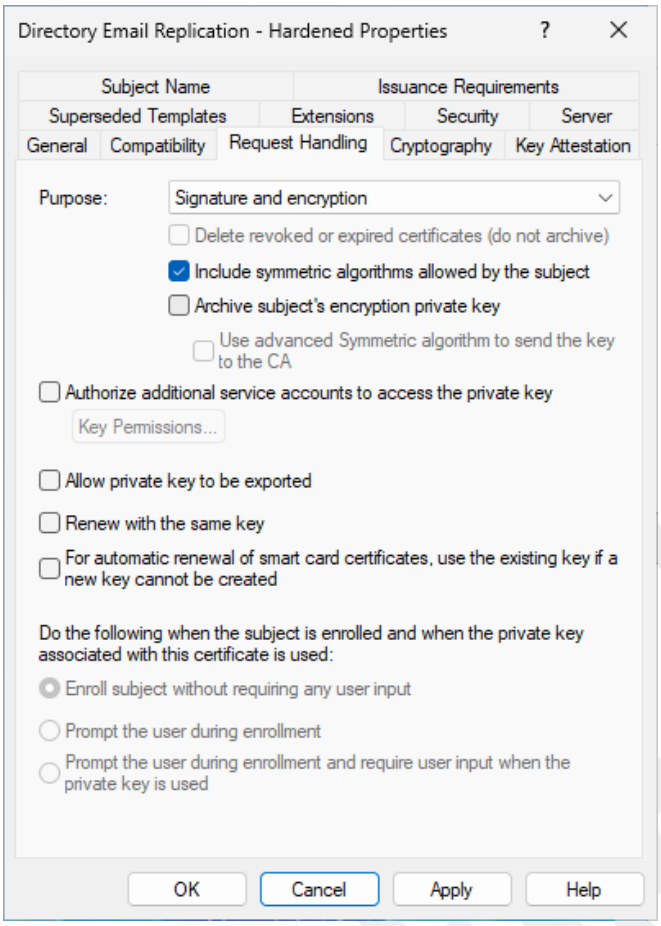
Template: Directory Email Replication – Hardened

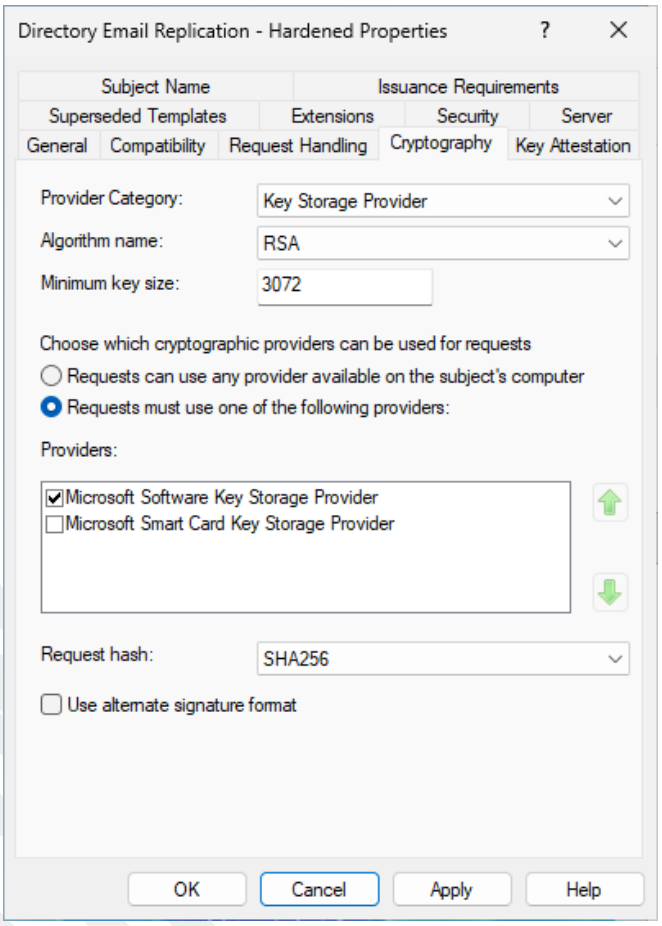
```yaml
based_on:
  - Directory Email Replication
purpose:
  - Issues certificates exclusively for Active Directory directory service email-style replication
  - Secures replication traffic used by directory-integrated services
  - Restricts certificate use to a single EKU to prevent misuse
compatibility:
  certification_authority: Windows Server 2016
  certificate_recipient: Windows Server 2016
cryptography:
  provider_category: Key Storage Provider
  provider: Microsoft Software Key Storage Provider
  algorithm: RSA
  minimum_key_size: 3072
  hash_algorithm: SHA256
  private_key_exportable: false
  requests_can_use_any_provider: false
subject_name:
  source: Active Directory
  subject_name_format: None
  subject_alternative_name:
    - DNS
extensions:
  application_policies:
    - Directory Service Email Replication (1.3.6.1.4.1.311.21.19)
```

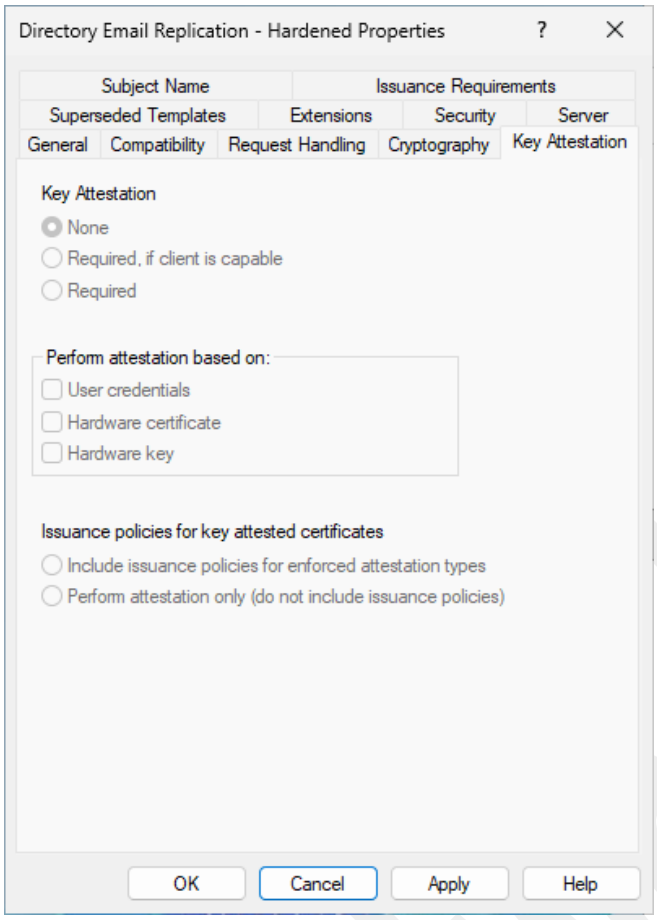
```yaml
  key_usage:
    - Digital Signature
    - Key Encipherment
  basic_constraints:
    ca: false
validity:
  validity_period: 1 year
  renewal_period: 6 weeks
  publish_to_active_directory: true
enrollment:
  domain_controllers:
    enroll: true
    autoenroll: true
  domain_admins:
    read: true
  enterprise_admins:
    read: true
  authenticated_users:
    read: true
issuance_requirements:
  ca_manager_approval: false
  authorized_signatures_required: 0
request_handling:
  purpose: Signature and Encryption
  archive_private_key: false
  allow_private_key_export: false

key_attestation:
  required: false
security:
  certificate_database_storage: enabled
  revocation_information: included
supersedes:
  - Directory Email Replication
notes:
  - Supersedes the default "Directory Email Replication" template
  - EKU scoping prevents certificate reuse for authentication or TLS
  - Required for secure replication behavior on modern domain controllers
```

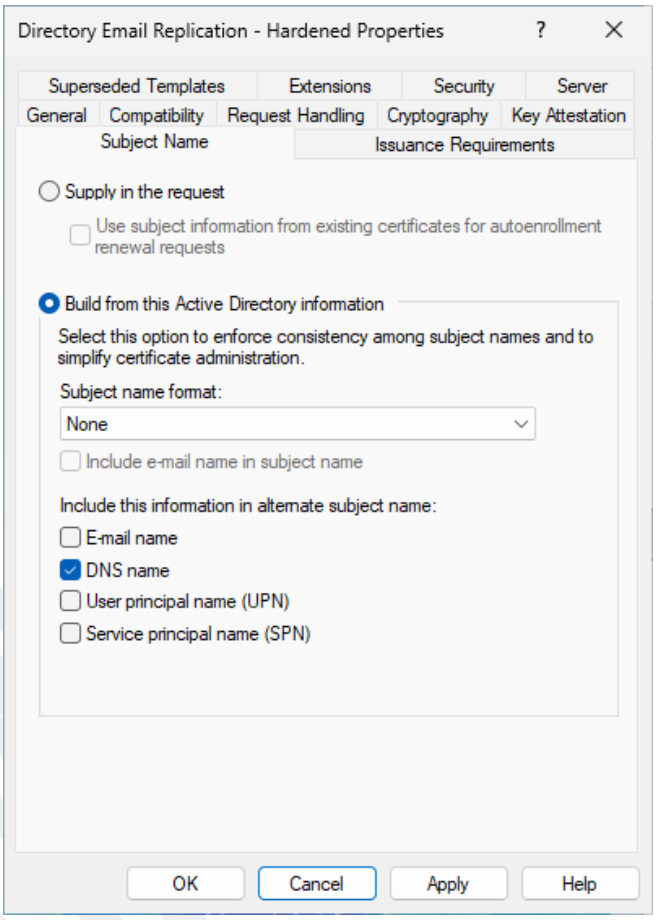

Directory Email Replication - Hardened Properties
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
General
Compatibility
Request Handling
Cryptography
Key Attestation
The template options available are based on the earliest operating system versions set in Compatibility Settings.
Show resulting changes
Compatibility Settings
Certification Authority
Windows Server 2016
Certificate recipient
Windows 10 / Windows Server 2016
These settings may not prevent earlier operating systems from using this template.
OK
Cancel
Apply
Help

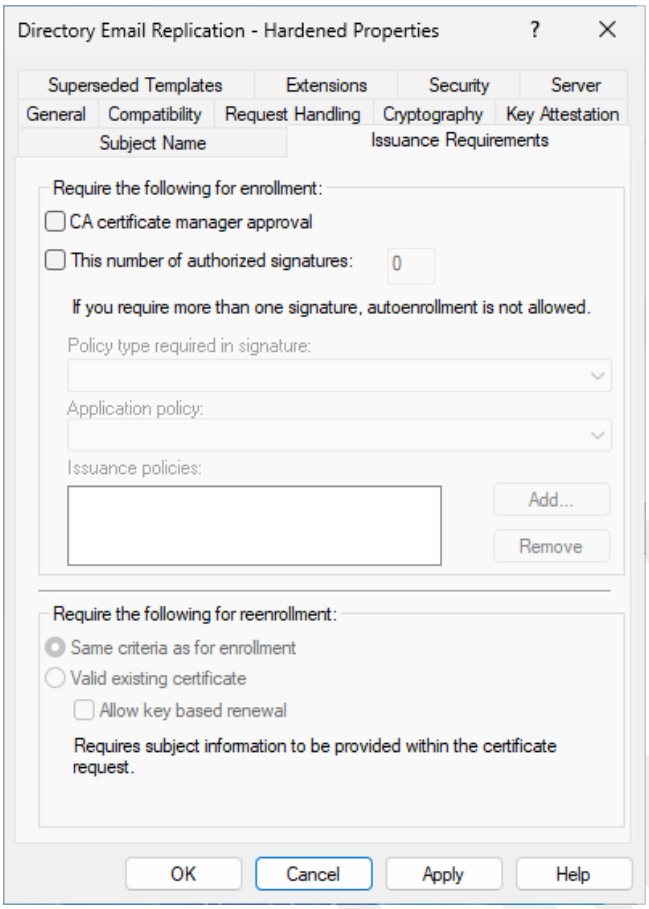

Directory Email Replication - Hardened Properties
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
General
Compatibility
Request Handling
Cryptography
Key Attestation
Template display name:
Directory Email Replication - Hardened
Template name:
DirectoryEmailReplication-Hardened
Validity period:
1
years
Renewal period:
6
weeks
Publish certificate in Active Directory
Do not automatically reenroll if a duplicate certificate exists in Active Directory
OK
Cancel
Apply
Help

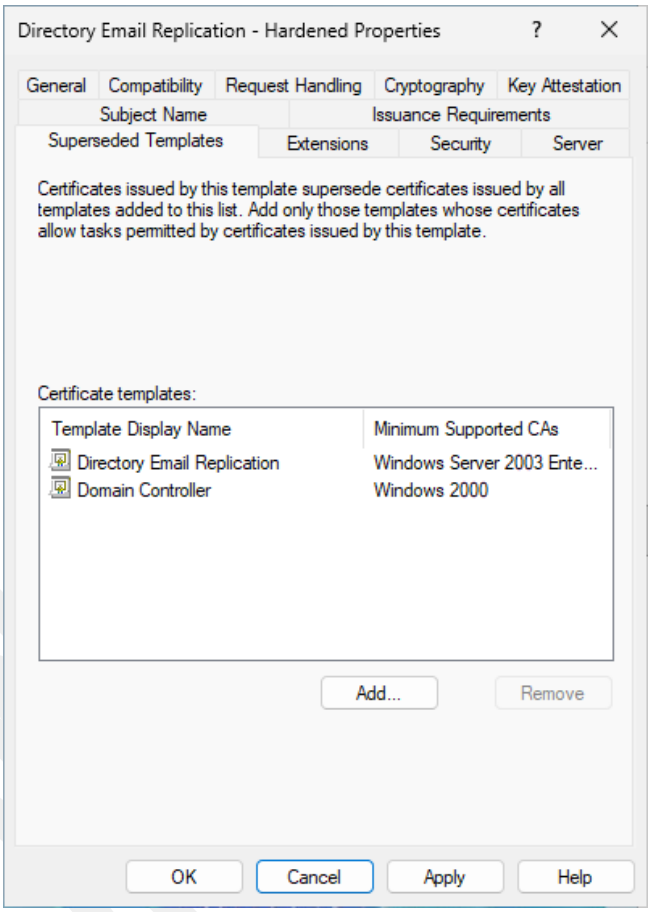

Directory Email Replication - Hardened Properties
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
General
Compatibility
Request Handling
Cryptography
Key Attestation
Purpose:
Signature and encryption
Delete revoked or expired certificates (do not archive)
Include symmetric algorithms allowed by the subject
Archive subject's encryption private key
Use advanced Symmetric algorithm to send the key to the CA
Authorize additional service accounts to access the private key
Key Permissions...
Allow private key to be exported
Renew with the same key
For automatic renewal of smart card certificates, use the existing key if a new key cannot be created
Do the following when the subject is enrolled and when the private key associated with this certificate is used:
Enroll subject without requiring any user input
Prompt the user during enrollment
Prompt the user during enrollment and require user input when the private key is used
OK
Cancel
Apply
Help

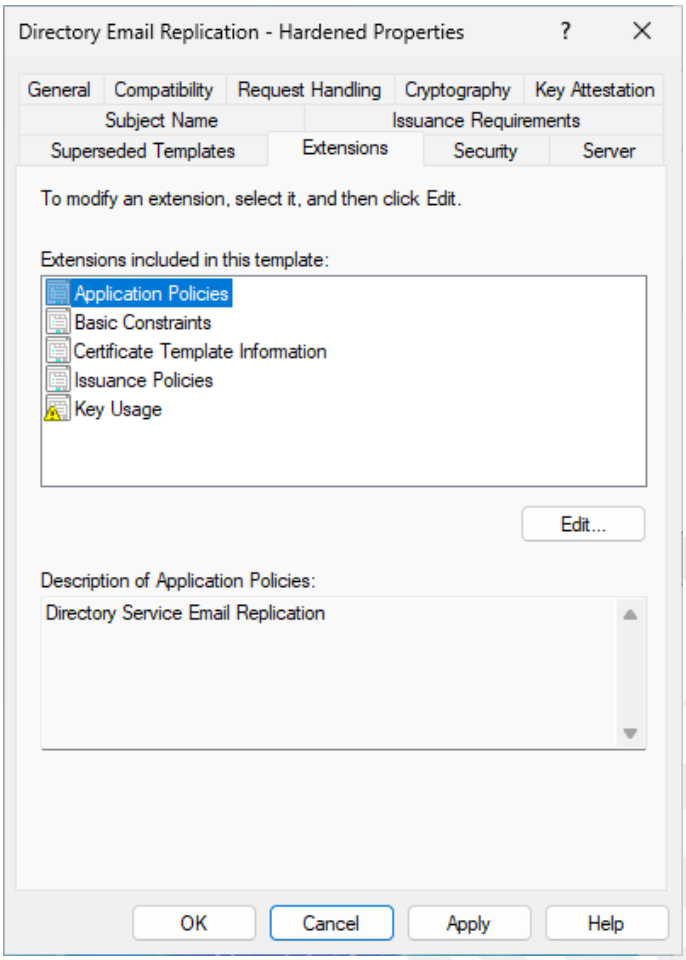

Directory Email Replication - Hardened Properties
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
General
Compatibility
Request Handling
Cryptography
Key Attestation
Provider Category:
Key Storage Provider
Algorithm name:
RSA
Minimum key size:
3072
Choose which cryptographic providers can be used for requests
Requests can use any provider available on the subject's computer
Requests must use one of the following providers:
Providers:
Microsoft Software Key Storage Provider
Microsoft Smart Card Key Storage Provider
Request hash:
SHA256
Use alternate signature format
OK
Cancel
Apply
Help

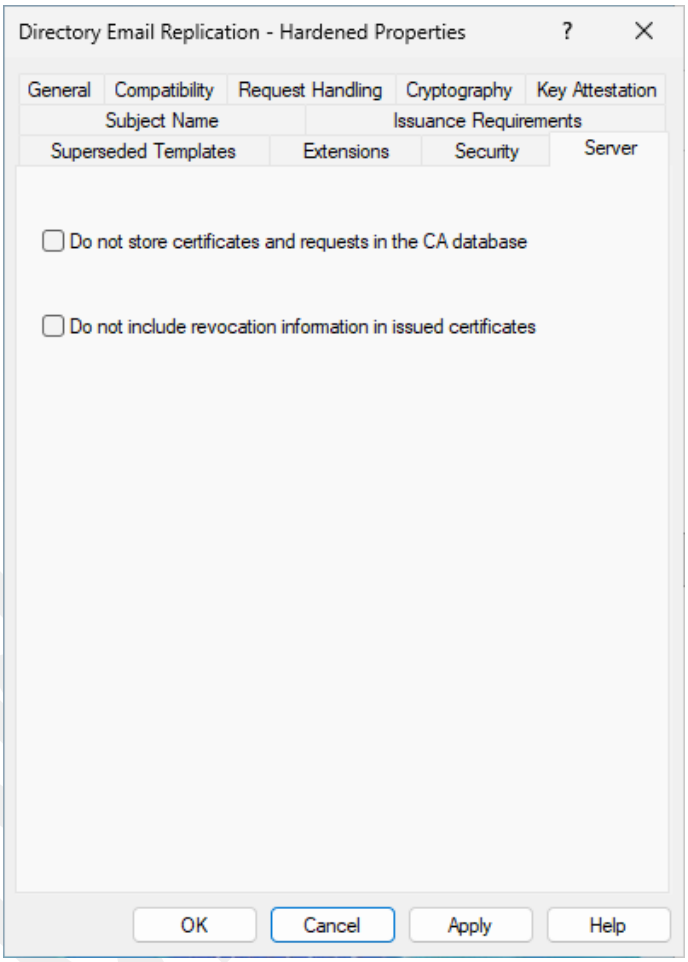

Directory Email Replication - Hardened Properties
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
General
Compatibility
Request Handling
Cryptography
Key Attestation
Key Attestation
None
Required, if client is capable
Required
Perform attestation based on:
User credentials
Hardware certificate
Hardware key
Issuance policies for key attested certificates
Include issuance policies for enforced attestation types
Perform attestation only (do not include issuance policies)
OK
Cancel
Apply
Help

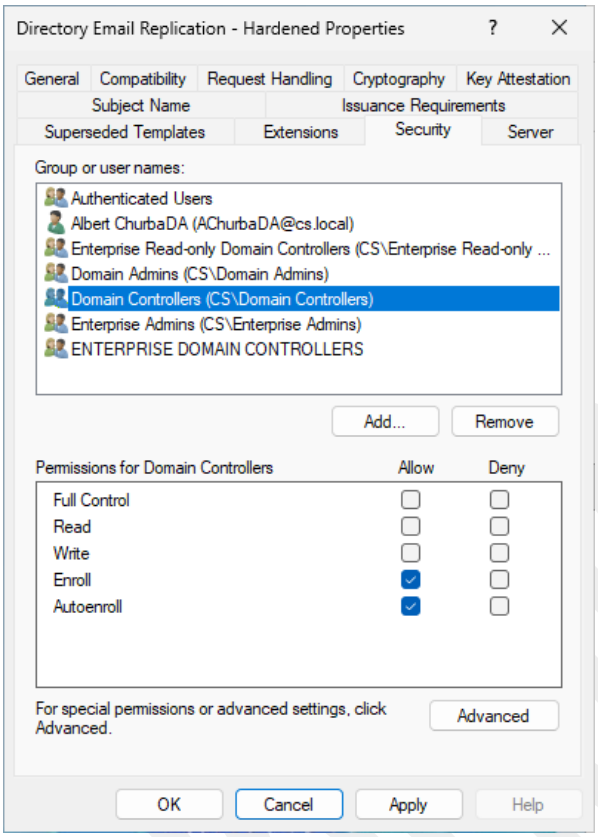

Directory Email Replication - Hardened Properties
Superseded Templates
Extensions
Security
Server
General
Compatibility
Request Handling
Cryptography
Key Attestation
Subject Name
Issuance Requirements
Supply in the request
Use subject information from existing certificates for autoenrollment renewal requests
Build from this Active Directory information
Select this option to enforce consistency among subject names and to simplify certificate administration.
Subject name format:
None
Include e-mail name in subject name
Include this information in alternate subject name:
E-mail name
DNS name
User principal name (UPN)
Service principal name (SPN)
OK
Cancel
Apply
Help


Directory Email Replication - Hardened Properties
Superseded Templates
Extensions
Security
Server
General
Compatibility
Request Handling
Cryptography
Key Attestation
Subject Name
Issuance Requirements
Require the following for enrollment:
CA certificate manager approval
This number of authorized signatures:
0
If you require more than one signature, autoenrollment is not allowed.
Policy type required in signature:
Application policy:
Issuance policies:
Add...
Remove
Require the following for reenrollment:
Same criteria as for enrollment
Valid existing certificate
Allow key based renewal
Requires subject information to be provided within the certificate request.
OK
Cancel
Apply
Help


Directory Email Replication - Hardened Properties
General
Compatibility
Request Handling
Cryptography
Key Attestation
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
Certificates issued by this template supersede certificates issued by all templates added to this list. Add only those templates whose certificates allow tasks permitted by certificates issued by this template.
Certificate templates:
Template Display Name
Minimum Supported CAs
Directory Email Replication
Windows Server 2003 Ente...
Domain Controller
Windows 2000
Add...
Remove
OK
Cancel
Apply
Help


Directory Email Replication - Hardened Properties
General
Compatibility
Request Handling
Cryptography
Key Attestation
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
To modify an extension, select it, and then click Edit.
Extensions included in this template:
Application Policies
Basic Constraints
Certificate Template Information
Issuance Policies
Key Usage
Edit...
Description of Application Policies:
Directory Service Email Replication
OK
Cancel
Apply
Help


Directory Email Replication - Hardened Properties
General
Compatibility
Request Handling
Cryptography
Key Attestation
Subject Name
Issuance Requirements
Superseded Templates
Extensions
Security
Server
Do not store certificates and requests in the CA database
Do not include revocation information in issued certificates
OK
Cancel
Apply
Help



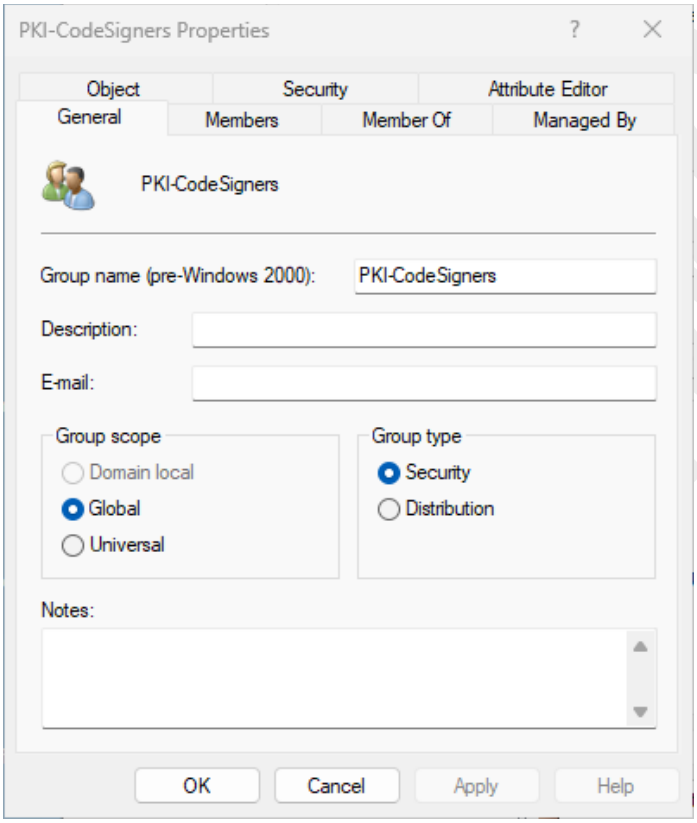
Appendix C – Active Directory Security Groups

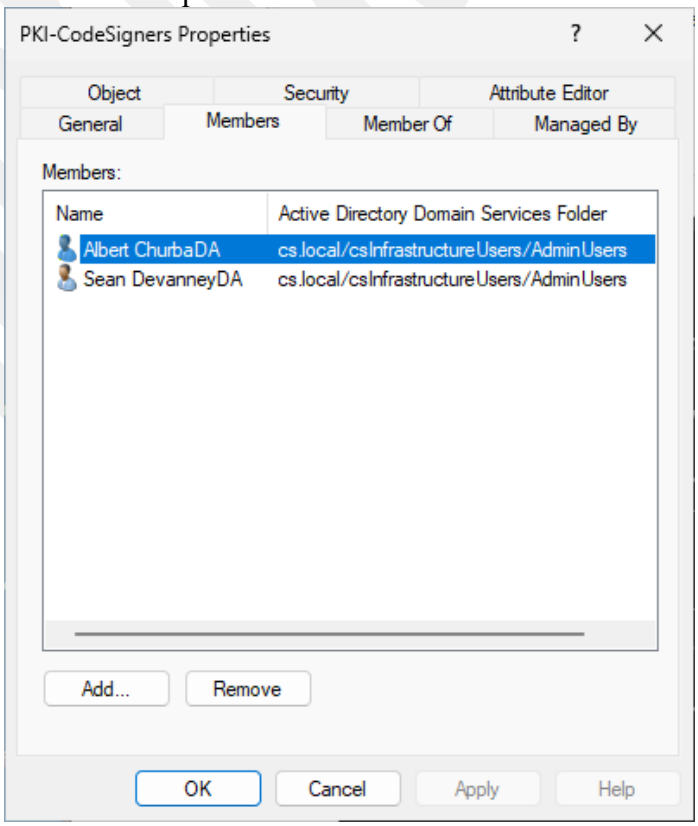
Group: PKI-CodeSigners

```
pki_security_group:
  name: PKI-CodeSigners
  object_type: ActiveDirectorySecurityGroup
  scope: Global
  category: Security
  purpose:
    Restricts enrollment of Code Signing certificates to explicitly authorized personnel. This prevents uncontrolled
     issuance of signing certificates that could be used to bypass application trust controls, execute malicious code,
     or undermine software integrity enforcement mechanisms.
  supported_templates:
    - Code Signing - Hardened
  membership_controls:
    membership_type: Explicit
    approved_members:
      - cs\[DOMAIN]\[PKI-AdminGroup]\AdminUsers
    prohibited_members:
      - Authenticated Users
      - Domain Users
  enrollment_rights:
```

```
  enroll: true
  autoenroll: false
  read: true
security_controls:
  - SeparationOfDuties
  - LeastPrivilege
  - ExplicitAuthorization
  - AuditTraceability
audit_alignment:
  nist_800_53:
    - IA-5
    - CM-5
    - CM-14
  iso_27001:
    - A.9.2
    - A.12.5
    - A.14.2
```

risk_statement: Without a dedicated Code Signing security group, unauthorized users could obtain signing certificates and introduce malicious or untrusted software into the enterprise environment.





PKI-CodeSigners Properties

Object | Security | Attribute Editor
General | Members | Member Of | Managed By

Member of:

Name	Active Directory Domain Services Folder

Add... | Remove

This list displays only groups from the current domain and groups maintained in the Global Catalog, such as universal groups.

OK | Cancel | Apply | Help

PKI-CodeSigners Properties

Object | Security | Attribute Editor
General | Members | Member Of | Managed By

Name:

Change... | Properties | Clear

Manager can update membership list

Office:

Street:

City:

State/province:

Country/region:

Telephone number:

Fax number:

OK | Cancel | Apply | Help

PKI-CodeSigners Properties

General | Members | Member Of | Managed By
Object | Security | Attribute Editor

Canonical name of object:

cs.local/csInfrastructureUsers/PKI-CodeSigners

Object class: Group

Created: 12/15/2025 10:21:16 PM

Modified: 12/15/2025 10:22:19 PM

Update Sequence Numbers (USNs):

Current: 5508308

Original: 5508287

Protect object from accidental deletion

OK | Cancel | Apply | Help

PKI-CodeSigners Properties

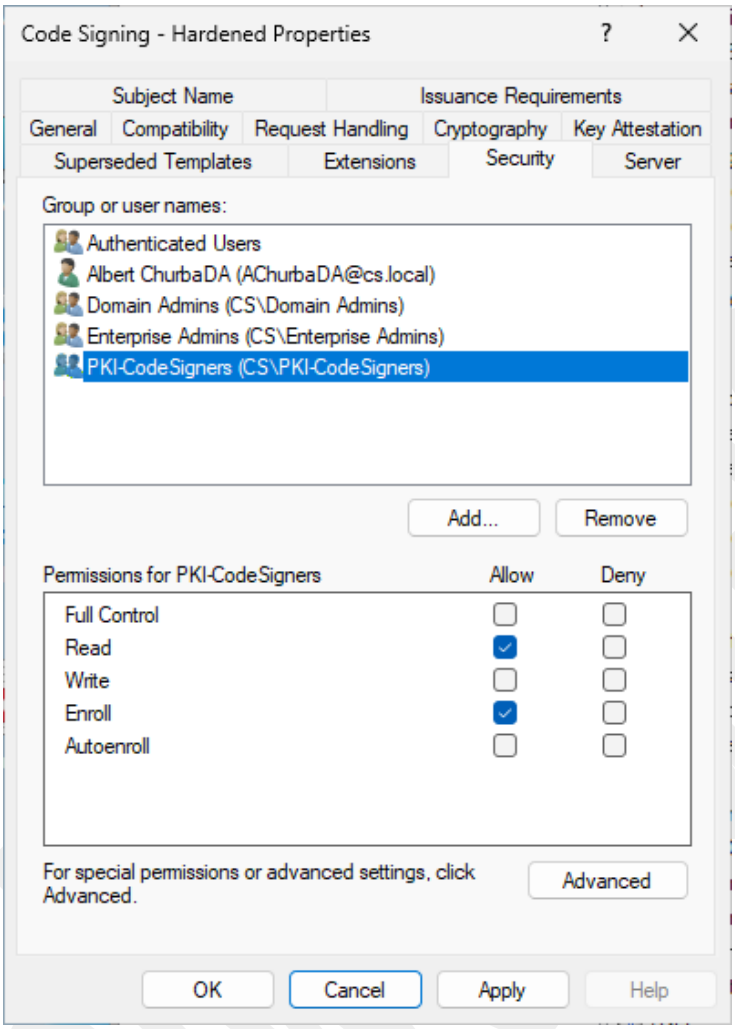
General | Members | Member Of | Managed By
Object | Security | Attribute Editor

Attributes:

Attribute	Value
accountNameHistory	<not set>
adminCount	<not set>
adminDescription	<not set>
adminDisplayName	<not set>
altSecurityIdentities	<not set>
cn	PKI-CodeSigners
controlAccessRights	<not set>
description	<not set>
desktopProfile	<not set>
displayName	<not set>
displayNamePrintable	<not set>
distinguishedName	CN=PKI-CodeSigners,OU=csInfrastructureUs
dSASignature	<not set>
dSCorePropagationD...	0x0 = ( )

Edit | Filter

OK | Cancel | Apply | Help



Group: PKI-ApplianceCerts

pki_security_group:
  name: PKI-ApplianceCerts
  object_type: ActiveDirectorySecurityGroup
  scope: Global
  category: Security
purpose:
  Controls issuance of certificates to non-domain-joined infrastructure appliances such as hypervisors, management controllers, network switches, load balancers, and storage platforms. Supports controlled CSR-based enrollment workflows.
supported_templates:
  - Web Server Appliance - Hardened
membership_controls:
  membership_type: Explicit
  approved_members:
    - PKI Administrators
  prohibited_members:
    - Authenticated Users

```
    - Domain Users
enrollment_model:
  csr_based: true
  autoenrollment: false
  key_generation: External
enrollment_rights:
  enroll: true
  autoenroll: false
  read: true
security_controls:
  - ManualCSRValidation
  - ExplicitApproval
  - AuditLogging
  - CertificateLifecycleControl
audit_alignment:
  nist_800_53:
    - IA-7
    - IA-12
    - SC-12
  iso_27001:
    - A.9.4
    - A.10.1
    - A.13.2

risk_statement:
  Infrastructure appliances often lack native domain security controls. Without a dedicated issuance group,
  certificates could be issued without proper validation, increasing exposure to impersonation or man-in-the-
  middle attacks.
```

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PKI-ApplianceCerts Properties

Object | Security | Attribute Editor
General | Members | Member Of | Managed By

PKI-ApplianceCerts

Group name (pre-Windows 2000): PKI-ApplianceCerts

Description:

E-mail:

Group scope
- Domain local
- Global
- Universal

Group type
- Security
- Distribution

Notes:

OK | Cancel | Apply | Help

PKI-ApplianceCerts Properties

Object | Security | Attribute Editor
General | Members | Member Of | Managed By

Members:

Name	Active Directory Domain Services Folder
Albert ChurbaDA	cs.local/csInfrastructureUsers/AdminUsers
Sean DevanneyDA	cs.local/csInfrastructureUsers/AdminUsers

Add... | Remove

OK | Cancel | Apply | Help

PKI-ApplianceCerts Properties

Object | Security | Attribute Editor
General | Members | Member Of | Managed By

Member of:

Name	Active Directory Domain Services Folder

Add... | Remove

This list displays only groups from the current domain and groups maintained in the Global Catalog, such as universal groups.

OK | Cancel | Apply | Help

PKI-ApplianceCerts Properties

Object | Security | Attribute Editor
General | Members | Member Of | Managed By

Name:

Change... | Properties | Clear

Manager can update membership list

Office:

Street:

City:

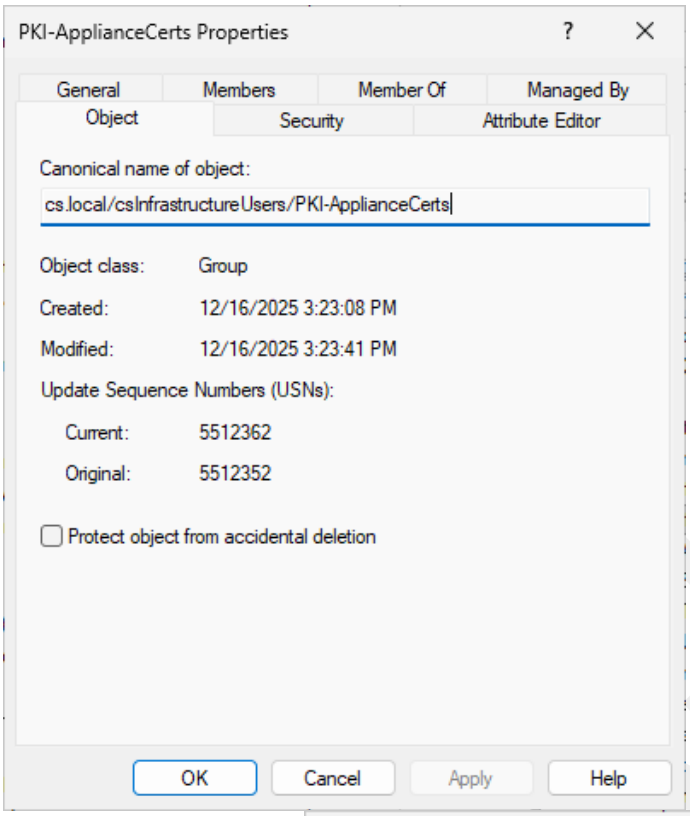
State/province:

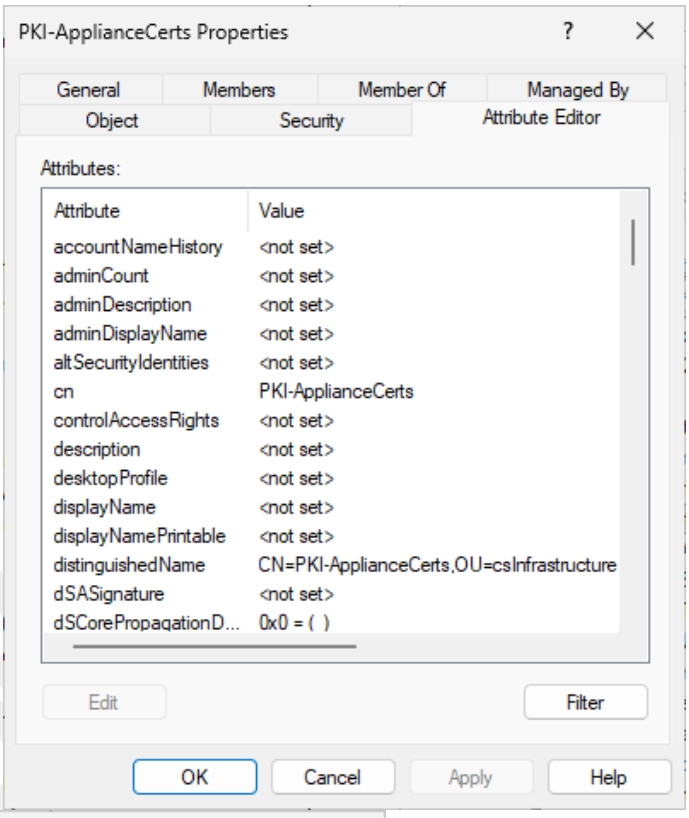
Country/region:

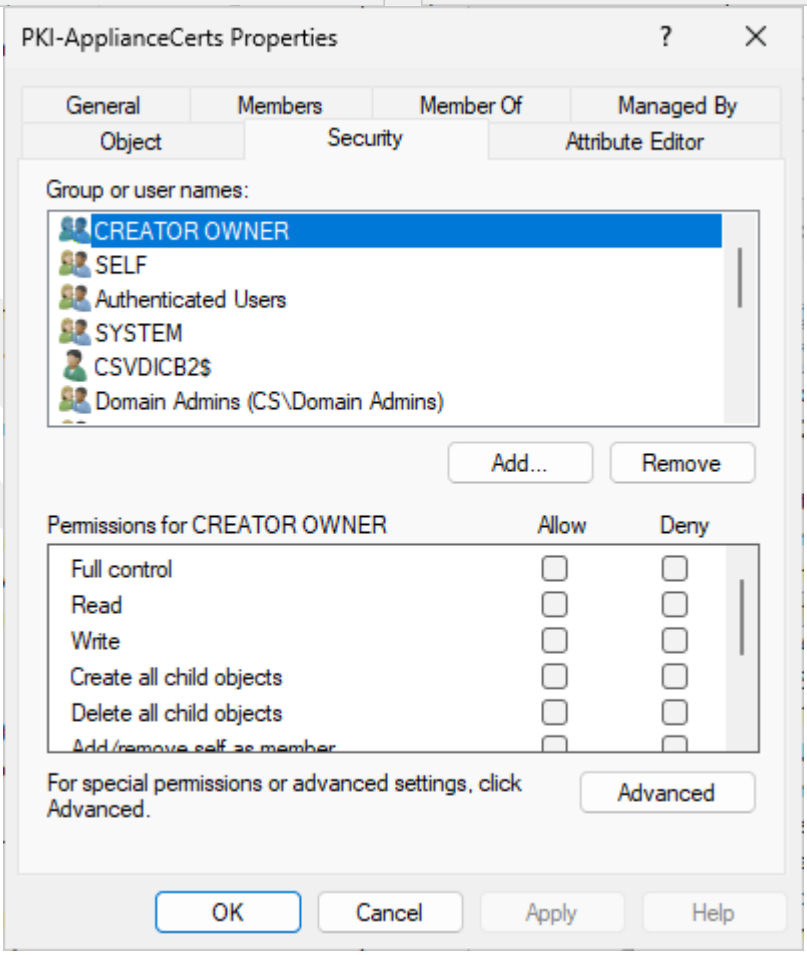
Telephone number:

Fax number:

OK | Cancel | Apply | Help


PKI-ApplianceCerts Properties
General
Members
Member Of
Managed By
Object
Security
Attribute Editor
Canonical name of object:
cs.local/csInfrastructureUsers/PKI-ApplianceCerts
Object class: Group
Created: 12/16/2025 3:23:08 PM
Modified: 12/16/2025 3:23:41 PM
Update Sequence Numbers (USNs):
Current: 5512362
Original: 5512352
Protect object from accidental deletion
OK
Cancel
Apply
Help


PKI-ApplianceCerts Properties
General
Members
Member Of
Managed By
Object
Security
Attribute Editor
Attributes:
Attribute
Value
accountNameHistory <not set>
adminCount <not set>
adminDescription <not set>
adminDisplayName <not set>
altSecurityIdentities <not set>
cn PKI-ApplianceCerts
controlAccessRights <not set>
description <not set>
desktopProfile <not set>
displayName <not set>
displayNamePrintable <not set>
distinguishedName CN=PKI-ApplianceCerts,OU=csInfrastructure
dSASignature <not set>
dSCorePropagationD... 0x0 = ( )
Edit
Filter
OK
Cancel
Apply
Help


PKI-ApplianceCerts Properties
General
Members
Member Of
Managed By
Object
Security
Attribute Editor
Group or user names:
CREATOR OWNER
SELF
Authenticated Users
SYSTEM
CSVDICB2$
Domain Admins (CS\Domain Admins)
Add...
Remove
Permissions for CREATOR OWNER
Allow
Deny
Full control
Read
Write
Create all child objects
Delete all child objects
For special permissions or advanced settings, click Advanced.
Advanced
OK
Cancel
Apply
Help

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Group: PKI-Admins

```yaml
pki_security_group:
  name: PKI-Admins
  object_type: ActiveDirectorySecurityGroup
  scope: Global
  category: Security
  purpose:
    PKI governance group. Holds read access on all hardened certificate
    templates. Membership is restricted to PKI administrators and
    security operations personnel. This group does not hold Enroll or
    AutoEnroll rights on any template.
  supported_templates:
    - All hardened certificate templates (Read only)
  membership_controls:
    membership_type: Explicit
    approved_members:
      - PKI Administrators
      - Security Operations
    prohibited_members:
      - Authenticated Users
      - Domain Users
      - Domain Computers
  enrollment_rights:
    enroll: false
    autoenroll: false
    read: true
  security_controls:
    - LeastPrivilege
    - SeparationOfDuties
    - ExplicitAuthorization
    - AuditTraceability
  audit_alignment:
    nist_800_53:
      - IA-5
      - CM-2
      - AC-6
    iso_27001:
      - A.5.15
      - A.8.24
      - A.5.17
  risk_statement:
    Without a dedicated PKI governance group, template ACL administration
    could be performed by principals with broader AD rights, undermining
    the least-privilege enrollment model enforced across all templates.
```

Appendix D – Active Directory Group Policy

Scheduled Task Enforcement

To enforce application of the certificate issuance process, a powershell module has been created. The powershell module is signed reducing risk of rouge application introduction. The scheduled task XML content should be

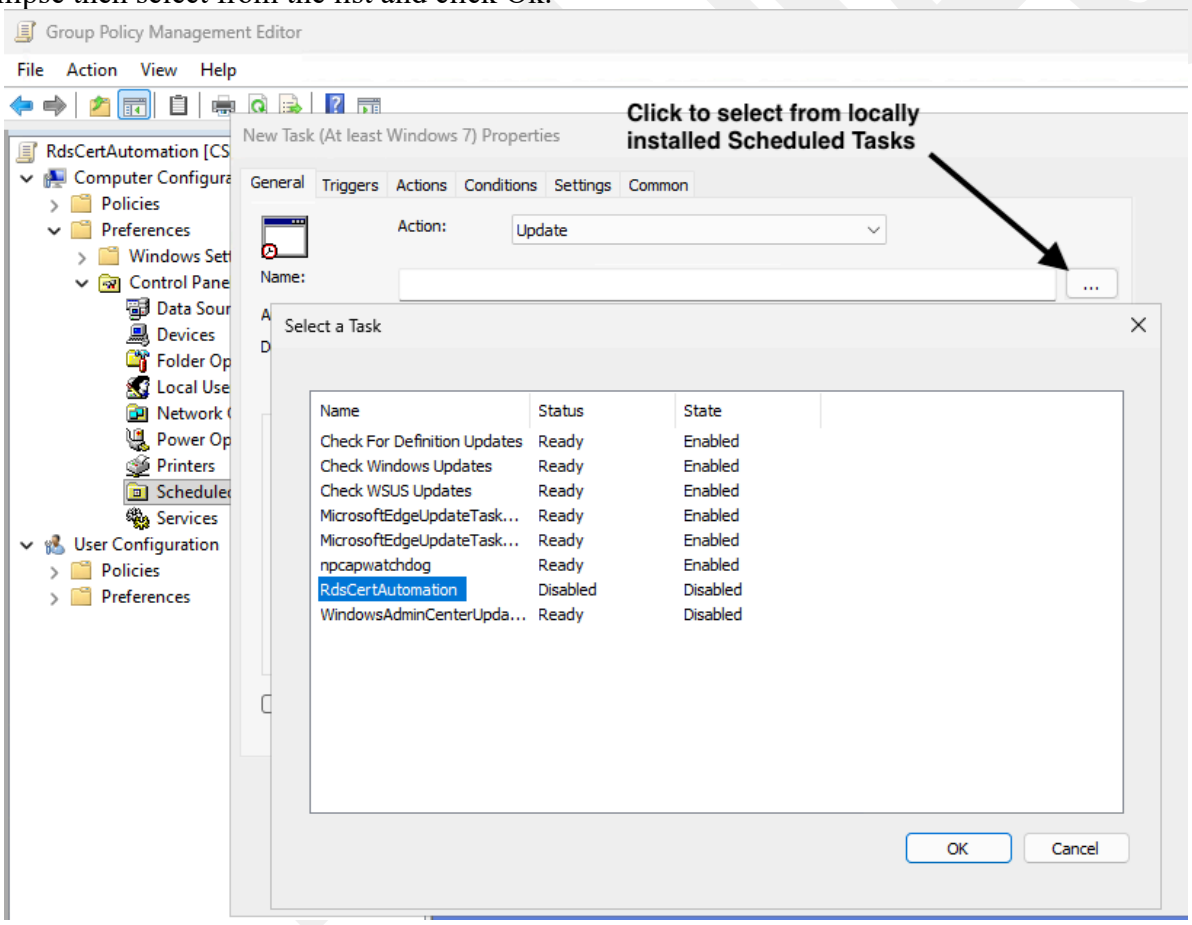
maintained in version control. The below operational notes are a guideline for creation. The RdsCertAutomation module is invoked by the RdsScheduledTask scheduled task to automate RDP and WinRM certificate lifecycle operations on domain-joined systems.

Creating the Schedule Task

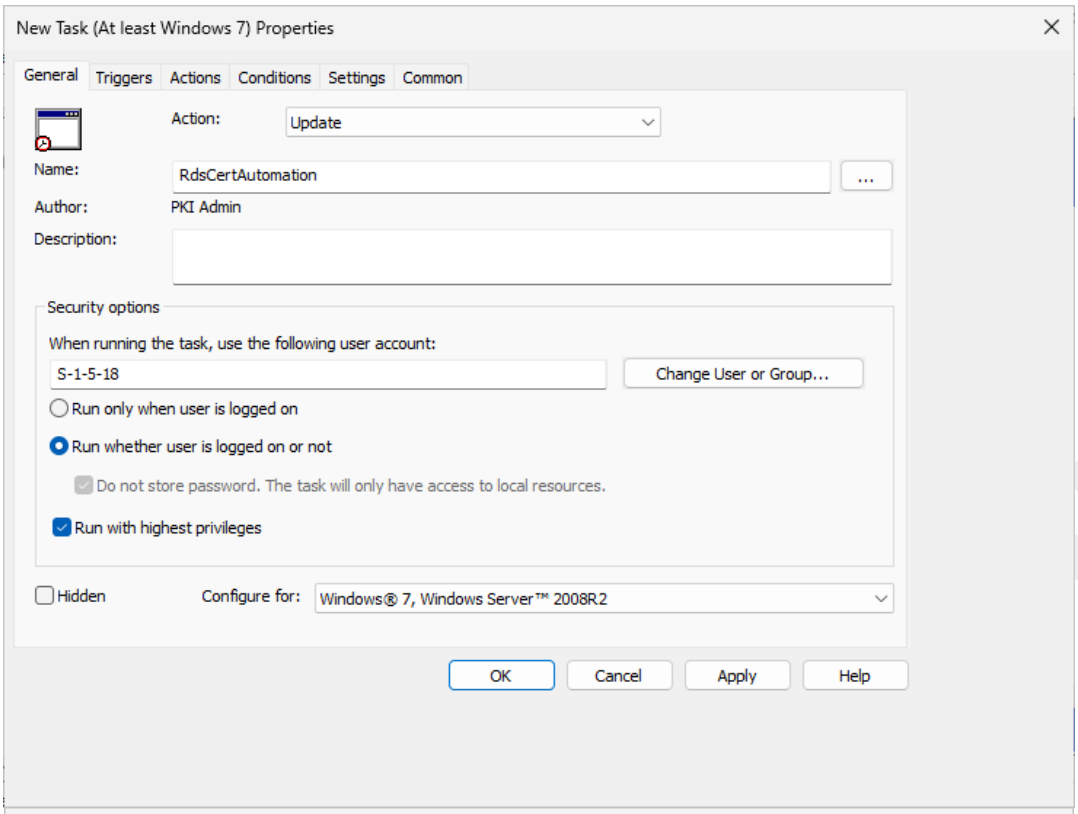
Before creating the Group Policy, import the version-controlled Scheduled Task XML file on the machine where Group Policy is being run. The version-controlled file is located at …\NETLOGON\ScheduledTasks named: RdsScheduledTask.xml.

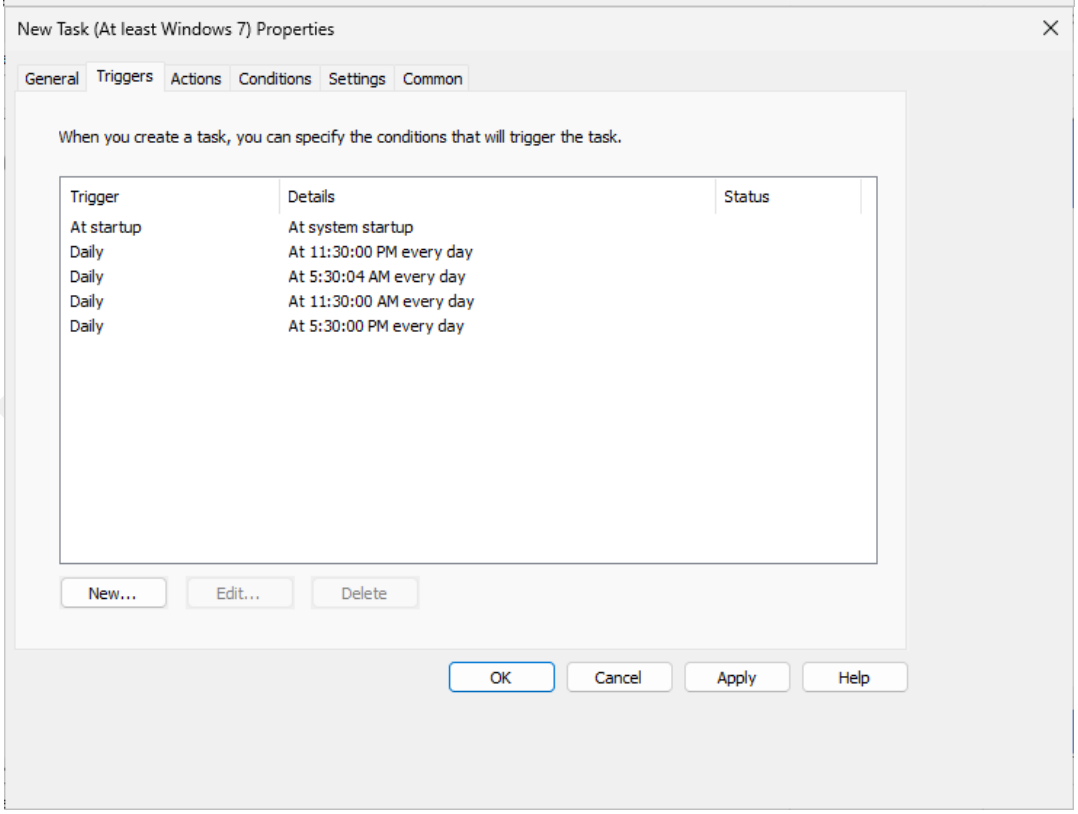
Create a New Group Policy

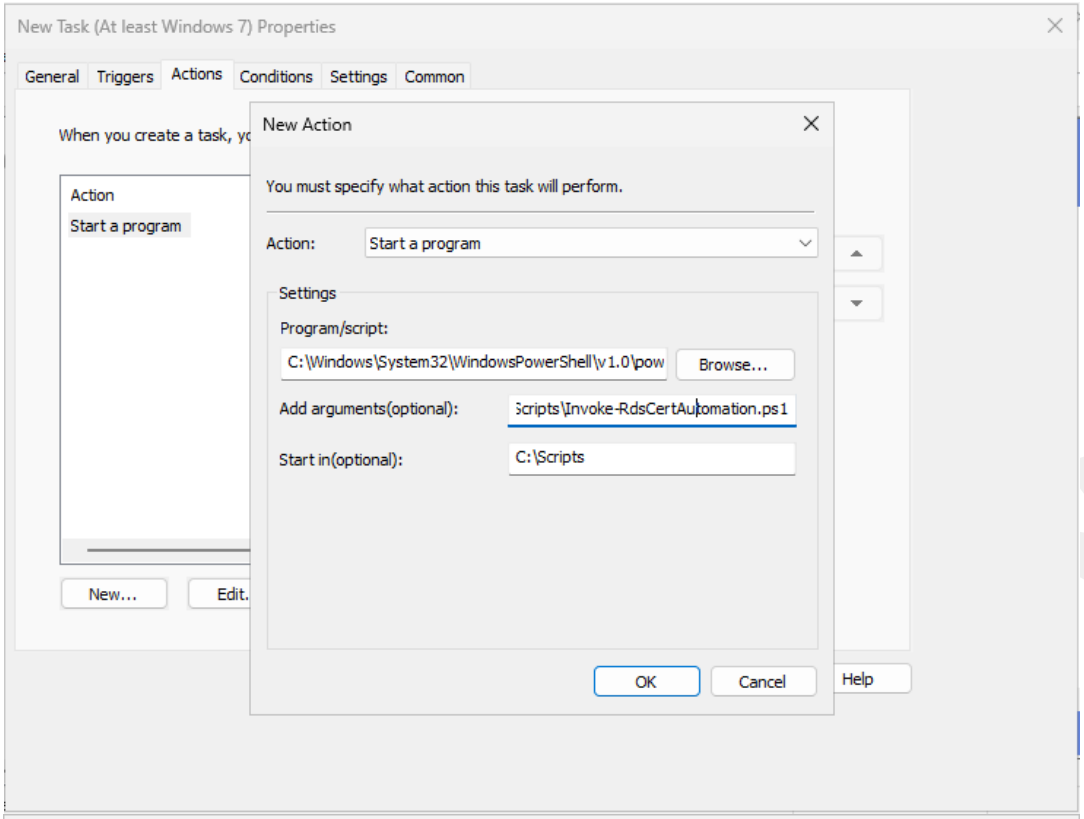
Create a new group policy named RdsScheduledTask. Edit the properties of the policy navigating to Computer Configuration/Preferences/Control Panel Settings/Scheduled Tasks
Right click in the content window selecting New/Scheduled Task (At least Windows 7). A blank task is displayed. Click the ellipse then select from the list and click Ok.

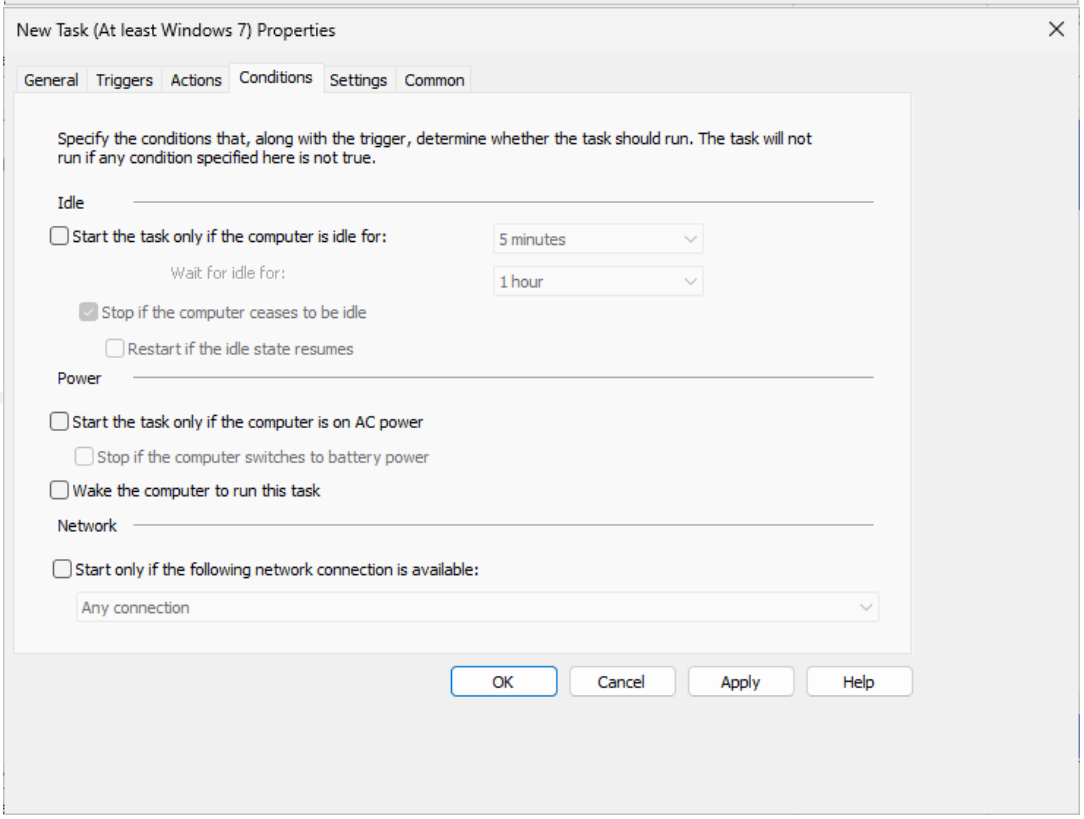


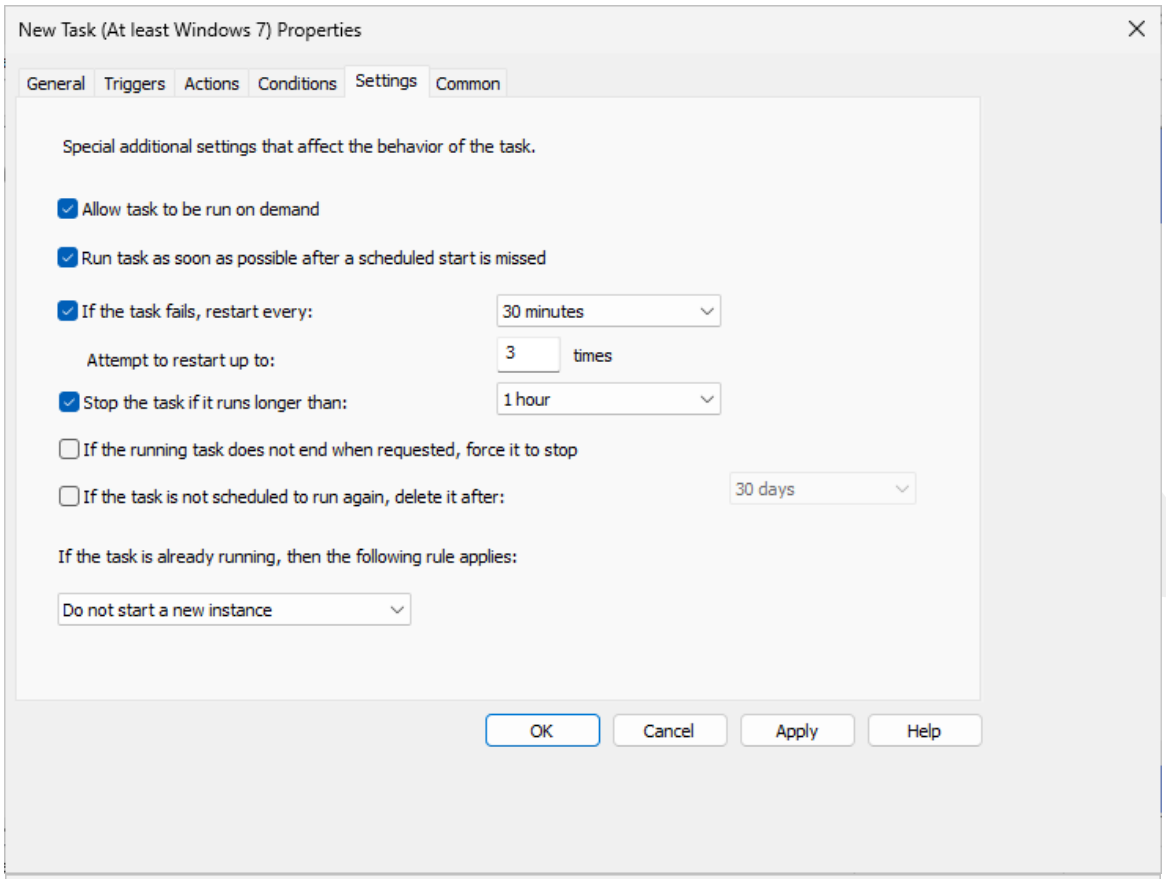
Validate the Scheduled Task properties. Keep in mind that the task needs individual triggers because GP does not fully support the repetition when random delay is enabled. To prevent swarm, every six hours with random delay of up to 1 hours overcoming the GP maximum repetition of 1 hour.

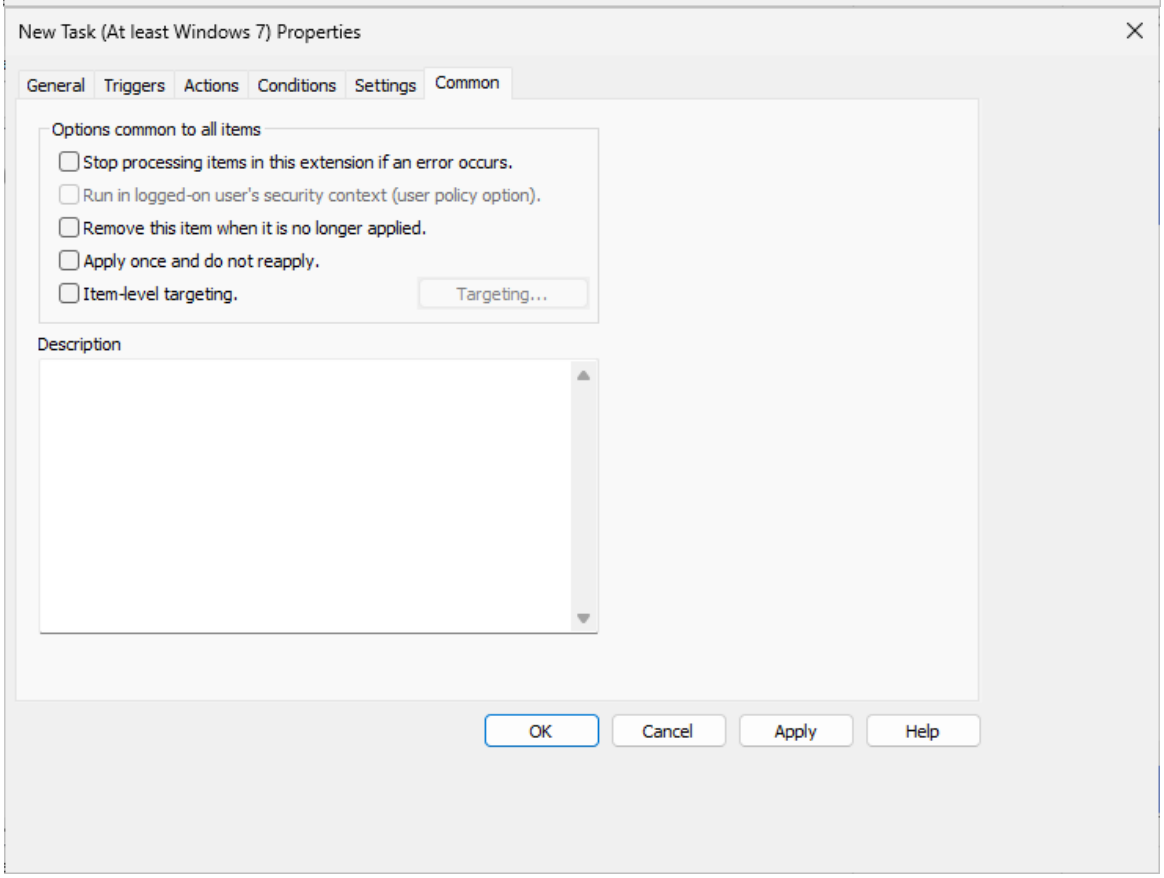

New Task (At least Windows 7) Properties
General
Triggers
Actions
Conditions
Settings
Common
Action:
Update
Name:
RdsCertAutomation
...
Author:
PKI Admin
Description:
Security options
When running the task, use the following user account:
S-1-5-18
Change User or Group...
Run only when user is logged on
Run whether user is logged on or not
Do not store password. The task will only have access to local resources.
Run with highest privileges
Hidden
Configure for:
Windows® 7, Windows Server™ 2008R2
OK
Cancel
Apply
Help


New Task (At least Windows 7) Properties
General
Triggers
Actions
Conditions
Settings
Common
When you create a task, you can specify the conditions that will trigger the task.
Trigger
Details
Status
At startup
At system startup
Daily
At 11:30:00 PM every day
Daily
At 5:30:04 AM every day
Daily
At 11:30:00 AM every day
Daily
At 5:30:00 PM every day
New...
Edit...
Delete
OK
Cancel
Apply
Help


New Task (At least Windows 7) Properties
General
Triggers
Actions
Conditions
Settings
Common
When you create a task, yo
Action
Start a program
New...
Edit...
New Action
You must specify what action this task will perform.
Action:
Start a program
Settings
Program/script:
C:\Windows\System32\WindowsPowerShell\v1.0\pow
Browse...
Add arguments(optional):
Scripts\Invoke-RdsCertAutomation.ps1
Start in(optional):
C:\Scripts
OK
Cancel
Help


New Task (At least Windows 7) Properties
General
Triggers
Actions
Conditions
Settings
Common
Specify the conditions that, along with the trigger, determine whether the task should run. The task will not run if any condition specified here is not true.
Idle
Start the task only if the computer is idle for:
5 minutes
Wait for idle for:
1 hour
Stop if the computer ceases to be idle
Restart if the idle state resumes
Power
Start the task only if the computer is on AC power
Stop if the computer switches to battery power
Wake the computer to run this task
Network
Start only if the following network connection is available:
Any connection
OK
Cancel
Apply
Help


New Task (At least Windows 7) Properties
General
Triggers
Actions
Conditions
Settings
Common
Special additional settings that affect the behavior of the task.
Allow task to be run on demand
Run task as soon as possible after a scheduled start is missed
If the task fails, restart every:
30 minutes
Attempt to restart up to:
3
times
Stop the task if it runs longer than:
1 hour
If the running task does not end when requested, force it to stop
If the task is not scheduled to run again, delete it after:
30 days
If the task is already running, then the following rule applies:
Do not start a new instance
OK
Cancel
Apply
Help


New Task (At least Windows 7) Properties
General
Triggers
Actions
Conditions
Settings
Common
Options common to all items
Stop processing items in this extension if an error occurs.
Run in logged-on user's security context (user policy option).
Remove this item when it is no longer applied.
Apply once and do not reapply.
Item-level targeting.
Targeting...
Description
OK
Cancel
Apply
Help

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Appendix E – Applied Certificate Sampling by Intent

Domain Controller Authentication

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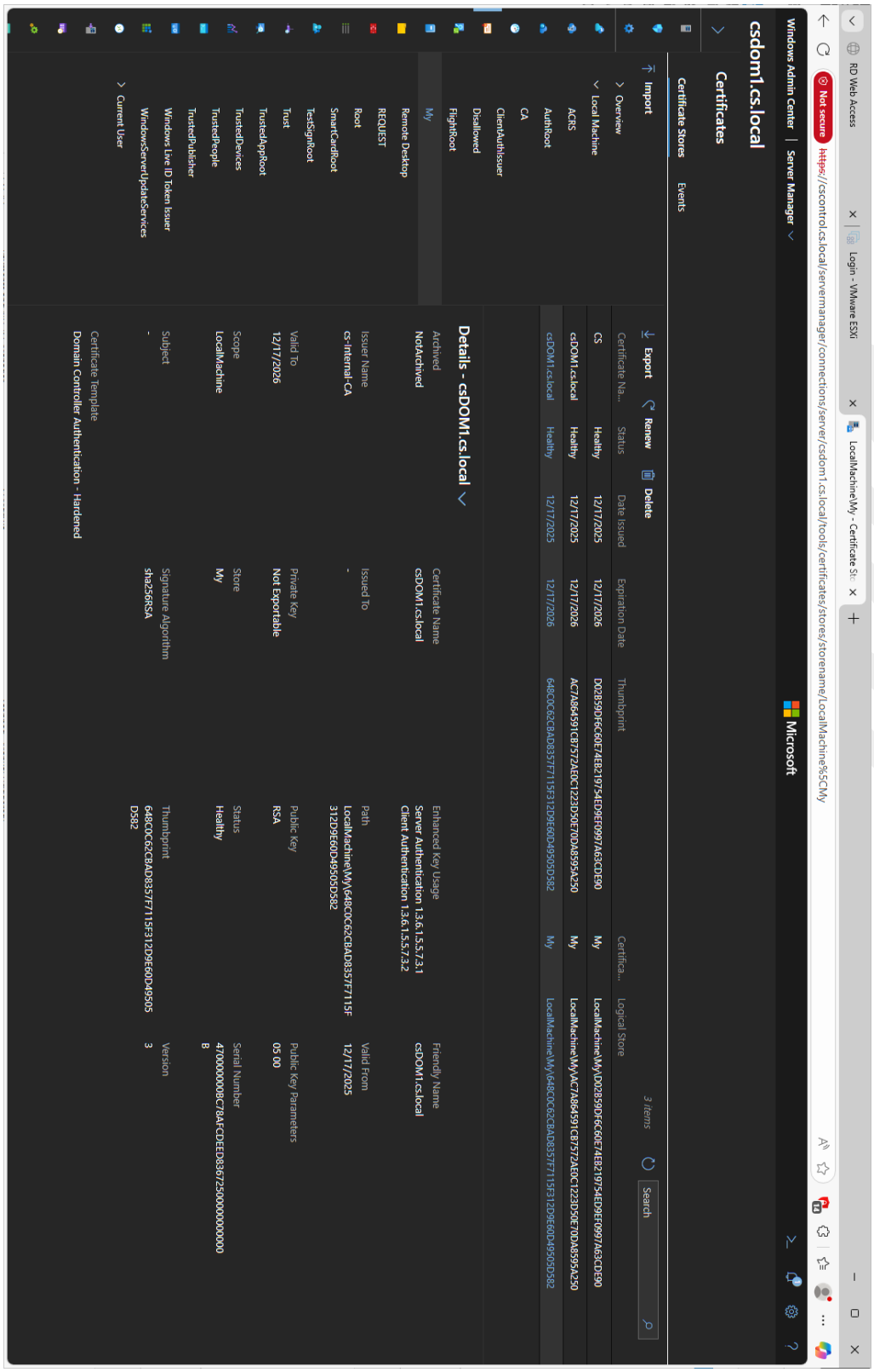
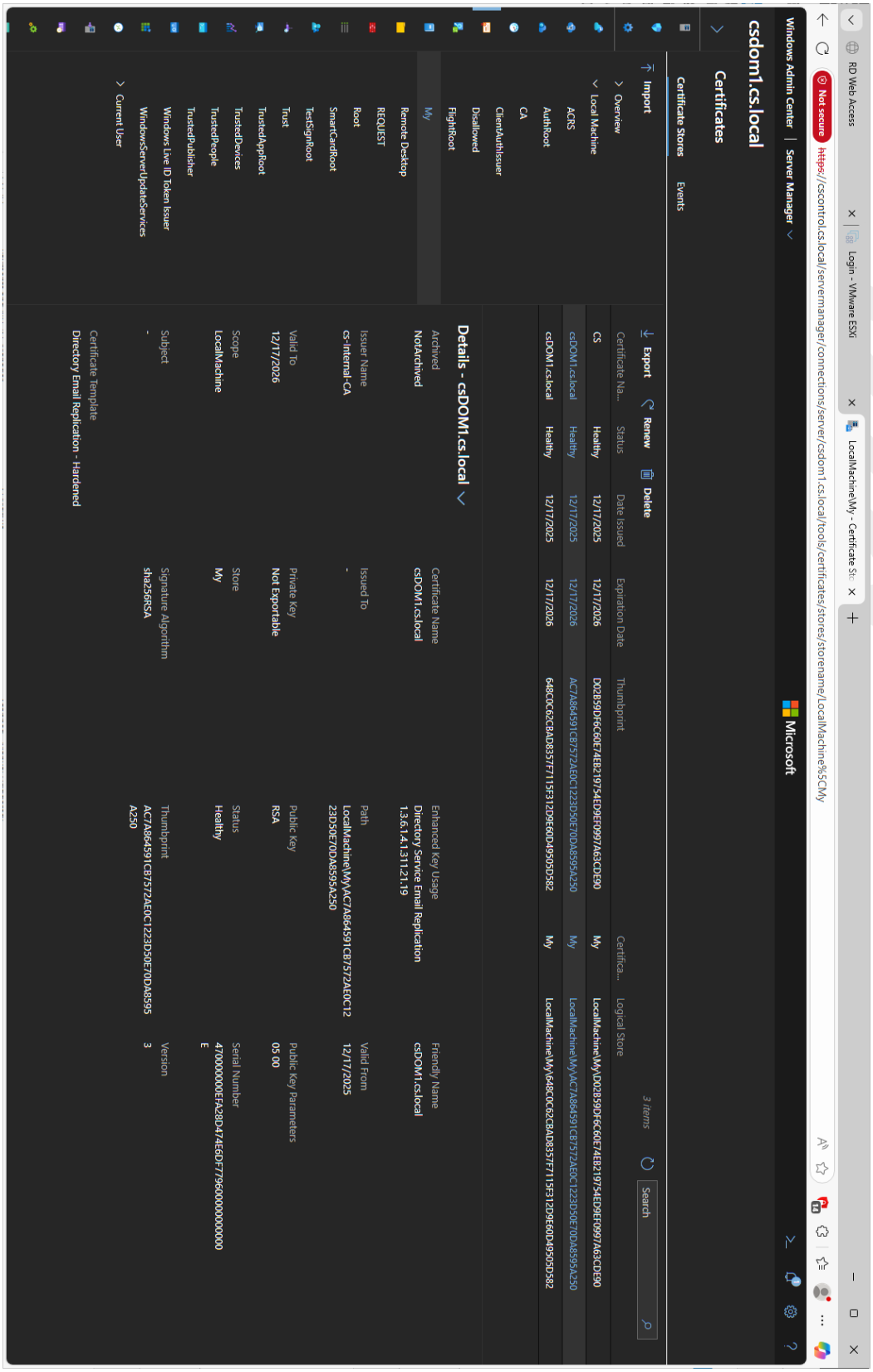

RD Web Access
Login - VMware ESXi
LocalMachine\My - Certificate St
Not secure
https://cscontrol.cs.local/servermanager/connections/server/csdom1.cs.local/tools/certificates/stores/storename/LocalMachine%5CMy
Windows Admin Center | Server Manager
Microsoft
csdom1.cs.local
Certificates
Certificate Stores
Events
Import
Export
Renew
Delete
3 items
Search
Overview
Local Machine
ACRS
AuthRoot
CA
ClientAuthIssuer
Disallowed
FlightRoot
My
Remote Desktop
REQUEST
Root
SmartCardRoot
TestSignRoot
Trust
TrustedAppRoot
TrustedDevices
TrustedPeople
TrustedPublisher
Windows Live ID Token Issuer
WindowsServerUpdateServices
Current User
Certificate Na...
Status
Date Issued
Expiration Date
Thumbprint
Certificate...
Logical Store
CS
Healthy
12/17/2025
12/17/2026
D02B59DF6C60E74EB21975ED9EF0997A63CDE90
My
LocalMachine\My
csDOM1.cs.local
Healthy
12/17/2025
12/17/2026
AC7A86459 1CB7572AE0C1223D50E70DA8595A250
My
LocalMachine\My
csDOM1.cs.local
Healthy
12/17/2025
12/17/2026
648C0C62CBAD8357F7115F312D9E60D49505D582
My
LocalMachine\My
Details - csDOM1.cs.local
Archived
NotArchived
Certificate Name
csDOM1.cs.local
Enhanced Key Usage
Server Authentication 1.3.6.1.5.5.7.3.1
Client Authentication 1.3.6.1.5.5.7.3.2
Friendly Name
csDOM1.cs.local
Issuer Name
cs-internal-CA
Issued To
-
Path
LocalMachine\My\648C0C62CBAD8357F7115F312D9E60D49505D582
Valid From
12/17/2025
Valid To
12/17/2026
Private Key
Not Exportable
Public Key
RSA
Public Key Parameters
05 00
Scope
LocalMachine
Store
My
Status
Healthy
Serial Number
470000000BC78AFCDEEDB3B72500000000000B
Subject
-
Signature Algorithm
sha256RSA
Thumbprint
648C0C62CBAD8357F7115F312D9E60D49505D582
Version
3
Certificate Template
Domain Controller Authentication - Hardened

Directory Email Replication



Kerberos Authentication

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RD Web Access | Login - VMware ESXi | LocalMachine\My - Certificate Sto

Not secure https://cscontrol.cs.local/servermanager/connections/server/csdom1.cs.local/tools/certificates/stores/storename/LocalMachine%5CMy

Windows Admin Center | Server Manager — Microsoft

csdom1.cs.local

Certificates

Certificate Stores | Events

Import

- Overview
- Local Machine
  - ACRS
  - AuthRoot
  - CA
  - ClientAuthIssuer
  - Disallowed
  - FlightRoot
  - My
  - Remote Desktop
  - REQUEST
  - Root
  - SmartCardRoot
  - TestSignRoot
  - Trust
  - TrustedAppRoot
  - TrustedDevices
  - TrustedPeople
  - TrustedPublisher
  - Windows Live ID Token Issuer
  - WindowsServerUpdateServices
- Current User

Export | Renew | Delete — 3 items — Search

Certificate Na...	Status	Date Issued	Expiration Date	Thumbprint	Certifica...	Logical Store
CS	Healthy	12/17/2025	12/17/2026	D02B59DF6C60E74EB219754ED9EF0997A63CDE90	My	LocalMachine\My\D02B59DF6C60E74EB219754ED9EF0997A63CDE90
csDOM1.cs.local	Healthy	12/17/2025	12/17/2026	AC7A864591CB7572AE0C1223D50E70DA8595A250	My	LocalMachine\My\AC7A864591CB7572AE0C1223D50E70DA8595A250
csDOM1.cs.local	Healthy	12/17/2025	12/17/2026	648C0C62CBAD8357F7115F312D9E60D49505D582	My	LocalMachine\My\648C0C62CBAD8357F7115F312D9E60D49505D582

Details - CS

Archived NotArchived	Certificate Name CS	Enhanced Key Usage KDC Authentication 1.3.6.1.5.2.3.5 Server Authentication 1.3.6.1.5.5.7.3.1 Client Authentication 1.3.6.1.5.5.7.3.2	Friendly Name CS
Issuer Name cs-Internal-CA	Issued To -	Path LocalMachine\My\D02B59DF6C60E74EB219754ED9EF0997A63CDE90	Valid From 12/17/2025
Valid To 12/17/2026	Private Key Not Exportable	Public Key RSA	Public Key Parameters 05 00
Scope LocalMachine	Store My	Status Healthy	Serial Number 470000000C875206C7BFDA27D200000000000C
Subject -	Signature Algorithm sha256RSA	Thumbprint D02B59DF6C60E74EB219754ED9EF0997A63CDE90	Version 3
Certificate Template Kerberos Authentication - Hardened			

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PKI Appliance Certs

PKI Code Signers

RDS WinRM

Web Server

Web Server Appliance

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Appendix F – Auditor Walkthrough Script

Use this script to guide an audit walkthrough when the PKI architects are not present.

1. Purpose and Scope: Internal PKI supports authentication and encryption for enterprise systems; external services use public CAs.

2. Design Intent: We enforce modern crypto, least-privilege enrollment, and automated lifecycle management while preserving compatibility.

3. Templates: Templates are separated by purpose (machine identity, DC/LDAPS, Kerberos PKINIT, web TLS, WinRM/RDP TLS) to prevent misuse.

4. Cryptography: RSA 3072/SHA-256 for critical identity and management; RSA 2048/SHA-256 for internal web services; keys non-exportable by default.

5. Enrollment: Enrollment and autoenrollment are role-scoped and reviewed; no broad user enrollment is granted.

6. Risk: Residual risk is low and mitigated by short lifetimes, automated renewal, revocation distribution, and reassessment triggers.

7. Evidence: Appendix A contains screenshots and configuration artifacts captured during implementation.

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